

A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

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Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick-slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

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Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

Symbol Summary

Symbols used throughout the derivation are summarized in [Table 1](#).

Symbol	Meaning
State and kinematics	
$\mathbf{x}(t)$	state vector
t	time
ω_p, ω_s	primary and secondary angular speed
$s, \dot{s}, \ddot{s}$	shift position, velocity, and acceleration
$R, \dot{R}$	CVT ratio and ratio rate
v_b	belt transport speed
v_b^*, T_b	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\text{outer}}(s), r_{s,\text{outer}}(s)$	primary and secondary outer radius
$r_{p,\text{eff}}(s), r_{s,\text{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\text{outer},\text{min}}$	minimum primary outer radius
$r_{s,\text{outer},\text{max}}$	maximum secondary outer radius
$s_{\text{dz}}, s_{\text{max}}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r, \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\text{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s), \frac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s), \frac{d\theta_s}{ds}$	secondary helix angle map and gradient

Symbol	Meaning
Forces and torques	
$F_{p,ax}, F_{s,ax}$	primary and secondary axial mechanism force
$F_{c,p}, F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
τ_{eng}, τ_{load}	engine input torque and reflected load torque
τ_p, τ_s	coupling torque seen at primary and secondary
τ, τ_c	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_{-}^{stick}, \tau_{+}^{stick}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_{-}^{slip}, \tau_{+}^{slip}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{stick}, \tau_{c,s,\pm}^{stick}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{slip}, \tau_{c,s,\pm}^{slip}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
I_p, I_s	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m}, m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt-pulley traction coefficient
$k_p, x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta}, \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x}, x_{s,0}$	secondary axial stiffness and preload displacement

Table 1: Core symbols used throughout the derivation.

Constants and Parameters

All parameters in [Table 2](#) are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

Symbol	Name	Unit	Description
Geometry and belt			
L_b	Belt length	m	Fixed total belt length
β	Sheave half-angle	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	m	Radial belt thickness
b_{out}	Belt outer width	m	Belt width at outer face
b_{in}	Belt inner width	m	Belt width at inner face
$r_{p,outer,min}$	Primary minimum outer radius	m	Radius at fully open primary
$r_{s,outer,max}$	Secondary maximum outer radius	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	m	Onset of effective radial motion

Symbol	Name	Unit	Description
$s_{\max}$	Maximum shift	m	Physical upper travel bound
Primary/secondary actuation			
m_f	Flyweight mass	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	rad	Initial torsional deflection preload
r_h	Helix reference radius	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	m	Initial compression of secondary axial spring
Inertia and drivetrain			
I_{eng}	Engine inertia	kg m ²	Engine rotational inertia
I_{prim}	Primary pulley inertia	kg m ²	Primary CVT pulley inertia
I_p	Effective primary inertia	kg m ²	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	kg m ²	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	kg m ²	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	kg m ²	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	kg m ²	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	1	Fixed ratio defined by $\omega_s = G \omega_w$
m	Vehicle mass	kg	Total vehicle mass
r_w	Wheel rolling radius	m	Effective tire rolling radius
Contact and environment			
ρ_b	Belt density	kg/m ³	Belt material density
μ_s	Belt-pulley static friction coefficient	1	Static traction coefficient used at impending slip
μ_k	Belt-pulley kinetic friction coefficient	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	1	Aerodynamic drag coefficient
A_f	Frontal area	m ²	Vehicle aerodynamic reference area
ρ	Air density	kg/m ³	Ambient air density used in drag model
g	Gravitational acceleration	m/s ²	Gravity constant

Table 2: Canonical non-varying parameters used by the CVT model equations.

Chapter 1

Introduction & Background

1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt-pulley contacts, and the vehicle drivetrain to which it is attached.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, while drivetrain inertia, road grade, and external vehicle load react back through the secondary side. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from vehicle response alone. A shift curve, engine-speed trace, or loss of acceleration may reflect the combined effect of primary engagement, secondary reaction, belt traction, shift motion, and changing drivetrain load. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during vehicle testing. This is especially consequential in racing-oriented applications such as Baja SAE, where launch behaviour, shift response, and torque transfer strongly affect performance, while changes to springs, flyweights, ramps, helices, or other hardware may require fabrication and repeated vehicle testing before their effect can be assessed.

Addressing this problem requires a physically interpretable dynamic framework that connects the internal CVT mechanisms to the vehicle-level response they produce. The model should be able to begin from varied initial conditions and respond to changing engine torque, road grade, external load, and other time-varying inputs, so that launch, clutching, shifting, backshift, traction loss, and vehicle acceleration emerge from the same coupled system response.

The remainder of this chapter develops the context for that framework. [Section 1.2](#) introduces the physical layout and operating principles of the belt-sheave CVT, including ratio change, traction,

mechanical actuation, and the surrounding drivetrain. [Section 1.3](#) contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, shift dynamics, and drivetrain coupling. Given this context, [Section 1.4](#) then states the aim, scope, assumptions, and contributions of the present formulation.

1.2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and why the CVT must be understood as part of a larger drivetrain system.

1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in [Figure 1.1](#). A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. [Figure 1.2](#) contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.

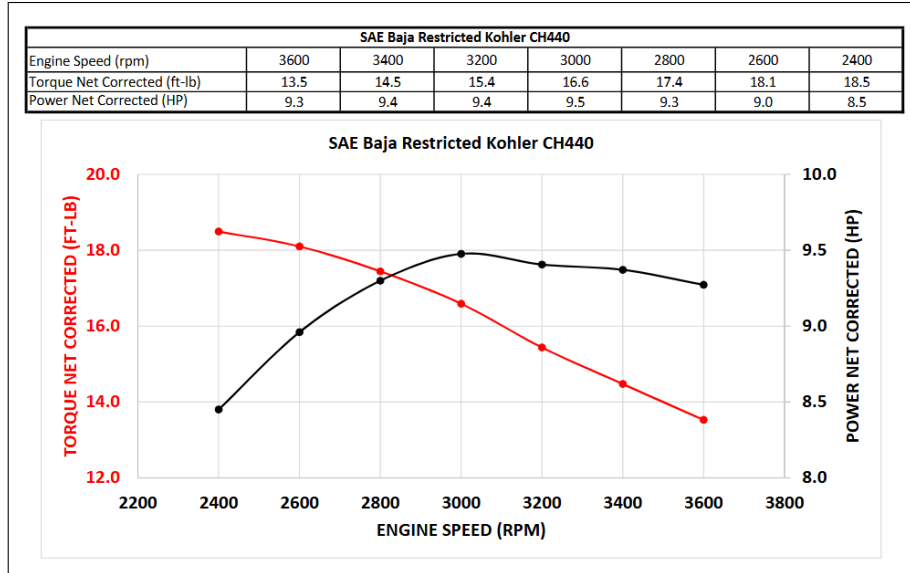


Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. ([Baja SAE, 2022](#))

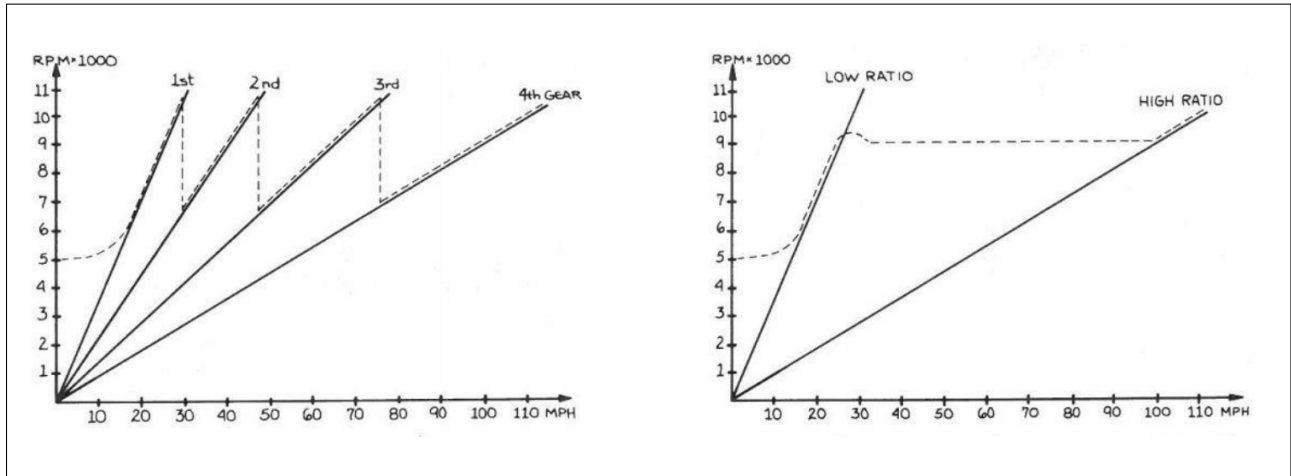


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. A stepped gearbox requires repeated engine-speed sweeps and drops between gears, whereas a CVT can vary its ratio continuously to keep the engine closer to a desired operating speed as vehicle speed increases. ([Aaen, 2007](#))

1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in [Figure 1.3](#). In this arrangement, two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

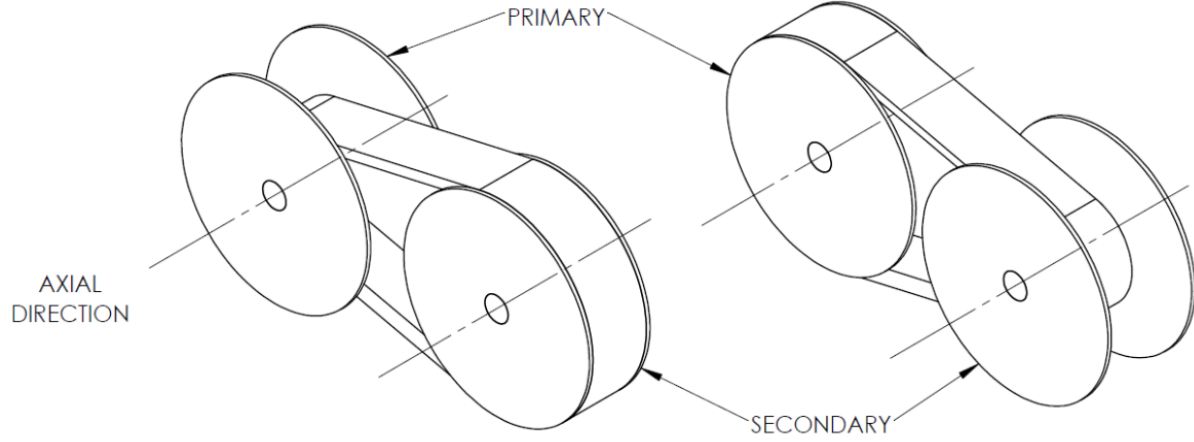


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in [Figure 1.4](#).

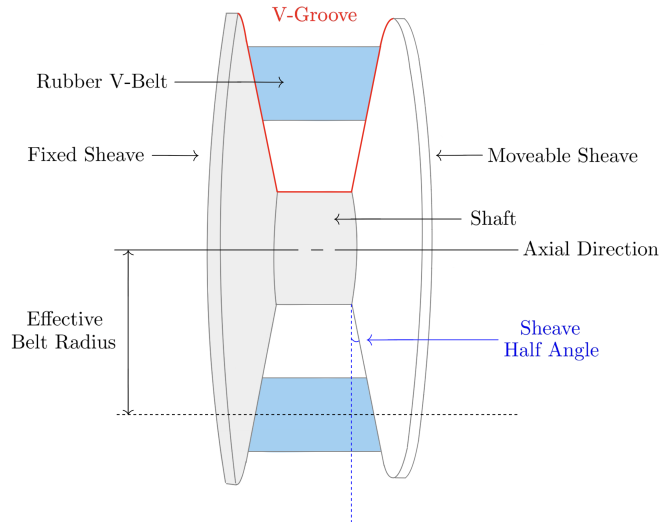


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower

groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in [Figure 1.5](#).

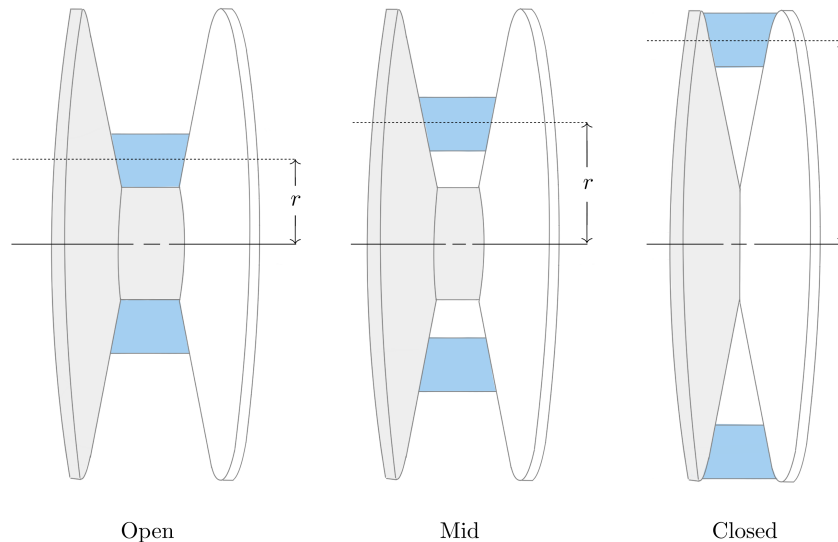


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its

launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in [Figure 1.6](#).

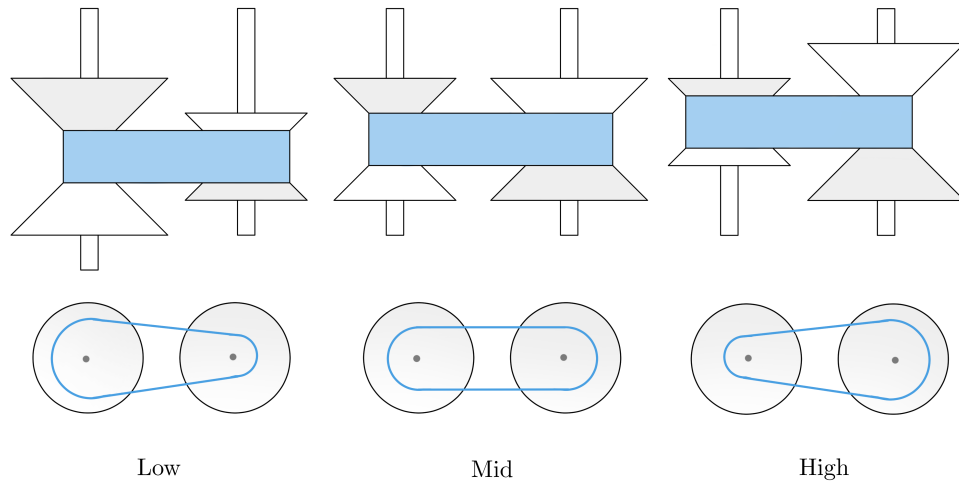


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt-sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. [Figure 1.7](#) shows how this happens at a small portion of the belt-pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt-pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without

excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

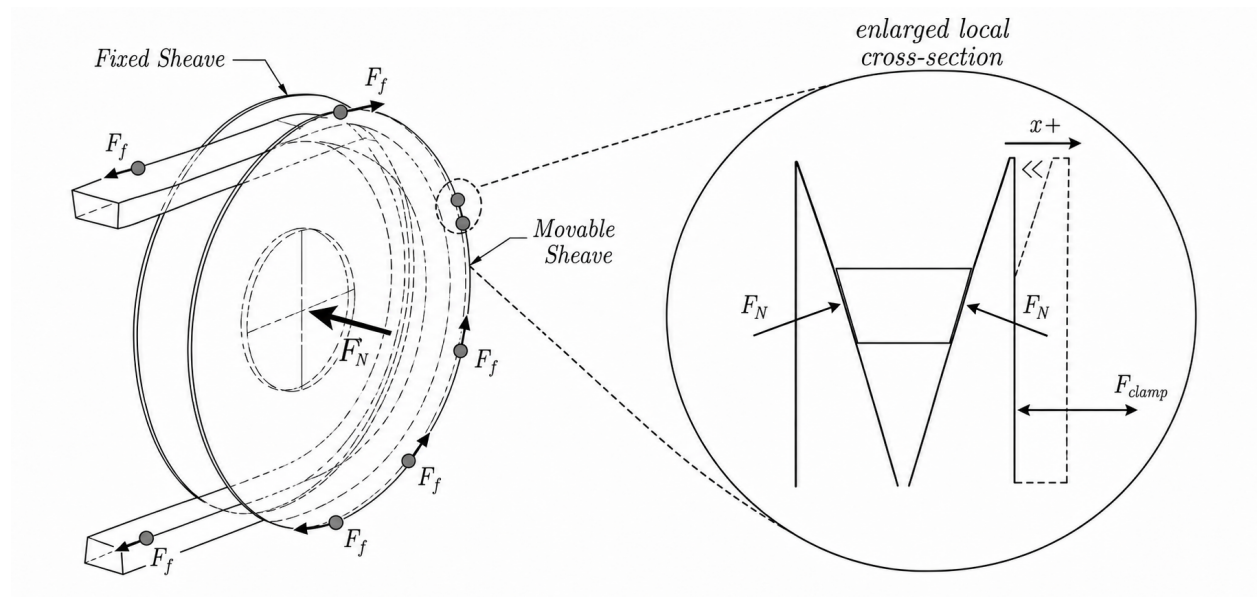


Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in [Figure 1.6](#). The primary actuation mechanism is shown conceptually on the left side of [Figure 1.8](#).

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of [Figure 1.8](#).

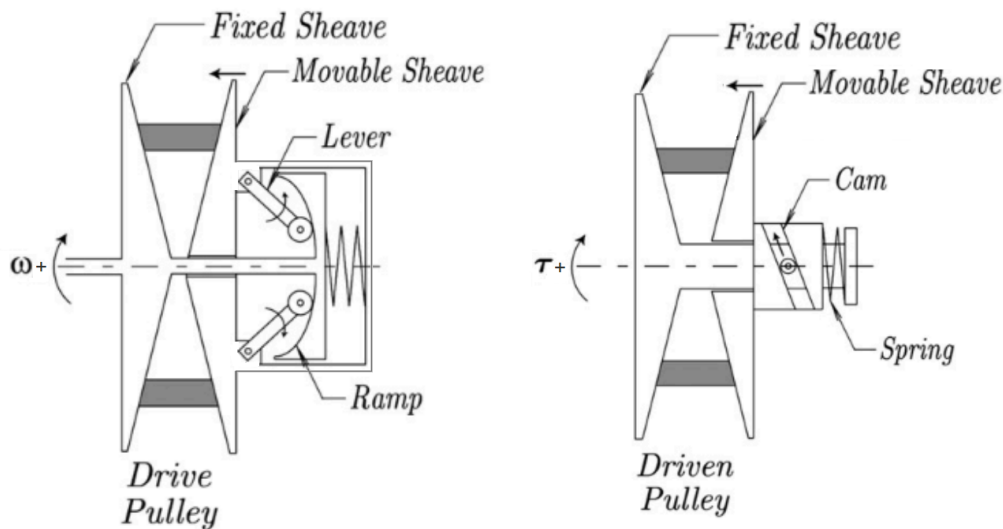


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. ([Juli'o and Plante, 2011](#))

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. The primary is mainly driven by speed, the secondary is mainly driven by transmitted torque, and the resulting CVT behavior depends on their combined interaction with the belt, pulley geometry, drivetrain inertia, external load, and available clamping force.

1.2.6 Full Drivetrain Architecture

The CVT does not operate in isolation. The physical drivetrain architecture considered in this work is shown in [Figure 1.9](#). The figure shows the CVT within the complete vehicle context, including the downstream gearbox, disconnect, shafts, bevel units, axles, and all four wheels.

Engine torque enters the primary pulley, passes through the belt to the secondary pulley, and then enters the downstream driveline. This driveline contains a fixed-ratio gearbox and the remaining shaft, bevel, and axle components that transmit torque to the wheels. At the tire-ground interface, tractive force accelerates the vehicle, while vehicle inertia, grade resistance, rolling resistance, and aerodynamic drag produce the load that feeds back into the drivetrain through the secondary side.

Although the full physical architecture is shown in [Figure 1.9](#), the dynamic model does not resolve the individual motion of every U-joint, bevel unit, axle, or wheel. Instead, their combined effect is represented through the downstream fixed reduction, equivalent drivetrain inertia, and vehicle longitudinal load. This preserves the coupling that matters to the CVT while avoiding unnecessary detail outside the scope of the present model.

Many drivetrain architectures are possible, but the engine-CVT-downstream driveline-wheel arrangement shown here is the system modeled in this work. This system-level view is important because the CVT ratio is not simply chosen independently of the rest of the vehicle. Engine speed affects primary actuation. Transmitted torque affects secondary clamping. Vehicle load affects the torque carried by the secondary. Belt traction determines whether that torque can be carried without slip. All of these quantities evolve together during a launch.

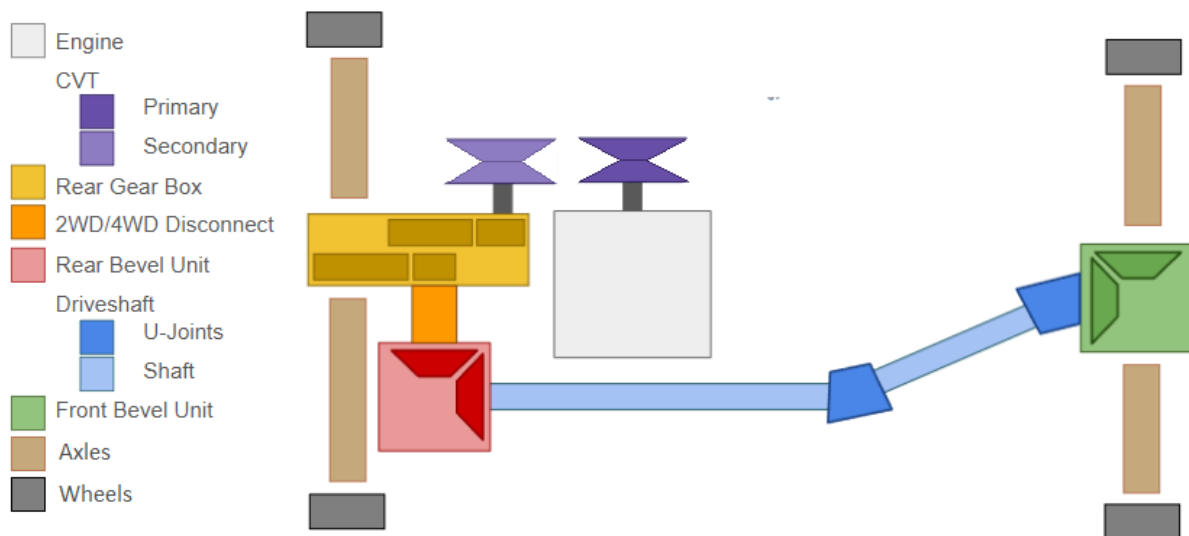


Figure 1.9: Physical drivetrain architecture considered in this work. The figure shows the CVT within the complete four-wheel vehicle layout, including the downstream gearbox, disconnect, shaft, bevel, and axle components. In the dynamic model, these downstream elements are represented by an equivalent fixed reduction, reflected inertia, and vehicle load acting through the secondary side of the CVT.

The rest of this document turns this physical picture into a dynamic model. The geometry describes how pulley radii change during shift. The force models describe how the primary and secondary sheaves are driven. The traction model describes when torque transfer can stick and when slip occurs. The drivetrain model closes the loop between engine torque, CVT behavior, vehicle load, and vehicle acceleration. Together, these pieces explain how the CVT behaves as a coupled mechanical system during launch.

1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict launch behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt-pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, to full transient launch behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt-sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system whose engine speed, belt state, shift position, and vehicle motion evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a drivetrain problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model tries to explain how the system moves from one condition to another.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but whether clamping force is prescribed from outside the model or emerges from the same system being simulated.

A third choice is how much of the belt-pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove. More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so

that local belt motion and contact behaviour are solved directly (Juli'o and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it also changes the purpose of the model: a model built for efficiency or local belt force prediction may not be the most transparent way to study the full drivetrain response.

A fourth choice is how the rotating system is coupled. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that input speed, output speed, and transmitted torque evolve together (Srivastava and Haque, 2006; Ballew, 2015; Duan et al., 2016). Extending this further, the input may be coupled to an engine model and the output may be coupled to a dynamometer or vehicle load. This determines whether the CVT is being studied as an isolated variator or as one part of a larger powertrain.

These choices build on one another. A model can have detailed belt contact but prescribed pulley speeds. Another can include vehicle dynamics but use a simplified belt law. Another can model mechanical actuation carefully but treat shifting as quasi-static. None of these choices is inherently wrong; they reflect the question being asked. The important point is that the assumptions define the model's domain of usefulness.

Launch and clutching sit near the intersection of these choices. In that regime, ratio change, actuation, belt contact, rotational dynamics, and external loading can all matter at once. This does not make launch the only goal of CVT modelling, but it does make launch a useful stress test of model closure: a formulation that prescribes too many of these quantities may still be valuable, but it is not resolving the same coupled problem. The following review uses this framework to move from practical and quasi-static design models, to mechanical actuation and shift mechanics, to belt contact and efficiency models, to transient rubber-belt formulations, and finally to related dynamic models in metal-belt and chain CVTs.

1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett (2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (1979) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the full system response under changing operating conditions.

1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models attempt to convert practical component knowledge into quantitative design procedures. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. They commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of [Arango and Mu noz Alzate \(2020\)](#), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. [Cammalleri \(2005\)](#) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. [La Battaglia et al. \(2022\)](#) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a full transient simulation of the drivetrain.

Within Baja-style applications, [Skinner \(2020\)](#) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally describe the complete time history of clutching, slip, shift motion, and vehicle acceleration.

1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

[Cammalleri \(2005\)](#) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are

treated as part of the variator design, rather than as external force inputs. This helps establish the link between actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT.

[Kim et al. \(2002\)](#) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their work is useful because it connects the mechanically generated driver and driven pulley forces to the resulting CVT behaviour. Although these earlier mechanical-actuator models include torque-reactive secondary behaviour, they simplify how torque is shared and reacted through the driven sheave assembly.

[Sorge and Cammalleri \(2011\)](#) address this secondary-side issue in more detail by studying the helical shift mechanics of rubber V-belt variators. Their work shows that the relative rotation between the driven sheaves is important to the secondary response, and that the helix should not be treated only as a simple torque-to-axial-force conversion. This makes the secondary a coupled torque-reactive mechanism whose axial force depends on helix geometry, transmitted torque, spring loading, and the internal motion of the driven pulley.

Together, these works show that mechanical actuation is more than a source of prescribed clamp force. The primary and secondary mechanisms are part of the dynamics of the variator: speed, torque, spring preload, ramp or helix geometry, and pulley motion all influence the axial forces applied to the belt. This establishes the actuator-side foundation needed before considering the belt contact and transient shift response.

1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. [Gerbert \(1996\)](#) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that “slip” is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. [Kim and Kim \(1989\)](#) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. [Sorge \(1996\)](#) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by [Kong and Parker \(2005\)](#), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. [Kong and Parker \(2006\)](#) extend this

framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick’s work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete drivetrain to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that efficiency depends on multiple interacting mechanisms, not just on whether the belt remains below a gross slip threshold.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also shows why contact fidelity must be chosen carefully. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or local belt mechanics. A lower-dimensional contact model may be more appropriate when the goal is to preserve physical intuition while coupling the belt to actuator dynamics, rotational dynamics, and drivetrain response.

1.3.6 Transient Rubber-Belt Models and the Launch Problem

Transient rubber-belt models address time-dependent behaviour more directly, but “transient” can mean several different levels of closure. A model may study how the belt migrates radially during shift, how local belt elements move through the pulley contact, how the variator responds to prescribed axial forces, or how the CVT interacts with an engine and vehicle load. Launch lies near the most coupled end of this spectrum. During launch, the belt may be slipping, the engine may be accelerating, clamping forces may be changing, the secondary torque reaction depends on transmitted torque, and the vehicle load evolves as the vehicle begins to move. A launch model therefore has to manage torque transfer and ratio change while the drivetrain state is still developing.

A useful bridge between steady force-balance models and fully transient belt simulations is the shift-mechanics work of Sorge (2008). Sorge shows that ratio change in a rubber belt variator is not simply a sequence of steady operating points. During shift, the belt moves radially through the sheave grooves while also travelling circumferentially and transmitting torque, so belt mass conservation, radial motion, sliding, and circumferential transport interact. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for this shift-mechanics problem, making the theory more accessible for engineering calculations. These works isolate the mechanics of ratio change itself, without attempting to close the full engine–vehicle launch problem.

Juli’o and Plante (2011) developed an experimentally validated transient model of rubber-belt CVT mechanics using a discretized belt formulation. This work represents the belt as a distributed mechanical body rather than relying only on lumped contact assumptions. Its strength is that local belt motion and contact response are resolved directly. The tradeoff is that the global mechanism chain becomes less compact, and the relationship between model behaviour and a small set of

practical tuning variables becomes less direct.

Ballew (2015) continues in the discretized-belt direction and expands the model to include engine torque output and vehicle loads such as rolling resistance and aerodynamic drag. Ballew also argues against prescribing input pulley speed, since doing so removes an important rotational degree of freedom. This makes the work especially relevant to drivetrain-level rubber-belt CVT simulation. At the same time, the reported simulations use a PI controller with feed-forward gain to generate primary axial force, fix the secondary axial force to simplify the simulation, and introduce damping between belt nodes to manage numerical behaviour. Thus, the model includes important engine and vehicle coupling, but it does not close the launch problem through a mechanically self-actuated flyweight and helix model.

The transient rubber-belt literature therefore provides important precedents, while also showing why the problem remains difficult. Shift-mechanics models clarify how ratio change occurs, discretized belt models resolve local belt behaviour, and drivetrain-coupled models show how engine and vehicle dynamics can be included. The remaining challenge is not simply to make the belt model more detailed, but to choose a level of fidelity that preserves the coupling between actuation, torque transfer, ratio evolution, and vehicle response without losing physical interpretability.

1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive dynamic modelling tradition for metal push-belt and chain CVTs, especially in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, but they are useful adjacent references because they show how other CVT architectures have handled shift dynamics, slip, clamping force, deformation, and validation.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs, covering modelling and control strategies for friction-limited transmissions. Their review is useful not only because it summarizes the field, but because it shows that CVT modelling has developed unevenly across architectures. Different CVT types are modelled with different assumptions, and fully dynamic, experimentally validated models remain less common than steady-state or control-oriented formulations. The review also reinforces an important distinction: rubber V-belts, metal push belts, and chains should not be treated interchangeably because their contact mechanics, actuation methods, deformation behaviour, torque capacity, and control objectives differ.

For metal belt CVTs, Carbone et al. (2005) study the influence of pulley deformation on shifting behaviour. Their model explains the distinction between creep-mode and slip-mode shifting and shows that pulley deformation can be essential to reproducing slow-shift behaviour under Coulomb-friction assumptions. Yildiz et al. (2017) later validate the Carbone–Mangialardi–Mantriota model under steady-state conditions and shift manoeuvres with torque load, supporting the usefulness of this modelling framework for controlled metal-belt or chain CVTs.

Other metal-belt studies provide useful analogues for traction-limited behaviour and ratio-change dynamics. Srivastava and Haque (2006) model the dynamics of metal V-belt CVTs, while Bonsen et al. (2004) analyze slip behaviour and the relationship between slip and transmitted torque. These works are relevant because they treat slip and dynamic ratio change as part of the variator mechanics, even though the transmission architecture differs from a dry rubber V-belt system.

Chain CVTs provide another adjacent comparison. Duan et al. (2016) develop a physics-based chain CVT model beginning from the kinematics and dynamics of an infinitesimal chain segment. Their model includes speed-dependent friction, pulley axial dynamics coupled with chain radial

movement, and pulley deformation. This makes the work a useful example of a physically coupled CVT formulation in which rotational dynamics, axial dynamics, contact mechanics, and friction state are treated together.

The important distinction is that these models are usually developed for lubricated automotive metal-belt or chain systems with hydraulic or electronic clamping. Their methods are therefore valuable, but their assumptions cannot be imported directly into dry rubber V-belt CVTs with mechanical flyweights, springs, and torque-sensitive helices. They show what is possible in adjacent CVT architectures, while also reinforcing the need for a formulation matched to rubber-belt contact and mechanical actuation.

1.3.8 Synthesis of Prior Work

The literature establishes a complementary set of foundations rather than a single preferred model family. Practical and design references identify the hardware variables and force relationships that govern rubber V-belt CVTs. Mechanical-actuation studies show how primary and secondary clamping forces can arise from the pulley mechanisms themselves. Contact and efficiency models resolve the belt–pulley interface in increasing detail, while transient formulations address ratio change, belt motion, and, in some cases, drivetrain response. Adjacent metal-belt and chain formulations further demonstrate how axial, rotational, and traction dynamics can be coupled when the transmission architecture and actuation system are different.

The remaining difficulty is not any single missing equation, but the closure of the whole system under assumptions appropriate to a mechanically actuated dry rubber V-belt CVT. Launch and clutching sit at the intersection of engine acceleration, primary engagement, secondary torque reaction, belt traction, shift motion, and vehicle load. Treating one or more of these quantities as a prescribed input can be appropriate for a focused study, but it changes the problem being solved. The central modelling challenge is therefore to retain the dominant couplings required for transient vehicle response without obscuring the system behind unnecessary local detail.

Table 1.1 compares selected formulations that most directly bracket the present modelling objective. The preceding sections discuss the individual models in detail; the table brings together their principal modelling choices and contributions. It therefore provides a comparison of modelling scope across the selected formulations.

Selected formulation	Ratio / shift	Actuation closure	Belt / contact treatment	Rotational vehicle coupling	Principal contribution
Cammalleri (2005)	Quasi-static equilibrium	Mechanical primary and secondary	Analytical belt-force model	Steady imposed conditions	Mechanical variator design with centrifugal and torque-sensitive actuation
Kim et al. (2002)	Steady plus empirical transient response	Mechanical primary and secondary	Simplified belt-force model	Externally supplied conditions	Empirical transient response of a mechanically actuated rubber-belt CVT
Sorge (2008)	Derived transient shift mechanics	Pulley loading prescribed	Radial belt motion and sliding during shift	No engine-vehicle model	Derived mechanics showing that ratio change is inherently transient
Sorge and Cammalleri (2011)	Secondary helix motion	Torque-reactive helix and spring	Belt loads supplied to secondary	No complete drivetrain model	Secondary helix resolved as a coupled torque-rotation-axial mechanism
Messick (2018)	Quasi-equilibrium operating states	Axial loading prescribed	Detailed distributed contact, thrust, and efficiency model	Dynamometer operating points	Experimentally validated rubber-belt contact, axial-thrust, and efficiency formulation
Juli'o and Plante (2011)	Transient discretized-belt response	Axial and kinematic inputs prescribed	Local belt inertia, elasticity, and contact	External speed and load conditions	Experimentally validated transient rubber-belt mechanics
Ballew (2015)	Transient discretized-belt response	Primary controlled; secondary force fixed	Nodal stick-sliding contact with damping	Engine torque and vehicle road loads included	Engine-vehicle coupling with controlled or prescribed pulley clamping
Duan et al. (2016)	Dynamic axial and radial shift	Applied clamping and axial dynamics	Chain elasticity, slip, and pulley deformation	Coupled pulley and chain dynamics	Coupled axial, rotational, contact, and friction dynamics in a chain CVT
Present formulation	Dynamic shift coordinate	Mechanical flyweight primary and helix secondary	Lumped geometry and gross-slip traction model	Engine, belt, pulleys, and vehicle coupled in time	Mechanically actuated, vehicle-coupled transient rubber-belt framework

Table 1.1: Selected CVT formulations most relevant to the present modelling objective. The table summarizes the principal closure choices and contribution of each formulation.

1.4 Aim, Scope, and Contributions

The literature review in Section 1.3 motivates a specific modelling objective: to develop a physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT coupled to a vehicle drivetrain. The formulation is intended to close the dominant system-level interactions within a single time-domain model, allowing the transmission and vehicle response to evolve together under changing operating conditions. It is not intended to provide the highest-fidelity representation of local rubber-belt contact or a complete efficiency-loss model; instead, it focuses on the coupled mechanisms needed to understand how actuation, shift behaviour, torque transfer, and vehicle loading produce transient CVT behaviour.

1.4.1 Aim

The aim of this work is to derive a coupled nonlinear dynamic model of a mechanically actuated rubber V-belt CVT and its longitudinal vehicle drivetrain. The model extends from the engine-side pulley, through the belt and downstream driveline, to the vehicle response. Pulley speeds, belt transport, torque transfer, axial shift position, and vehicle motion evolve from the governing equations rather than being prescribed independently.

The formulation is intended to describe how the drivetrain responds from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Primary actuation, secondary torque reaction, belt traction, ratio change, rotational inertia, and vehicle loading are treated as interacting parts of the same system: a change in one mechanism alters the conditions under which the others act. This allows launch, clutching, shifting, backshift, gross slip, engine braking, and vehicle acceleration to be studied as outcomes of a common coupled response.

A further aim is to preserve the physical interpretation of that response. The derivation is organized so that the role of pulley geometry, primary speed-sensitive actuation, secondary torque-reactive actuation, belt traction, rotational inertia, and vehicle loading remains visible throughout the model. Its assumptions are stated explicitly, together with the mechanisms they exclude, so that simulation results can be interpreted in terms of both the physics retained and the limits of the formulation. The goal is therefore not only to predict system behaviour, but also to clarify why a particular response occurs and how a mechanical or operating change propagates through the drivetrain.

1.4.2 Scope

The system considered in this work is a mechanically actuated dry rubber V-belt CVT coupled to a simplified longitudinal vehicle drivetrain. The CVT consists of a speed-sensitive primary pulley, a torque-sensitive secondary pulley, a rubber V-belt, and a downstream fixed-ratio driveline that transmits power to the vehicle wheels. The vehicle is represented through one-dimensional longitudinal motion.

The model includes the following effects:

- variable pulley geometry, belt-length closure, and ratio kinematics;
- a dynamic shift coordinate relating sheave motion to pulley radii, wrap angles, and transmission ratio;
- mechanically generated primary and secondary axial forces;

- flyweight, ramp, spring, helix, and torque-reactive effects within the pulley actuation mechanisms;
- effective inertia associated with axial shift motion;
- explicit rotational dynamics of the primary pulley, belt, and secondary pulley;
- belt–pulley torque transfer subject to traction limits and gross-slip conditions;
- engine torque and engine-side inertia;
- downstream fixed reduction, reflected driveline inertia, and longitudinal vehicle road load; and
- transient response to specified initial conditions and time-varying input histories.

The belt is represented through a reduced fixed-length geometry and wrap-level contact description rather than a fully distributed deformable-belt formulation. The principal consequences of this and the other model idealizations are collected in [Section 2.2](#); they include omitted local belt deformation and migration, detailed actuator and contact tribology, flexible-hardware response, and thermal or wear evolution. Three-dimensional vehicle dynamics are also outside the present scope. Tire–ground contact is assumed to provide sufficient traction for the longitudinal force demanded by the drivetrain; wheel slip and tire-force limits are not represented.

The formulation retains these assumptions explicitly so that their consequences remain clear when interpreting a simulation result. Its structure also permits individual closures to be refined in later work. For example, the traction relation could be replaced by a distributed contact model, belt elasticity or pulley deformation could be introduced, or thermal, efficiency-loss, and tire-traction models could be incorporated while preserving the surrounding drivetrain framework.

1.4.3 Contributions

The primary contribution of this work is the closure of a coupled mechanism chain for a mechanically actuated dry rubber V-belt CVT operating within a vehicle drivetrain. Individual elements of the formulation build on established ideas in CVT and belt-drive modelling. The contribution lies in combining these elements into a physically interpretable time-domain framework in which the response of the transmission and vehicle emerges from their interaction.

The main contributions are as follows:

1. A vehicle-coupled dynamic formulation that closes the speed-sensitive primary mechanism, torque-reactive secondary mechanism, belt-mediated torque transfer, ratio change, pulley dynamics, and longitudinal vehicle response within a single system of governing equations.
2. A dynamic ratio-change and belt-transport treatment in which axial sheave motion, pulley radii, belt geometry, and belt speed evolve as physical states of the system rather than being imposed through a prescribed ratio map or instantaneous equilibrium condition.
3. A traction-limited torque-transfer formulation that compares the torque required for no-slip motion with the torque supportable by the belt–pulley contacts. This permits gross-slip behaviour to be represented when the sticking response is not admissible.
4. A transient drivetrain framework that evolves from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Launch, clutching, shifting, backshift, engine braking, traction loss, and vehicle acceleration can therefore be studied as responses of the same coupled system.
5. A derivation structure that preserves the physical role of the major mechanisms and states the modelling assumptions explicitly. This supports interpretation of the predicted response

and provides a modular basis for future extensions, including distributed belt contact, belt elasticity, pulley deformation, thermal effects, efficiency losses, and tire-traction limits.

Together, these contributions provide a framework for tracing how a change in component geometry, spring loading, actuator parameters, operating conditions, or vehicle load propagates through the CVT. The resulting effect on clamping, traction state, shift behaviour, torque transfer, engine speed, and vehicle acceleration can then be examined within the assumptions of the model.

Chapter 2

Model Definition and Simulation Problem

Chapter 1 established the need for a first-principles formulation in which shaft motion, belt motion, and ratio evolution emerge from the mechanics of the CVT. This chapter turns that objective into a simulation problem. It defines what is included in the model, how systems outside the CVT act at its two shaft boundaries, the assumptions and sign conventions used throughout the formulation, and the state variables advanced through time.

At any instant, the current state and the mechanical conditions at the two shaft boundaries are known. The formulation developed in Chapter 3 determines the corresponding state derivative, and numerical integration advances the solution. This chapter therefore defines the system to be simulated; Chapter 3 derives the mechanics that govern its evolution.

2.1 Modeled System and External Boundaries

The modeled system is the complete mechanically actuated rubber V-belt CVT. It contains the primary and secondary pulley assemblies, the complete belt loop, and both belt–pulley contacts. The primary flyweights, ramp, and spring, together with the secondary spring and helix, are included because they act directly on the pulley assemblies. The masses and rotational inertias of the belt, pulleys, movable components, and actuation hardware also lie within the model boundary.

The boundary is cut through the primary and secondary shafts, as shown in [Figure 2.1](#). Systems attached beyond those cuts are represented through their mechanical action at the corresponding shaft. This keeps the CVT formulation independent of a particular vehicle, engine, or test apparatus.

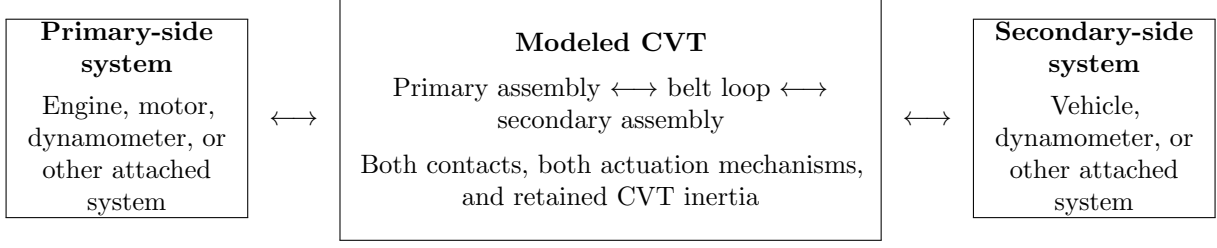


Figure 2.1: System boundary of the CVT formulation. The primary and secondary shaft cuts separate the modeled CVT from interchangeable attached systems. At either cut, the attached system is represented by the torque and equivalent inertia that it presents to the shaft. The shaft motion remains part of the CVT simulation.

2.1.1 Torque–Inertia Boundary Contract

At either shaft cut, everything beyond the modeled CVT is reduced to the mechanical load it presents at that shaft: an applied torque and an attached or reflected inertia. The two boundary descriptions are written as

$$\mathcal{B}_p = (\tau_{B,p}, I_{B,p}), \quad \mathcal{B}_s = (\tau_{B,s}, I_{B,s}), \quad (2.1.1)$$

and are summarized in [Table 2.1](#).

Table 2.1: Mechanical information supplied at the external shaft boundaries.		
Boundary	Applied torque	Inertia seen at shaft
Primary-side shaft	$\tau_{B,p}$	$I_{B,p}$
Secondary-side shaft	$\tau_{B,s}$	$I_{B,s}$

The shaft speeds are retained and advanced by the CVT simulation. During an evaluation, the current primary and secondary speeds are made available to the corresponding attached systems. Each attached system may use that speed, time, and its own internal information to compute the torque and equivalent inertia that it presents at the shaft cut. Only this torque–inertia pair is returned to the CVT formulation. The pair enters the coupled shaft dynamics, which determine the shaft acceleration subsequently used by the time integrator to update the speed.

A prescribed torque curve or load law therefore prescribes the boundary relation, not the shaft motion. Neither ω_p nor ω_s is imposed by an attached system; both remain dynamic results of the CVT formulation.

A positive boundary torque acts in the positive rotational direction defined for that shaft, and the equivalent inertia is expressed about the same shaft coordinate. Pulley, sheave, flyweight, helix, and belt inertias are assigned to the CVT; $I_{B,p}$ and $I_{B,s}$ account only for inertia beyond the two shaft cuts.

Engine, vehicle, electric-dynamometer, and hydraulic-dynamometer implementations are given separately in the appendices as particular realizations of this common contract.

The two cuts are called the primary-side and secondary-side boundaries so the terminology remains valid under either direction of power flow. Any attached system that can supply the torque–inertia description in [Equation \(2.1.1\)](#) can be coupled to the same CVT equations.

2.2 Modeling Assumptions

The assumptions below define the physical resolution of the formulation rather than a particular operating case. Each assumption is stated together with its principal consequence for the mechanics that can and cannot be represented. The consequence summaries are not intended to enumerate every secondary effect of an idealization; they identify the main physical resolution removed by that choice so that later results can be interpreted against a clear model boundary.

2.2.1 Material and Mechanism Idealizations

Assumption 1 (Time-Invariant Material and Surface State)

Model statement

The material properties and surface condition assigned to the CVT remain unchanged during a simulation. Intrinsic component dimensions are likewise not allowed to evolve through wear or thermal growth. The selected properties describe the hardware for the duration of the simulated event rather than evolving with its operating history.

Consequences for model fidelity

Temperature-dependent stiffness or friction, thermal state evolution, wear, glazing, conditioning, degradation, and other feedback from accumulated contact history are not represented. The formulation therefore describes short-term mechanical response for a specified material and surface state rather than the progressive evolution of that state.

Assumption 2 (Rigid Pulley and Actuation Hardware)

Model statement

The pulleys, shafts, actuation hardware, guides, and supporting CVT structure are represented as rigid bodies. Their prescribed geometry does not deform under the loads developed during operation.

Consequences for model fidelity

Pulley bending, shaft bending or torsion, guide and housing deflection, load-dependent groove geometry, and structural vibration generated by those compliances are outside the model. More detailed CVT formulations show that pulley deformation can alter local belt mechanics and axial loading; those effects would require compliant component models rather than changes to the rigid-body balances alone ([Carbone et al., 2005](#); [Duan et al., 2016](#)).

Assumption 3 (Idealized Actuator Mechanics)

Model statement

The primary ramp and flyweight mechanism, secondary helix, guides, and springs follow their prescribed kinematic and force laws without additional contact friction, backlash, or local compliance unless such an effect is introduced explicitly by the selected component law.

Consequences for model fidelity

Ramp or roller friction, pin-helix friction, roller or pin spin, guide friction, backlash, helix-flank transitions, contact compliance, and unmodelled spring hysteresis or internal damping are omitted. Relaxing this assumption would add actuator-specific tribology and compliance to the same mechanism free bodies; for example, contact friction appears explicitly in more detailed centrifugal and helix treatments ([Kim et al., 2002](#); [Sorge and Cammalleri, 2011](#)).

Assumption 4 (Fixed Belt Cross-Section)

Model statement

The belt retains its prescribed cross-sectional shape and dimensions while it runs and shifts. It may bend along the modeled belt path, but its local cross-section does not deform under centrifugal or sheave loading.

Consequences for model fidelity

Lateral and radial compression, bulging, cross-sectional shear and warping, centrifugal growth, and local cog or rib deformation are not resolved. The resulting load-dependent changes in belt width, height, contact geometry, and internal strain energy therefore lie outside the present formulation; detailed belt models retain some of these compliance effects explicitly ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

2.2.2 Belt Geometry and Kinematics

Assumption 5 (Aligned Pulley Axes and Fixed Center Distance)

Model statement

The primary and secondary pulley axes remain parallel at a fixed spacing. The pulley centers and shaft directions therefore do not change as the CVT shifts.

Consequences for model fidelity

Shaft misalignment, eccentricity, geometric runout, bearing-center motion, and the associated changes in belt path and face loading are not represented. Removing this assumption would require the belt geometry and contact loading to respond to the additional shaft and bearing

motion.

Assumption 6 (Fixed Belt Length)

Model statement

The modeled belt has a fixed total length. Longitudinal extension is not treated as an independent deformation of the belt, so every admissible pulley configuration must be compatible with the same installed length.

Consequences for model fidelity

Tension-dependent elongation, longitudinal strain energy, extension waves, permanent elongation, and the longitudinal component of elastic creep are not represented. Detailed traction-belt models can retain these effects through a longitudinal compliance field; the present formulation instead assigns ratio change to the pulley and contact motions of a fixed-length belt ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

Assumption 7 (Single Belt Transport Coordinate)

Model statement

The circumferential motion of the complete belt is represented by one global transport coordinate. Individual material points are not tracked around the loop, and the local tangential belt speed is not given an independent spatial field.

Consequences for model fidelity

The geometry-induced redistribution of local tangential material speed during shift, material-point migration through the wraps, and the additional momentum transport associated with that spatially varying velocity field are not resolved. The model still retains acceleration of the global belt transport and the motion of the prescribed wrap geometry. Shift-mechanics and discretized belt formulations resolve the omitted local transport in greater detail ([Sorge, 2008](#); [Cammalleri and Sorge, 2009](#); [Juli'o and Plante, 2011](#)).

Assumption 8 (Prescribed Planar Belt Path)

Model statement

At each instant, the belt centerline follows the path prescribed by the current pulley geometry: a circular wrap on each pulley joined by straight tangent spans in one common plane. The belt path is therefore determined kinematically rather than solved as a flexible dynamic curve.

Consequences for model fidelity

Progressive or spiral radial migration within a wrap, seating and unseating trajectories, bending-stiffness-dependent entry and exit geometry, span sag, transverse belt vibration, and transverse inertia associated with straight-span reorientation are omitted. The same planar description also excludes the out-of-plane span skew, twist, and axial belt inertia that would arise when the two physical groove centerplanes move differently during shift. More detailed belt and shift-mechanics models resolve subsets of these path effects explicitly ([Sorge, 2008](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

2.2.3 Belt–Pulley Contact Representation

Assumption 9 (Gross Tangential Coulomb Traction)

Model statement

Each belt–pulley interface is represented by a dry gross stick–slip law with fixed static and kinetic traction capacities for the selected surface state. The frictional contact action retained by the model acts in the circumferential belt-transport direction.

Consequences for model fidelity

The formulation can distinguish torque-carrying stick from gross kinetic slip, but it does not resolve radial friction, local creep or microslip, detailed presliding behaviour, lubrication or Stribeck effects, or friction variation with pressure, temperature, conditioning, or local sliding speed. Friction associated specifically with radial migration, seating, and unseating is also outside the retained contact direction. These mechanisms are treated separately in higher-fidelity rubber-belt contact models ([Gerbert, 1996](#); [Kong and Parker, 2006](#)).

Assumption 10 (Wrap-Level Contact Resolution)

Model statement

The model may resolve how belt tension and normal loading vary around a pulley wrap, but each wrap is assigned one overall traction state at an instant. The contact is not resolved through the belt width or depth, and no separate local traction state is carried for different portions of the same wrap.

Consequences for model fidelity

Local adhesion and sliding subregions, spatial traction reversals, detailed pressure concentrations, edge loading, and cross-sectional contact-stress fields cannot be predicted. Resolving those effects requires a distributed or discretized contact description of the type used in higher-fidelity belt models ([Kong and Parker, 2006](#); [Messick, 2018](#); [Juli'o and Plante, 2011](#)).

Assumption 11 (Symmetric Sheave-Face Load Sharing)

Model statement

The two sheave faces at a pulley are assumed to share the total normal contact load equally. On the secondary pulley, the belt torque is also divided equally between the fixed and movable faces for the purpose of the movable-sheave balance.

Consequences for model fidelity

Face-to-face differences in normal loading and transmitted torque are not resolved. On the secondary, this removes the possibility that unequal local sliding on the fixed and movable faces changes the torque reacted by the helix. Face-resolved shift models show that the two secondary face torques can depart substantially from an equal split during shifting ([Sorge and Cammalleri, 2011](#)).

2.2.4 Other Omitted Losses

Assumption 12 (Negligible Unmodeled Parasitic Losses)

Model statement

Bearing, seal, windage, and auxiliary-hardware losses are neglected unless they are introduced explicitly by a component or external boundary model. No global empirical efficiency factor is inserted to represent these omitted mechanisms.

Consequences for model fidelity

Energy loss and damping arise only from mechanisms that are represented explicitly by the formulation. Absolute efficiency may therefore be optimistic, and real hardware may exhibit additional damping or resisting torque when these parasitic losses are significant. Adding such losses does not require changing the kinematic structure of the CVT; it requires supplying the corresponding physical torque or force laws.

Resulting fidelity Taken together, these assumptions retain the dominant rigid-body actuation, pulley rotation, sheave-shift motion, global belt transport, wrap tension and contact resultants, clamping, traction limitation, and gross stick-slip mechanics of the CVT. They do not attempt to reproduce the complete deformable belt and local contact problem, actuator tribology, flexible-structure response, thermal or wear evolution, or miscellaneous parasitic losses. The assumptions are separated by physical mechanism so that a higher-fidelity model can replace one of these closures without obscuring which additional behaviour that change is intended to recover.

2.3 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions. Coordinates belonging only to a particular engine, vehicle, or dynamometer are defined with the corresponding boundary model.

2.3.1 Rotational Sign Convention

All rotating components in the reference drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion. Their positive directions are chosen such that positive drivetrain rotation produces forward vehicle motion.

As illustrated in [Figure 2.2](#), positive wheel angular velocity $\omega_w > 0$ corresponds to wheel rotation that drives the vehicle forward. The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley, $\omega_s > 0$, is the direction that produces this wheel motion through the final reduction and axle. Positive rotation of the primary pulley, $\omega_p > 0$, is the direction that drives the secondary pulley in this same sense through the CVT belt.

Torque follows the same sign convention. A torque is positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity. The coordinate signs remain fixed when a torque resists the current motion or when power flow reverses.

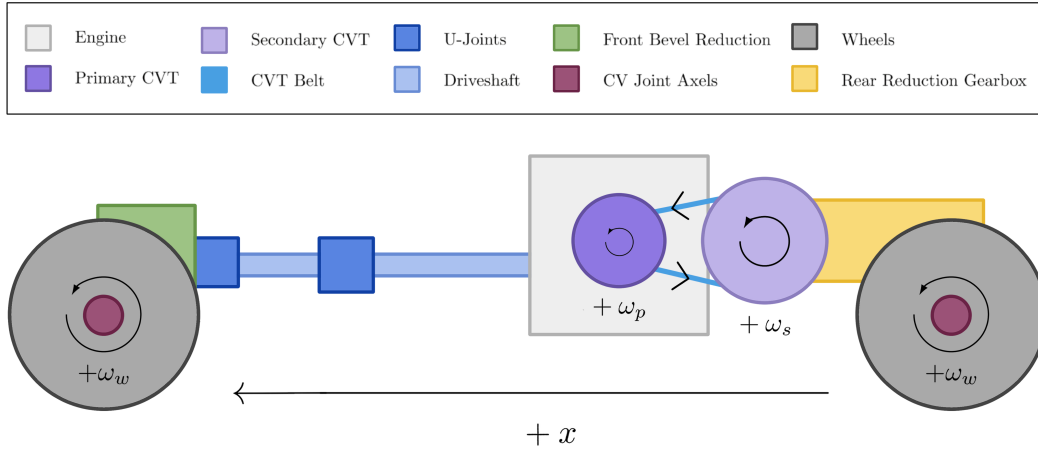


Figure 2.2: Rotational sign convention for the reference drivetrain. Positive angular velocities for the wheels (ω_w), secondary pulley (ω_s), and primary pulley (ω_p) are defined such that drivetrain rotation produces forward vehicle motion.

The belt loop is assigned a positive path direction consistent with transport produced by positive primary rotation. The belt transport speed v_b is positive in this direction and negative in the opposite direction. This sign choice defines belt motion independently of whether either belt-pulley contact is sticking or slipping.

2.3.2 Axial Sheave Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in [Figure 2.3](#) (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate. The axial variable s measures the translation of the movable sheave relative to this open configuration:

$s = 0$: fully open configuration.

A positive value of s corresponds to the movable sheave translating toward the fixed sheave along the pulley axis:

$s > 0$: sheave-closing motion.

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission. The maximum displacement $s = S_{\max}$ corresponds to the fully closed configuration shown in [Figure 2.3](#) (right).

Although s is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open. The precise geometric relationship between these motions will be derived later. The primary pulley is used here as the example; the same local convention—positive axial motion closes the sheave pair—applies when the secondary pulley is considered.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the primary movable sheave translating toward its fixed sheave.

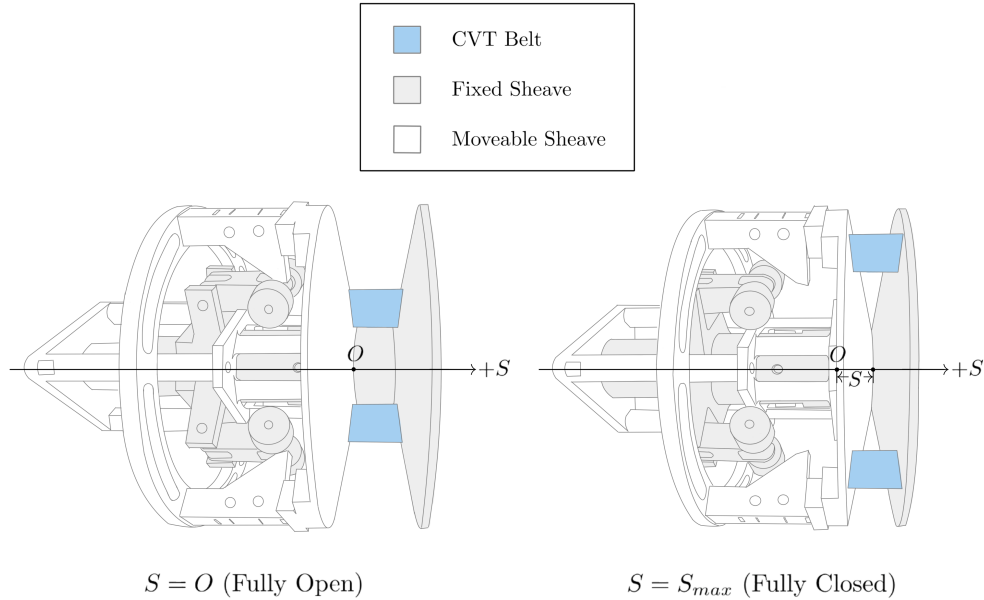


Figure 2.3: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

2.3.3 Local Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction $\hat{e}_r$ and the tangential direction $\hat{e}_\theta$. These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus, $F_r > 0$ corresponds to a force directed outward along $\hat{e}_r$.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction $\hat{e}_\theta$ is defined to be consistent with the positive rotational convention introduced in [Section 2.3.1](#). Tangential motion in this direction corresponds to motion produced by positive angular velocity.

[Figure 2.4](#) illustrates these directions using examples from the primary pulley. The centrifugal flyweights generate outward radial forces F_r , while the belt moves along the tangential direction with velocity v_t . These quantities serve as representative examples of forces and motion that act along the radial and tangential directions. The same local basis applies at the secondary pulley.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

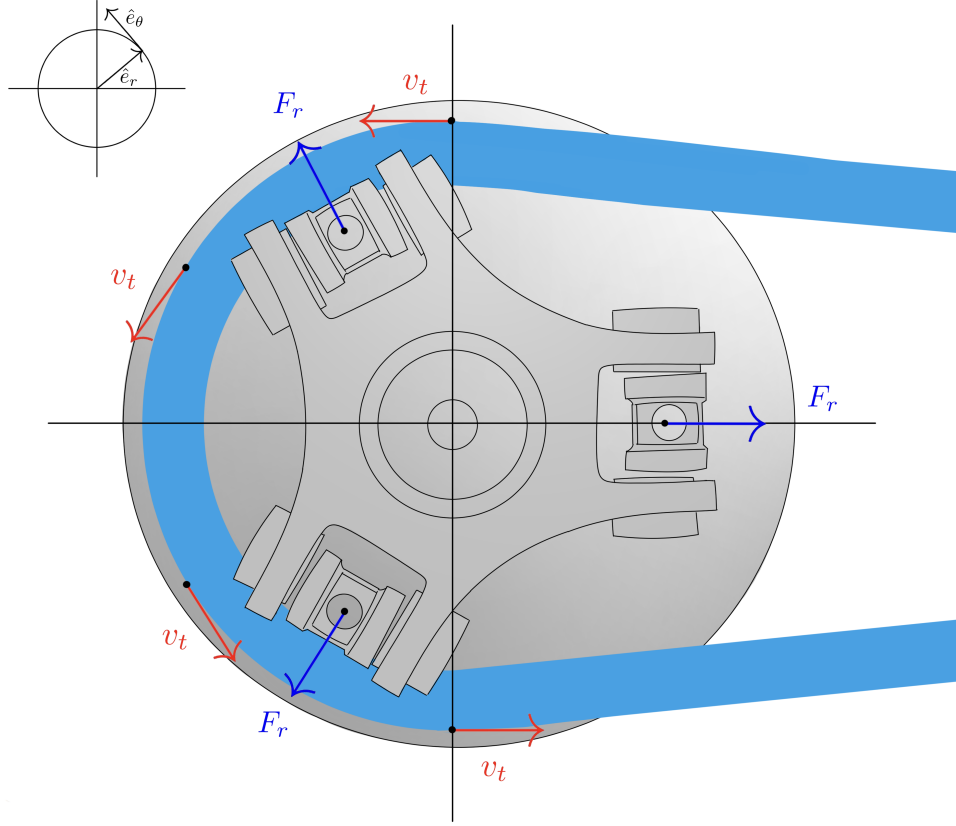


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The radial direction $\hat{e}_r$ points outward from the center of the pulley, while the tangential direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$ and aligned with positive rotation. Example quantities illustrated on the primary pulley include outward radial forces F_r generated by the flyweights and tangential belt velocity v_t . The same local directions apply at the secondary pulley. For reference, the axial coordinate s is perpendicular to the page in this view and points along the pulley shaft, into the page.

2.4 System State Definition

The simulation advances a state vector through time. At the beginning of each model evaluation, these state variables are known and describe the current motion and shift configuration of the CVT.

2.4.1 State Vector

State Vector

(2.4.1)

$$\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

Here, ω_p and ω_s are the primary and secondary shaft speeds, s is the primary axial shift position, $\dot{s}$ is its velocity, and v_b is the belt transport speed.

These are the quantities whose current values must be carried into the next instant. The rotational mechanics depend on shaft speed rather than absolute shaft angle, so absolute angular positions need not be stored. Similarly, under [Assumption 7 \(Single Belt Transport Coordinate\)](#), the belt state records its global transport motion rather than the identity or phase of individual material points, so no belt-phase coordinate is required. Shift position is retained because the current geometry and actuation depend on where the pulleys are in their travel, and $\dot{s}$ is retained because the shift motion is governed by second-order dynamics.

2.4.2 Initial-Value Problem and Time Integration

At a chosen starting time t_0 , the simulation is initialized with

$$\mathbf{x}(t_0) = \mathbf{x}_0. \quad (2.4.2)$$

The initial state may be any mechanically admissible condition within the modeled regimes. From that point onward, the formulation evaluates the time derivative of the state from the current state, the fixed CVT data, and the torque–inertia pairs returned at the two shaft boundaries. At the highest level, this evolution law is written as

$$\dot{\mathbf{x}}(t) = \mathcal{F}(t, \mathbf{x}(t); \boldsymbol{\theta}, \mathcal{B}_p, \mathcal{B}_s), \quad (2.4.3)$$

where $\boldsymbol{\theta}$ collects the fixed physical and configuration data, and $\mathcal{B}_p$ and $\mathcal{B}_s$ are the boundary pairs evaluated using the current shaft speeds. Chapter 3 constructs $\mathcal{F}$ from the CVT geometry, pulley dynamics, belt mechanics, and contact laws.

A numerical time integrator repeatedly evaluates [Equation \(2.4.3\)](#) and advances the state according to

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_{t_0}^t \dot{\mathbf{x}}(\xi) d\xi. \quad (2.4.4)$$

Contact transitions, engagement, and travel boundaries may change the particular form of $\mathcal{F}$ during the simulation. Their event conditions and regime-specific equations are developed after the corresponding mechanics have been derived.

This construction is fully dynamic. The boundary models define the external torque and inertia seen by the CVT, while the coupled mechanics and time integration determine the evolving shaft speeds, belt speed, shift position, and transmission ratio.

2.5 Model Quantity Roles and Evaluation Flow

The constants and parameters collected in ?? and ?? serve different computational roles from the state and boundary data. [Table 2.2](#) places these quantities in the high-level simulation workflow.

Table 2.2: High-level roles of quantities used in the simulation.

Quantity role	Source	Use during simulation
Fixed model data	Reference data and selected CVT configuration	Define the modeled hardware and remain fixed for the selected simulation.
Initialized data	Fixed model data	Established once for the selected configuration and reused during time integration.
State variables	Initial condition and integrator	Record the current CVT motion and shift configuration at time t .
Boundary data	Attached-system models evaluated at the current shaft speeds	Give the applied torque and attached or reflected inertia seen at each shaft cut for the current evaluation.
Instantaneous results	Current model evaluation	Provide the state derivative and event information requested by the integrator.

The simulation proceeds at two levels. Initialization checks the selected CVT data, establishes quantities fixed by that configuration, assigns the initial state, and identifies the initial mechanical regime. Time integration then repeats the following high-level evaluation:

1. The integrator supplies a trial time and the corresponding current state $\mathbf{x}(t)$, including ω_p and ω_s .
2. Each attached system uses the corresponding current shaft speed to evaluate the torque and equivalent inertia it presents at the shaft cut.
3. The coupled CVT mechanics use the current state, fixed model data, and both boundary pairs to evaluate $\dot{\mathbf{x}}(t)$ and the event conditions for the current mechanical regime.
4. The integrator advances every state variable, including both shaft speeds. If an event is encountered, the applicable regime is updated and integration continues from that event state.

This process can be summarized as

$$\begin{aligned}
\omega_p(t) &\longrightarrow \mathcal{B}_p, & \omega_s(t) &\longrightarrow \mathcal{B}_s & \text{boundary evaluation,} \\
\{t, \mathbf{x}(t), \boldsymbol{\theta}, \mathcal{B}_p, \mathcal{B}_s\} &\longrightarrow \dot{\mathbf{x}}(t) & \text{coupled CVT mechanics,} \\
\dot{\mathbf{x}}(t) &\longrightarrow \mathbf{x}(t') & \text{time integration.}
\end{aligned} \tag{2.5.1}$$

Chapter 3 supplies the mechanics contained in the first arrow. Geometry is developed first because the current shift state fixes the instantaneous pulley configuration. The axial, rotational, belt-loop, and contact relations are then exposed in sequence before they are assembled into the evolution law. This ordering supports the derivation; the completed relations are evaluated together as one coupled instantaneous problem.

Chapter 3

Derivation

3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley, and the resulting pair of pulley radii determines the transmission ratio.

Both pulleys contain a movable sheave. Let x_p denote the axial position of the movable primary sheave and x_s the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in [Section 2.3.2](#) and illustrated in [Figure 2.3](#). The local coordinates x_p and x_s follow the same convention on their respective pulleys.

For either pulley $j \in \{p, s\}$, $x_j = 0$ denotes the fully open configuration. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. The corresponding velocity follows the same sign convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion.

The primary therefore begins at $x_p = 0$ and moves in the positive direction as it closes. The secondary begins in a closed position, with $x_s > 0$, and moves toward smaller values of x_s as it opens.

Consider a belt seated between two straight sheave faces with half-angle β , as shown in [Figure 3.1](#). Let b_{in} denote the width of the belt at its inner surface. As assumed in [Assumption 4 \(Fixed Belt](#)

Cross-Section), this width remains constant.

Suppose the movable sheave closes the pulley by a distance Δx_j . At the original belt radius, the available space between the sheaves is therefore reduced from b_{in} to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance Δr_j . At its new position, the angled sheave faces provide an additional axial distance d on each side.

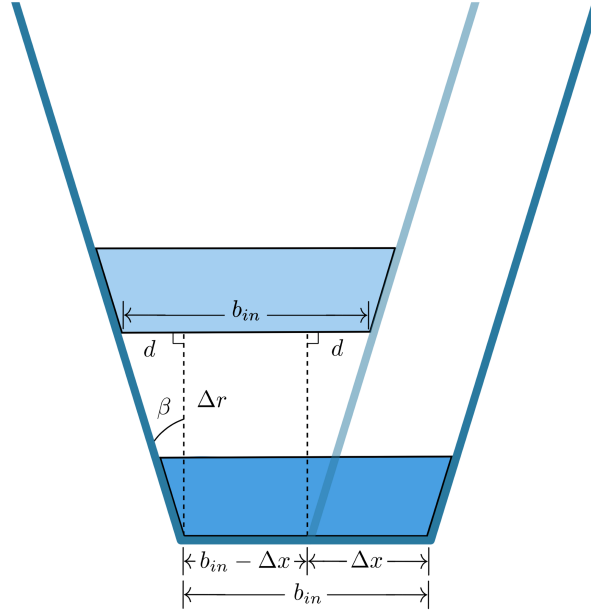


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley j . Closing the pulley by Δx reduces the available width at the original belt position to $b_{\text{in}} - \Delta x$. Moving outward by Δr provides an additional axial distance d at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from [Figure 3.1](#),

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j). \quad (3.1.1)$$

The right triangle beside either sheave face relates d to the radial movement. For the sheave half-angle β ,

$$\tan \beta = \frac{d}{\Delta r_j}, \quad (3.1.2)$$

and therefore

$$d = \Delta r_j \tan \beta. \quad (3.1.3)$$

Substituting Equation (3.1.3) into Equation (3.1.1) gives

$$b_{\text{in}} = 2\Delta r_j \tan \beta + (b_{\text{in}} - \Delta x_j). \quad (3.1.4)$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2 \tan \beta}. \quad (3.1.5)$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2 \tan \beta}. \quad (3.1.6)$$

Because the sheave faces are straight, this slope is constant.

Radius convention The derivation above follows the inner surface of the belt, where b_{in} is defined. The belt also has the finite radial height h_b , so its inner and outer surfaces occupy different radii. Let $r_{j,\text{inner}}$ and $r_{j,\text{outer}}$ denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \quad (3.1.7)$$

The fixed-cross-section assumption in Assumption 4 (Fixed Belt Cross-Section) also makes h_b constant. Consequently, the inner and outer surfaces undergo the same radial displacement, and the axial-to-radial relationship derived above applies to either one.

An absolute pulley radius must nevertheless identify which belt surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \quad (3.1.8)$$

The derivative in Equation (3.1.6) describes how this radius changes, but one known configuration is needed to establish its absolute value. Let $x_{j,\text{ref}}$ denote a known axial position at which the belt is seated, and let $r_{j,\text{ref}}$ denote the outer pulley radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2 \tan \beta}. \quad (3.1.9)$$

The pair $(x_{j,\text{ref}}, r_{j,\text{ref}})$ simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing x_s , [Equation \(3.1.9\)](#) produces the corresponding decrease in r_s .

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At $x_p = 0$, the distance between the primary sheaves can therefore exceed b_{in} . Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let $x_{p,\text{dz}}$ denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \leq x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At $x_p = x_{p,\text{dz}}$, the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\text{ref}} = x_{p,\text{dz}}, \quad r_{p,\text{ref}} = r_{p,\text{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\text{min}}, & 0 \leq x_p < x_{p,\text{dz}}, \\ r_{p,\text{min}} + \frac{x_p - x_{p,\text{dz}}}{2 \tan \beta}, & x_{p,\text{dz}} \leq x_p \leq x_{p,\text{max}}. \end{cases} \quad (3.1.10)$$

The linear axial-to-radial relationship in [Equation \(3.1.9\)](#) is standard in reduced CVT geometry models ???. It applies to both pulleys whenever the belt is seated, while [Equation \(3.1.10\)](#) extends the primary description through the deadzone before contact.

The local geometry now tells us the pulley radius associated with either movable-sheave position. To determine which primary and secondary positions can occur together, the two local descriptions must next be joined through the geometry of the complete belt.

3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the radius on each pulley. The belt, however, passes around both pulleys. To connect the two radii, the belt must now be followed around the complete loop.

Under [Assumption 8 \(Prescribed Planar Belt Path\)](#), the modeled belt path consists of a circular wrap on each pulley joined by two straight tangent spans. The aligned, fixed-center geometry of [Assumption 5 \(Aligned Pulley Axes and Fixed Center Distance\)](#) makes the two spans equal. If each straight span has length l , and the primary and secondary wrap angles are ϕ_p and ϕ_s , the total geometric length of the loop is

$$L_{\text{geo}} = r_p \phi_p + r_s \phi_s + 2l. \quad (3.1.11)$$

The quantities l , ϕ_p , and ϕ_s follow directly from the geometry shown in [Figure 3.2](#).

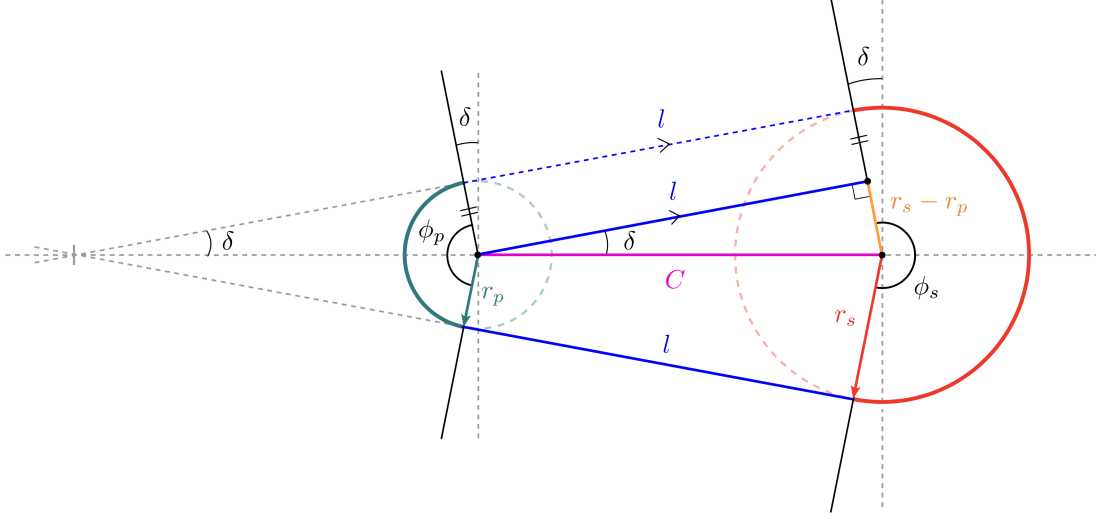


Figure 3.2: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length l . Unequal pulley radii incline the straight spans by the angle δ , producing the primary and secondary wrap angles ϕ_p and ϕ_s .

Consider the right triangle in the upper span in [Figure 3.2](#). The line joining the pulley centres forms its hypotenuse and has length C . One side lies along the straight belt span and therefore has length l .

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown by the orange construction in [Figure 3.2](#), the difference between their lengths gives the remaining side of the triangle, $r_s - r_p$. The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \quad (3.1.12)$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \quad (3.1.13)$$

This expression requires

$$|r_s - r_p| < C. \quad (3.1.14)$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring $C > r_s + r_p$, and $r_s + r_p$ is necessarily greater than $|r_s - r_p|$.

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to π .

When the radii differ, the straight spans rotate through the angle δ shown in [Figure 3.2](#). The same right triangle gives

$$\sin \delta = \frac{r_s - r_p}{C}, \quad (3.1.15)$$

and therefore

$$\delta = \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \quad (3.1.16)$$

The angle δ is signed. For the configuration shown, $r_s > r_p$, so $\delta > 0$. The primary wrap loses an angle δ at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.17)$$

If $r_p > r_s$, then $\delta < 0$, and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting [Equations \(3.1.13\)](#), [\(3.1.16\)](#) and [\(3.1.17\)](#) into [Equation \(3.1.11\)](#) gives

Exact geometric belt length (3.1.18) $ \begin{aligned} L_{\text{geo}}(r_p, r_s, C) = & \pi (r_p + r_s) \\ & + 2 (r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) \\ & + 2 \sqrt{C^2 - (r_s - r_p)^2} \end{aligned} $
--

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of r_p , r_s , and C , [Equation \(3.1.18\)](#) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under [Assumption 6 \(Fixed Belt Length\)](#), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \quad (3.1.19)$$

where L_b is the fixed belt length expressed using the same outer-surface reference adopted for r_p and r_s in Equation (3.1.8). This is the geometric compatibility condition for the belt loop. Once any two of r_p , r_s , and C are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance C . The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled, C remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by $r_{p,\text{ref}}$ and $r_{s,\text{ref}}$, the only unknown in Equation (3.1.19) is C .

In the general unequal-radius case, C appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for C . Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \quad (3.1.20)$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L(r_{p,\text{ref}}, r_{s,\text{ref}}, C) = 0. \quad (3.1.21)$$

The root solver varies C until this condition is satisfied.

Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{(r_s - r_p)^2 + \frac{1}{4}[L_b - \pi(r_p + r_s)]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in [Chapter A](#).

Once the initial root solve has determined C , both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius r_p is known. Substituting r_p , C , and L_b into [Equation \(3.1.20\)](#) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_s. \quad (3.1.22)$$

The resulting r_s is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If r_s is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_p. \quad (3.1.23)$$

Thus, once C has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. [Budynas and Nisbett \(2015\)](#) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations $r_p(x_p)$ and $r_s(x_s)$ now describe how each pulley produces its radius, while [Equation \(3.1.19\)](#) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley: x_p determines $r_p(x_p)$, and x_s determines $r_s(x_s)$. A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length L_b . The remaining problem is therefore to identify the pairs (x_p, x_s) that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \quad (3.1.24)$$

Every pair that makes [Equation \(3.1.24\)](#) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take x_p simply as an arbitrary choice. For each value of x_p , the primary relation gives $r_p(x_p)$. The shaft spacing fixes C , and L_b remains fixed by [Assumption 6 \(Fixed Belt Length\)](#), so the belt-length residual can then be solved for the required secondary radius r_s . The local secondary relation finally gives the corresponding position x_s :

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in [Figure 3.3](#).

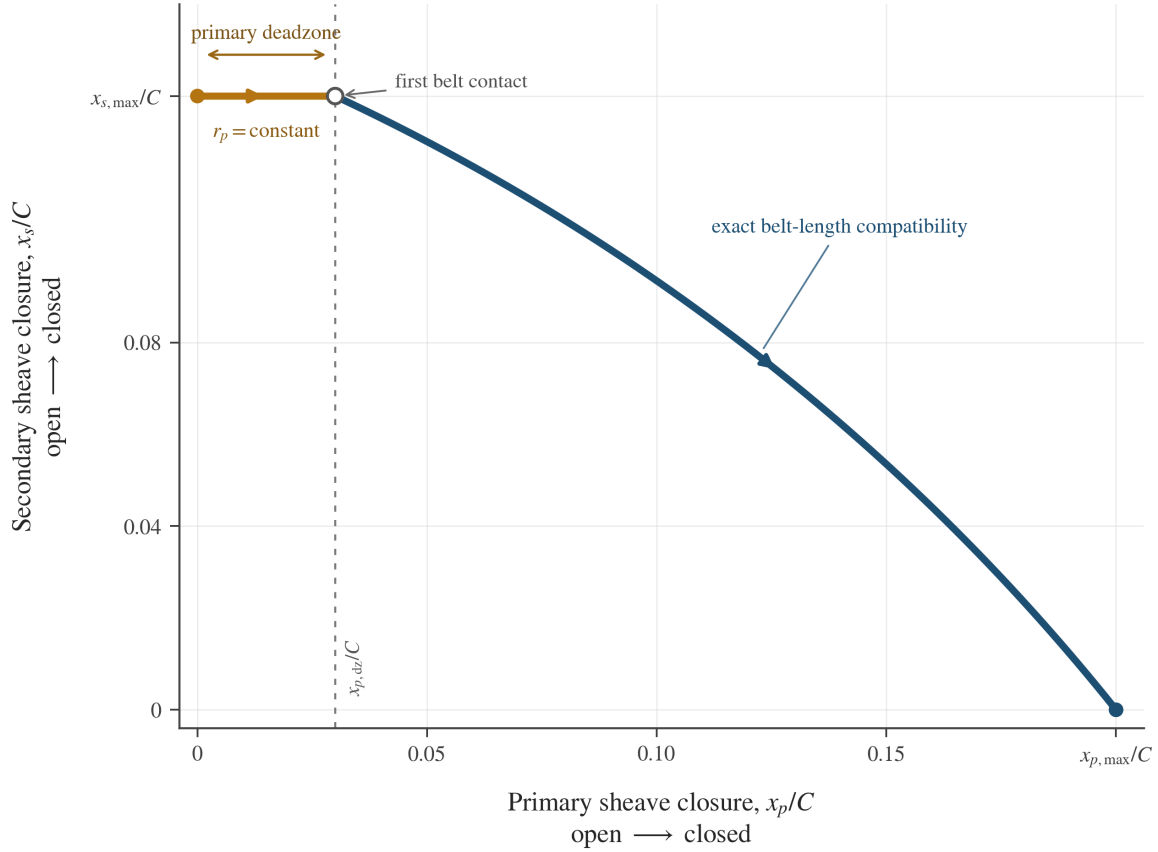


Figure 3.3: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance C , so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While $0 \leq x_p < x_{p,dz}$, the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases r_p . The fixed belt loop then requires r_s to decrease, which means that the secondary opens and x_s decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any x_p meets the curve once, giving one compatible x_s . At the deadzone height, however, a horizontal line meets every x_p between 0 and $x_{p,dz}$. Thus x_s is a unique function of x_p over the full travel, whereas x_p is not a unique function of x_s .

The primary coordinate is therefore adopted as the system shift coordinate. This is the same s included in the state vector of [Section 2.4.1](#):

System shift coordinate

(3.1.25)

$$s \equiv x_p, \quad x_p(s) = s, \quad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \leq s < x_{p,\text{dz}}, \\ \text{root}_{0 \leq x_s \leq x_{s,\max}} [\mathcal{R}_x(s, x_s) = 0], & x_{p,\text{dz}} \leq s \leq x_{p,\max}. \end{cases} \quad (3.1.26)$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give $x_{s,\max}$ at $s = x_{p,\text{dz}}$, so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when s changes with time.

3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount ds , how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit s . Throughout this subsection, a prime denotes differentiation with respect to s . For either pulley $j \in \{p, s\}$, the corresponding time rates are then

$$\dot{x}_j = x'_j(s)\dot{s}, \quad \dot{r}_j = r'_j(s)\dot{s}. \quad (3.1.27)$$

Primary radius rate The primary side follows immediately from the definition $x_p(s) = s$:

$$x'_p(s) = 1, \quad \dot{x}_p = \dot{s}. \quad (3.1.28)$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating [Equation \(3.1.10\)](#) therefore gives

$$r'_p(s) = \begin{cases} 0, & 0 \leq s < x_{p,\text{dz}}, \\ \frac{1}{2 \tan \beta}, & x_{p,\text{dz}} < s \leq x_{p,\max}. \end{cases} \quad (3.1.29)$$

Multiplying this slope by $\dot{s}$ gives

Primary Radius Rate

(3.1.30)

$$\dot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

Secondary coordinate and radius rates The secondary requires one additional step. Its local pulley relation tells us how r_s changes when x_s changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\text{geo}}(r_p(s), r_s(s), C) = L_b. \quad (3.1.31)$$

The centre distance C is fixed by the pulley shafts, while L_b is constant under **Assumption 6 (Fixed Belt Length)**. A small change in s therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\text{geo}}}{ds} = 0. \quad (3.1.32)$$

Both radii change during this motion. The change caused by r_p is the sensitivity of the belt length to r_p , multiplied by the actual rate r'_p . The secondary contributes in the same way. Applying the chain rule to **Equation (3.1.31)** gives

$$\frac{\partial L_{\text{geo}}}{\partial r_p} r'_p + \frac{\partial L_{\text{geo}}}{\partial r_s} r'_s = 0. \quad (3.1.33)$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in **Equations (3.1.13)** and **(3.1.16)**. Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1} \left(\frac{r_s - r_p}{C} \right), \quad l = \sqrt{C^2 - (r_s - r_p)^2}, \quad (3.1.34)$$

Here, δ sets the wrap angles through **Equation (3.1.17)**, while l is the length of either straight span. Using these quantities, the exact path length from **Equation (3.1.18)** is

$$L_{\text{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \quad (3.1.35)$$

First, hold r_s and C fixed and increase r_p . The radius difference $r_s - r_p$ then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \quad (3.1.36)$$

Using this result in the definitions of δ and l gives

$$\begin{aligned} \frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(-\frac{1}{C}\right) \\ &= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} = -\frac{1}{l}, \end{aligned} \quad (3.1.37)$$

$$\begin{aligned} \frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [2(r_s - r_p)] \\ &= \frac{r_s - r_p}{l}. \end{aligned} \quad (3.1.38)$$

We can now differentiate the complete path length in [Equation \(3.1.35\)](#). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned} \frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2 \left[(-1)\delta + (r_s - r_p) \left(-\frac{1}{l}\right) \right] + 2 \left(\frac{r_s - r_p}{l} \right) \\ &= \pi - 2\delta \\ &= \phi_p. \end{aligned} \quad (3.1.39)$$

The two terms containing $(r_s - r_p)/l$ cancel. The remaining sensitivity is the primary wrap angle.

The secondary calculation follows the same path, but increasing r_s increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \quad (3.1.40)$$

The corresponding derivatives of δ and l are

$$\begin{aligned}\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{1}{C}\right) \\ &= \frac{1}{l},\end{aligned}\tag{3.1.41}$$

$$\begin{aligned}\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [-2(r_s - r_p)] \\ &= -\frac{r_s - r_p}{l}.\end{aligned}\tag{3.1.42}$$

Differentiating the full path length with these signs gives

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2 \left[(1)\delta + (r_s - r_p) \left(\frac{1}{l}\right) \right] + 2 \left(-\frac{r_s - r_p}{l} \right) \\ &= \pi + 2\delta \\ &= \phi_s.\end{aligned}\tag{3.1.43}$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \quad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s.\tag{3.1.44}$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition [Equation \(3.1.33\)](#), the two radial changes must therefore satisfy

$$\phi_p r'_p + \phi_s r'_s = 0.\tag{3.1.45}$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since $x'_p = 1$,

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{1}{2 \tan \beta} x'_s.\tag{3.1.46}$$

Substituting these two slopes into [Equation \(3.1.45\)](#) gives

$$\phi_p \left(\frac{1}{2 \tan \beta} \right) + \phi_s \left(\frac{1}{2 \tan \beta} x'_s \right) = 0.\tag{3.1.47}$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x'_s = 0. \quad (3.1.48)$$

Solving for the slope of the secondary mapping gives

Secondary Coordinate Slope (3.1.49) $x'_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor x_s changes. After contact, $x'_s < 0$: primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in [Figure 3.3](#) is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by $\dot{s}$:

$$\dot{x}_s = x'_s(s)\dot{s} = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\dot{s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases} \quad (3.1.50)$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

Secondary Radius Rate (3.1.51) $\dot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$

The two pulley-radius rates are now determined by s and $\dot{s}$. However, the secondary multiplier $-\phi_p/\phi_s$ changes as the radii and wrap angles change. Even if $\dot{s}$ is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in [Equations \(3.1.30\)](#) and [\(3.1.51\)](#) were obtained by multiplying the slope of each geometric mapping by the shift velocity $\dot{s}$. Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in [Equation \(3.1.49\)](#) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

Primary radius acceleration The primary coordinate requires no additional geometry. The system-coordinate definition in Equation (3.1.25) gives $x_p(s) = s$, so

$$x'_p(s) = 1, \quad x''_p(s) = 0, \quad (3.1.52)$$

and therefore

$$\dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.1.53)$$

The general shift-to-time relation in Equation (3.1.27) gives the primary radius rate in the form

$$\dot{r}_p = r'_p(s)\dot{s}. \quad (3.1.54)$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt} [r'_p(s)\dot{s}] \\ &= \frac{dr'_p}{dt}\dot{s} + r'_p(s)\ddot{s}. \end{aligned} \quad (3.1.55)$$

Because r'_p is itself a function of s ,

$$\frac{dr'_p}{dt} = r''_p(s)\dot{s}. \quad (3.1.56)$$

Substituting Equation (3.1.56) into Equation (3.1.55) gives

$$\ddot{r}_p = r''_p(s)\dot{s}^2 + r'_p(s)\ddot{s}. \quad (3.1.57)$$

Within either smooth branch of the primary radius map, $r'_p(s)$ is constant. Hence

$$r''_p(s) = 0. \quad (3.1.58)$$

Using the primary slopes from Equation (3.1.29), the primary radius acceleration is

Primary Radius Acceleration (3.1.59) $\ddot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{\ddot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.29).

Secondary coordinate curvature The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.48):

$$\phi_p + \phi_s x'_s = 0.$$

The balance in Equation (3.1.48) contains the information needed for the second derivative. We will differentiate it once more with respect to s , which first requires the rates at which the two wrap angles change with s .

The required geometric quantities were defined in Equations (3.1.16) and (3.1.17). Written together, they are

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \quad \phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.60)$$

As s changes, both r_p and r_s change, so the radius difference inside δ changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r'_s - r'_p. \quad (3.1.61)$$

Applying the chain rule to δ , and using the straight-span length from Equation (3.1.13) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{r'_s - r'_p}{C}\right) \\ &= \frac{r'_s - r'_p}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \frac{r'_s - r'_p}{l}. \end{aligned} \quad (3.1.62)$$

After primary contact, the local pulley slopes are given by Equation (3.1.46):

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{x'_s}{2 \tan \beta}.$$

Substituting these slopes into Equation (3.1.62) gives

$$\begin{aligned} \delta' &= \frac{\frac{x'_s}{2 \tan \beta} - \frac{1}{2 \tan \beta}}{l} \\ &= \frac{x'_s - 1}{2l \tan \beta}. \end{aligned} \quad (3.1.63)$$

The wrap-angle derivatives now follow directly from Equation (3.1.60):

$$\phi'_p = -2\delta', \quad \phi'_s = 2\delta'. \quad (3.1.64)$$

We can now differentiate the secondary slope balance in Equation (3.1.48). The primary term contributes ϕ'_p , while the secondary term requires the product rule because both ϕ_s and x'_s change with s :

$$\begin{aligned} 0 &= \frac{d}{ds} (\phi_p + \phi_s x'_s) \\ &= \phi'_p + \phi'_s x'_s + \phi_s x''_s. \end{aligned} \quad (3.1.65)$$

Substituting the wrap-angle derivatives from Equation (3.1.64) gives

$$-2\delta' + 2\delta' x'_s + \phi_s x''_s = 0. \quad (3.1.66)$$

Solving first for x''_s ,

$$\begin{aligned} \phi_s x''_s &= 2\delta'(1 - x'_s), \\ x''_s &= \frac{2\delta'(1 - x'_s)}{\phi_s}. \end{aligned} \quad (3.1.67)$$

Using the expression for δ' from Equation (3.1.63),

$$\begin{aligned} x''_s &= \frac{2}{\phi_s} \left(\frac{x'_s - 1}{2l \tan \beta} \right) (1 - x'_s) \\ &= -\frac{(1 - x'_s)^2}{\phi_s l \tan \beta}. \end{aligned} \quad (3.1.68)$$

On the engaged branch, the first-derivative result in Equation (3.1.49) gives $x'_s = -\phi_p/\phi_s$. The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x'_s &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \quad (3.1.69)$$

where $\phi_p + \phi_s = 2\pi$ follows from Equation (3.1.17). Substituting this result into Equation (3.1.68) gives the curvature of the secondary coordinate mapping:

Secondary Coordinate Curvature

(3.1.70)

$$x_s''(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{4\pi^2}{\phi_s^3 l \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The deadzone branch in Equation (3.1.26) is horizontal, so both x_s' and x_s'' are zero there. On the engaged branch, $x_s'' < 0$: as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

Secondary axial and radial accelerations The curvature in Equation (3.1.70) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.50):

$$\dot{x}_s = x_s'(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt} [x_s'(s)\dot{s}] \\ &= \frac{dx_s'}{dt}\dot{s} + x_s'(s)\ddot{s}. \end{aligned} \tag{3.1.71}$$

Because x_s' changes only through s ,

$$\frac{dx_s'}{dt} = x_s''(s)\dot{s}. \tag{3.1.72}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x_s'(s)\ddot{s} + x_s''(s)\dot{s}^2. \tag{3.1.73}$$

Substituting the slope from Equation (3.1.49) and the curvature from Equation (3.1.70) gives

Secondary Axial Acceleration

(3.1.74)

$$\ddot{x}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\ddot{s} - \frac{4\pi^2}{\phi_s^3 l \tan \beta}\dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant $1/(2 \tan \beta)$. Applying it to the secondary gives

$$\ddot{r}_s = \frac{1}{2 \tan \beta} \ddot{x}_s. \quad (3.1.75)$$

Substituting Equation (3.1.74) into Equation (3.1.75) gives

$$\text{Secondary Radius Acceleration} \quad (3.1.76)$$

$$\ddot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\ddot{s}}{2 \tan \beta} - \frac{2\pi^2}{\phi_s^3 l \tan^2 \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.49) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.70): even when $\ddot{s} = 0$, moving through the engaged branch changes $-\phi_p/\phi_s$, so the secondary velocity continues to change.

Remark (Deadzone Boundary)

The idealized maps in Equations (3.1.10) and (3.1.26) change slope when the primary first reaches the belt at $s = x_{p,dz}$. Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.59) and (3.1.76) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

Remark (Practical Consequences of the Pulley Geometry)

The radius, rate, and acceleration relations derived in this section describe how the pulley geometry evolves as the CVT shifts. Chapter D explores the practical consequences of these results in more detail. In particular, it examines the resulting geometric CVT ratio and ratio-change rate across the shift travel, together with how the current pulley geometry and sheave angle shape the available ratio range and ratio-change sensitivity.

3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in Equation (3.1.25) sets the primary position directly, while Equation (3.1.26) gives the unique compatible secondary position. The local pulley maps in Equations (3.1.9) and (3.1.10) convert those two positions into pulley radii. Finally, the rate and acceleration relations in Equations (3.1.30), (3.1.51), (3.1.59) and (3.1.76) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \longmapsto (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s).$$

These results follow from the geometry derived above under the idealizations stated in Section 2.2. The belt cross-section and total length remain fixed under Assumption 4 (Fixed Belt Cross-Section) and Assumption 6 (Fixed Belt Length), while Assumption 8 (Prescribed Planar Belt Path) constrains the belt to circular wraps and straight tangent spans. Together with the aligned pulley axes and

fixed centre distance of [Assumption 5 \(Aligned Pulley Axes and Fixed Center Distance\)](#), these conditions leave one modeled belt path for each admissible value of s .

Relation to prior transient models These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

[Duan et al. \(2016\)](#) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through [Assumption 6 \(Fixed Belt Length\)](#) and prescribes the arc-and-tangent belt path through [Assumption 8 \(Prescribed Planar Belt Path\)](#). Consequently, elastic path-length change and deformation-driven departure from that prescribed path are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of [Sorge \(2008\)](#) and [Cammalleri and Sorge \(2009\)](#), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of [Juli'o and Plante \(2011\)](#) and [Ballew \(2015\)](#), instead obtain the evolving belt geometry from the motion and contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in [Equation \(3.1.19\)](#) to find the representative pulley-radius pair and its derivatives directly from s , $\dot{s}$, and $\ddot{s}$. The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

This closure determines which sheave and radius motions are compatible with the modeled belt path, but it does not determine which compatible acceleration occurs. In particular, every acceleration relation above still contains the shift acceleration $\ddot{s}$. Its value must be consistent with the forces and inertias acting on the two movable pulley assemblies.

3.2 Axial Actuation and Sheave Balances

The preceding section established how the two movable sheaves must move together. It did not determine the motion itself. That requires the forces acting on each movable assembly and the inertia that those forces must accelerate.

The primary and secondary remain separate mechanisms even though the belt geometry couples their positions. Each has its own actuator, spring, moving hardware, and belt reaction, so each must satisfy its own axial equation of motion. We therefore complete the primary balance in its local coordinate x_p , then do the same for the secondary in x_s . Once each mechanism has been interpreted and every local force has been derived, the geometric maps developed above in [Section 3.1](#) express

both balances through the shared shift coordinate s . This final step shows how the two local free bodies act on one shift motion.

3.2.1 Primary Movable-Sheave Balance

The primary is easiest to understand by separating rotation about the shaft from motion along it. The entire primary rotates with the shaft at ω_p , but its parts do not all move axially. The fixed sheave remains at one axial position. Opposite it, the movable sheave, ramp carrier, and the hardware attached rigidly to them slide together along the shaft. Motion in the closing direction narrows the pulley groove and clamps the belt; motion in the opening direction widens the groove.

The flyweights provide the closing drive. They are attached to the fixed sheave and carried around the shaft with it, but each can also pivot about its attachment to the fixed sheave. As the primary rotates, centrifugal loading tends to swing each flyweight radially outward until it presses against a ramp carried by the movable sheave. Because the ramp surface is inclined, its contact reaction has an axial component that drives the movable sheave in the closing direction. The primary spring pushes the movable sheave in the opening direction, and the belt pushes back through its normal reaction on the movable face.

This leaves two distinct moving bodies in the model. The movable sheave, ramp carrier, and attached hardware form one translating assembly with axial coordinate x_p . The flyweights rotate with the primary and pivot separately, but do not translate rigidly with x_p ; they deliver their closing action to the movable sheave through the ramp contacts. During a shift, the net axial force must accelerate the translating assembly, while the flyweights require their own rotational balance. We therefore begin by isolating the translating assembly, then determine the force transmitted to it by the flyweights. This mechanism and the resulting axial free body are shown together in [Figure 3.4](#).

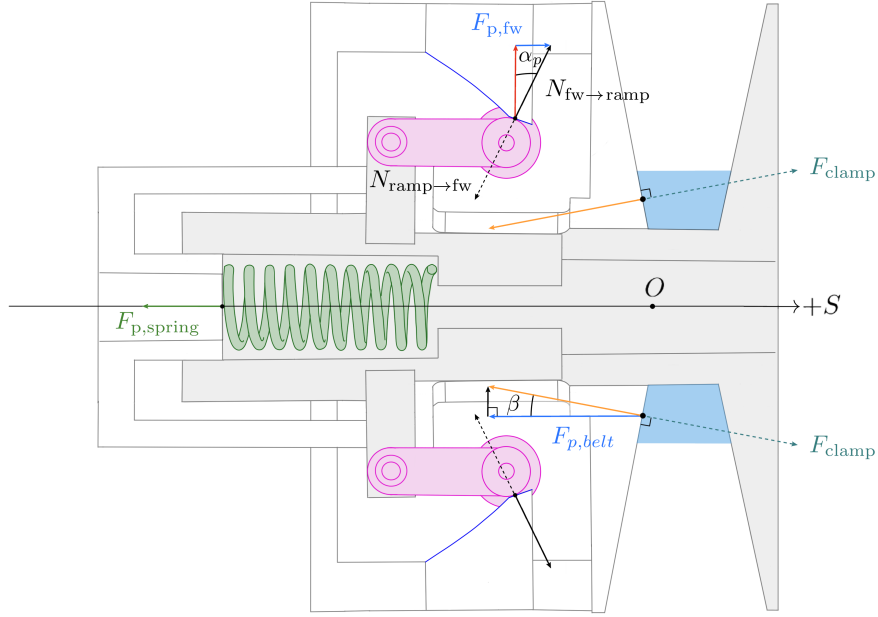


Figure 3.4: Axial free-body diagram of the primary movable assembly (unshaded). The symmetric sides of the cross-section are used to show the flyweight–ramp and belt–sheave force resolutions separately. Dotted arrows show the equal-and-opposite forces on the flyweight and belt, with F_{clamp} indicating the movable sheave’s squeeze on the belt.

As established in Section 3.1, positive x_p closes the primary. Let m_p be the total mass of the movable sheave, ramp carrier, and every other component that translates rigidly with them. The three axial forces shown in Figure 3.4 are:

- $F_{p, \text{fly}}$, the closing force delivered to the ramp carrier by the flyweight contacts;
- $F_{p, \text{spring}}$, the opening force exerted by the primary spring; and
- $F_{p, \text{belt}}$, the opening force exerted by the belt on the movable face.

Newton’s second law along the shaft is therefore

$$m_p \ddot{x}_p = F_{p, \text{fly}} - F_{p, \text{spring}} - F_{p, \text{belt}}. \quad (3.2.1)$$

This free body tells us exactly what remains to be found. The spring force will follow from its compression, and the belt force from the normal reaction on the conical face. The flyweight force is harder. It is the axial result of a contact on a body that pivots rather than translating rigidly with the sheave. We therefore determine $F_{p, \text{fly}}$ first.

Force Transmitted by the Flyweights

Flyweight free body Figure 3.5 isolates one flyweight in the radial–axial plane fixed to the primary. The flyweight is pinned at P and pivots about the circumferential axis through that point. Its pose is described by q_f , measured from $+x_p$ to the body-fixed reference line shown in the figure.

At $q_f = 0$, this line is aligned with $+x_p$; at $q_f = \pi/2$, it points radially outward. Increasing q_f therefore describes the outward-pivoting direction.

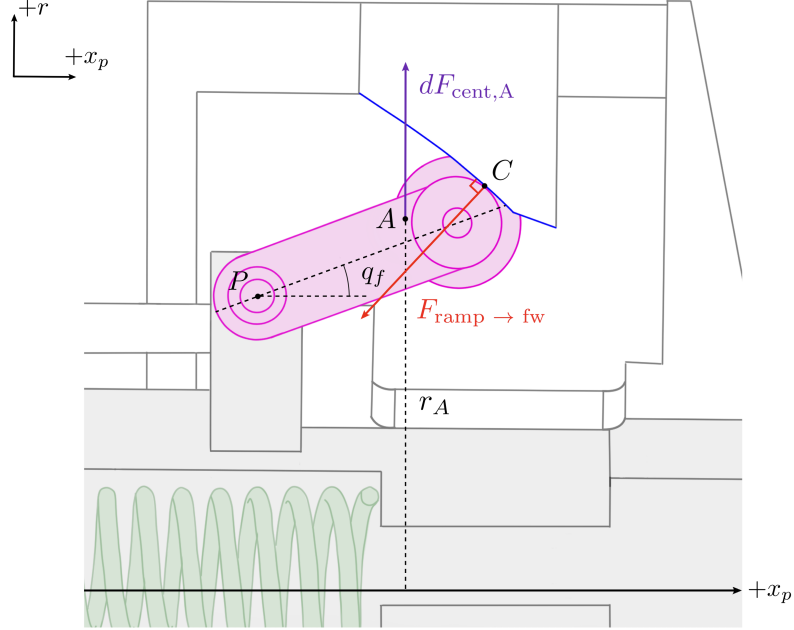


Figure 3.5: Free body of one primary flyweight in the frame rotating with the primary. Centrifugal loading acts throughout the flyweight mass; the marked element A identifies one infinitesimal mass dm at shaft radius r_A and its outward load. The ramp-contact normal at C and the pivot reaction at P complete the force system.

As the primary rotates at ω_p , the flyweight experiences centrifugal loading throughout its mass. Each infinitesimal element is loaded radially outward according to its distance from the primary shaft. The marked element A is one example: for mass dm at shaft radius r_A , its outward load is

$$dF_{\text{cent},A} = \omega_p^2 r_A dm. \quad (3.2.2)$$

Every other element is loaded in the same way at its own shaft radius. Together, these forces form the distributed centrifugal loading of the flyweight.

Because the flyweight's relative motion is confined to the radial-axial plane, the Euler and Coriolis inertial forces are circumferential and have no moment about the circumferential pivot axis. The q_f balance therefore depends on ω_p , but not on $\dot{\omega}_p$.

Taking moments about P separates the roles of these loads. The pivot reaction passes through P and therefore contributes no moment. The radial lines of action of the centrifugal loads generally do not pass through P , so together they produce a moment in the direction of increasing q_f . The ramp normal produces the opposing moment.

The figure shows one flyweight, whereas the primary contains a symmetric set whose members follow the same q_f . Let $\tau_{f,\text{cent}}$ and $\tau_{f,\text{contact}}$ denote the respective magnitudes of the centrifugal and contact moments summed over this complete set. Since their shared coordinate q_f is a rotation about the flyweight pivots, let I_f denote the sum of each flyweight's mass moment of inertia about its own pivot axis. With positive moment defined in the direction of increasing q_f , rotational Newton's second law gives

$$\tau_{f,\text{cent}} - \tau_{f,\text{contact}} = I_f \ddot{q}_f. \quad (3.2.3)$$

Mechanism geometry The moment balance is written in the flyweight coordinate q_f , whereas the force required by the movable-sheave balance must be expressed through the axial coordinate x_p . The ramp geometry connects the two: each sheave position fixes the corresponding flyweight pose.

The centrifugal moment in Equation (3.2.3) comes from the distributed loading shown in the figure, so its value depends on how the complete flyweight mass is distributed about the primary shaft. The marked element A contributes $r_A^2 dm$ to the mass moment of inertia about that shaft. Adding the corresponding contributions from every element of every flyweight gives the complete shaft-axis inertia $J_{f,p}$. Since x_p fixes the flyweight pose, it fixes every element's shaft radius and therefore fixes $J_{f,p}$. This shaft-axis inertia should not be confused with the pivot-axis inertia I_f in Equation (3.2.3), which is constant for rigid flyweights.

The mechanism therefore supplies two configuration maps:

$$q_f = q_f(x_p), \quad J_{f,p} = J_{f,p}(x_p). \quad (3.2.4)$$

Together with the constant I_f , these maps abstract the physical geometry of the flyweight mechanism into the quantities required by the general force balance.

Remark (Constructing the flyweight maps)

The mechanism-specific construction is not carried out here. Chapter E works through two common flyweight arrangements, showing how their geometry and mass distribution produce $q_f(x_p)$, $J_{f,p}(x_p)$, and I_f .

With these mechanism properties specified, the remaining task is to express the two moments and $\ddot{q}_f$ in Equation (3.2.3) through the primary motion. We begin with the moments.

Both could be found by resolving the forces and constructing their moment arms for the specific flyweight geometry. A more convenient approach is equivalent virtual work. Hold the mechanism at its current position and imagine a small compatible motion. During this motion, a moment τ_f acting through dq_f performs the work

$$\tau_f dq_f = \delta W_f. \quad (3.2.5)$$

We can therefore find each moment by calculating the work of its underlying load during the same motion.

For the centrifugal loading, at fixed instantaneous angular speed, the work associated with a change dJ in shaft-axis inertia is $\frac{1}{2}\omega^2 dJ$. Applying this relation to the flyweights gives

$$\tau_{f,\text{cent}} dq_f = \delta W_{f,\text{cent}} = \frac{1}{2} \omega_p^2 dJ_{f,p}. \quad (3.2.6)$$

For the contact moment, the corresponding work is the force–displacement work $F d\ell$. The movable sheave moves axially through dx_p , so only the axial component of the ramp-contact force performs work. This component is $F_{p,\text{fly}}$, the closing force introduced by the movable-sheave free body, and its work is therefore $F_{p,\text{fly}} dx_p$. Under **Assumption 3 (Idealized Actuator Mechanics)**, the ramp contact is treated as an ideal frictionless constraint, so no work is dissipated at the contact. The work of the contact moment therefore appears entirely as axial work on the movable assembly. Using the positive magnitudes defined above,

$$\tau_{f,\text{contact}} dq_f = \delta W_{f,\text{contact}} = F_{p,\text{fly}} dx_p. \quad (3.2.7)$$

Flyweight acceleration The two moment terms are now written in forms that can be returned to the rotational balance. Its remaining term is the flyweight angular acceleration. Ramp contact makes this acceleration a consequence of the sheave motion rather than an independent state. Since the contact geometry already gives q_f as a function of x_p , its angular velocity is

$$\dot{q}_f = \frac{dq_f}{dx_p} \dot{x}_p, \quad (3.2.8)$$

and one more derivative gives

$$\ddot{q}_f = \frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2. \quad (3.2.9)$$

The second term appears when the amount of flyweight rotation required per unit sheave travel changes along the ramp. Even at constant $\dot{x}_p$, the flyweight then changes angular speed as it moves.

Axial closing force There is now an expression for every term in the rotational balance. Multiplying **Equation (3.2.3)** by dq_f and substituting **Equations (3.2.6)**, **(3.2.7)** and **(3.2.9)** brings them together,

$$\frac{1}{2} \omega_p^2 dJ_{f,p} - F_{p,\text{fly}} dx_p = I_f \left(\frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 \right) dq_f. \quad (3.2.10)$$

Rearranging for the force transmitted to the movable sheave gives

$$\boxed{F_{p,\text{fly}} = \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \left(\frac{dq_f}{dx_p} \right)^2 \ddot{x}_p - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2.} \quad (3.2.11)$$

The result now has the form required by the movable-sheave free body. The first term is the centrifugal drive and therefore uses the shaft-axis property $J_{f,p}$. The final two terms arise from

acceleration about the flyweight pivots and therefore use the pivot-axis property I_f : the first reflects $\ddot{x}_p$ through the motion ratio, while the second appears when that motion ratio changes with position.

Remark (Admissible flyweight maps)

The completed force law uses $q_f''(x_p)$ and $J_{f,p}'(x_p)$, while the selected coordinate directions require primary closure to correspond to outward flyweight rotation. Over the admissible shift range, the supplied mechanism maps must therefore satisfy

$$q_f \in C^2, \quad J_{f,p} \in C^1, \quad 0 < \frac{dq_f}{dx_p} < \infty.$$

These are requirements on the fitted or constructed maps, not additional equations of motion.

The one-sided flyweight contact can transmit compression but not tension, so the force predicted by [Equation \(3.2.11\)](#) must satisfy $F_{p,\text{fly}} \geq 0$.

Spring and Belt Forces

Returning to the axial force balance, the two remaining forces in [Equation \(3.2.1\)](#) act directly on the translating assembly.

Primary spring Let k_p be the spring stiffness and $x_{p,\text{pre}}$ its installed compression when the primary is open at $x_p = 0$. Closing the sheave by x_p adds the same amount of compression, so the spring opening force is

$$F_{p,\text{spring}} = k_p (x_{p,\text{pre}} + x_p). \quad (3.2.12)$$

Belt reaction Let N_p be the total integrated normal load over both primary sheave faces. This load is coupled to the belt and the secondary balance, so it remains an unknown at this stage. Under [Assumption 11 \(Symmetric Sheave-Face Load Sharing\)](#), the movable face carries one half of this total load. Its normal is inclined by the sheave half-angle β from the axial direction, so the opening force on the movable face is

$$F_{p,\text{belt}} = \frac{N_p}{2} \cos \beta. \quad (3.2.13)$$

Completed Primary Balance

Each force in the original free body now has a mechanism-level expression. Substitute [Equations \(3.2.11\)](#) to [\(3.2.13\)](#) into [Equation \(3.2.1\)](#) and collect the terms multiplying $\ddot{x}_p$:

$$\left[m_p + I_f \left(\frac{dq_f}{dx_p} \right)^2 \right] \ddot{x}_p = \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 - k_p (x_{p,\text{pre}} + x_p) - \frac{N_p}{2} \cos \beta. \quad (3.2.14)$$

The flyweights now appear through the same two axes introduced by their free body. Rotation about the primary shaft, described by $J_{f,p}$, provides the speed-dependent closing drive. Rotation about the flyweight pivots, described by I_f , adds to the inertia that the axial motion must accelerate. If the motion ratio dq_f/dx_p changes along the ramp, the flyweights also accelerate even when $\dot{x}_p$ is constant, producing the term proportional to $\dot{x}_p^2$. The spring and belt forces act directly on the movable assembly and oppose closure exactly as shown in the first free body.

This is the primary axial equation of motion. It states how the flyweight action is divided among spring compression, belt loading, and acceleration of the translating and pivoting hardware.

Point-Mass Limit and Relation to Existing Models

The general force law becomes the familiar classical actuator model when the distributed flyweight mass is collapsed to its centre of mass and the flyweight's pivot dynamics are neglected. Let m_f be the total mass of the symmetric flyweight set and let $r_f(x_p)$ be the common instantaneous centre-of-mass radius. Under the point-mass assumption, the shaft-axis inertia reduces to

$$J_{f,p} = m_f r_f^2. \quad (3.2.15)$$

Substituting this point-mass inertia into the centrifugal term of [Equation \(3.2.11\)](#), while omitting the pivot-acceleration terms, gives

$$F_{p,\text{fly}}^{\text{pm}} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}. \quad (3.2.16)$$

This is the familiar point-mass result. The factor $m_f r_f \omega_p^2$ is the total centrifugal loading, while dr_f/dx_p is the local radial-to-axial motion ratio that converts that loading into sheave force. The path need not be straight, and its motion ratio need not be constant. A sliding mass on a straight ramp is one simple case; for a pivoted flyweight, $r_f(x_p)$ instead follows from the flyweight pose and its mass geometry.

If the movable sheave is also in axial equilibrium, the point-mass actuator force balances the opening forces from the spring and belt:

$$m_f r_f \omega_p^2 \frac{dr_f}{dx_p} = k_p (x_{p,\text{pre}} + x_p) + \frac{N_p}{2} \cos \beta. \quad (3.2.17)$$

This reduction appears directly in the quasi-static model of [Messick \(2018\)](#). Their three wedge-shaped flyweights slide along linear ramps without changing orientation. Measured CAD geometry supplies the centre-of-mass radius and the radial-to-axial displacement ratio, giving a force of the same form as [Equation \(3.2.16\)](#). Friction is neglected, and the resulting primary force is supplied to the belt model as a boundary condition; shifting is treated as a sequence of equilibrium configurations rather than a time-dependent response. [Arango and Mu noz Alzate \(2020\)](#) use the same instantaneous force conversion to construct both sliding-weight ramps and pivoted flyweight-cam profiles. Their contact geometry is designed so that the available centrifugal force supplies the required clamping force through the local mechanical advantage.

Other treatments retain more of the contact geometry while keeping the same instantaneous-equilibrium closure. [Kim et al. \(2002\)](#) resolve the two contacts of a centrifugal roller and include

contact friction when converting roller centrifugal force into driver thrust. The current roller radius and contact angles vary with ratio, but no roller acceleration enters the thrust equation. Skinner (2020) resolve a flyweight, its supporting link, and the ramp reactions in a complete free body, including the centrifugal loading of both the flyweight and the link. In the present notation, those separate mass contributions could be collected in $J_{f,p}(x_p)$. Skinner’s derivation therefore retains a substantially richer equilibrium force path than a single fixed ramp factor, but still closes the flyweight system by force equilibrium at each configuration.

Despite their different levels of geometric detail, none of these actuator relations includes the acceleration of the flyweight mechanism. Messick does not model shift time, Skinner solves algebraically for the speed required at each configuration, and Kim places the time dependence in a separate experimentally identified first-order ratio law, using different fitted coefficients for upshift and downshift. Conversely, the transient belt model of Ballew (2015) imposes primary axial force through a controller rather than resolving the mechanical actuator. Across these examples, transient shift motion is therefore obtained without returning the flyweights’ own acceleration to the primary axial balance.

The distinction of the present formulation is therefore not that it can describe a curved ramp or a detailed flyweight shape; the equilibrium models above can already retain substantial geometric detail. Here, the complete mass distribution enters through $J_{f,p}(x_p)$, so the centrifugal drive is not restricted to a centre-of-mass point model. More importantly, the flyweight moment balance is not closed at instantaneous equilibrium. Mapping $I_f\ddot{q}_f$ into x_p produces both the reflected pivot inertia and the changing-motion-ratio term in Equation (3.2.14). Consequently, the flyweight drive, spring force, belt reaction, translating mass, and pivoting mass all determine the same shift acceleration rather than being divided between an equilibrium actuator law and a separate shift-response model. No equivalent explicit reflection of the flyweights’ pivoting inertia was identified in the reviewed mechanically actuated rubber-belt CVT formulations.

This extension does not retain every effect represented in the earlier models. Structural deformation is excluded by Assumption 2 (Rigid Pulley and Actuation Hardware), while Assumption 3 (Idealized Actuator Mechanics) treats the ramp contact as an ideal constraint rather than resolving its tribology or local compliance. The roller-contact friction retained by Kim et al. (2002), for example, therefore lies outside the present force law; roller spin and contact compliance are likewise not resolved. The added fidelity is in the coupled mechanism dynamics, not in frictional or compliant contact detail.

3.2.2 Secondary Movable-Sheave Balance

While the primary closes mainly in response to rotational speed, the secondary is torque reactive: torque transmitted through the pulley can also increase the axial force clamping the belt.

To see how, begin with the two secondary sheaves rotationally disconnected. The fixed sheave rotates with the secondary shaft. The movable sheave can slide along that shaft and rotate relative to the fixed sheave. Because the belt contacts the two faces separately, it applies torque to each body. With no connection between them, the two bodies can accelerate differently and rotate relative to one another.

Now place a pin on the movable sheave in a straight slot running along the shaft. The movable sheave remains free to slide, but the pin prevents relative rotation. Whenever the two sheaves tend to rotate differently, the pin presses against the side of the slot and a normal reaction develops. Because the slot is axial, this normal acts entirely in the circumferential direction. It transfers

torque between the sheaves but produces no axial force.

A helix is the same slot tilted around the shaft. The contact normal remains perpendicular to the slot surface, but now has both circumferential and axial components. The circumferential component continues to couple the rotational motion of the sheaves, while the axial component acts on the movable sheave. Relative rotation is therefore allowed only when accompanied by axial travel along the helix.

These three cases and their contact reactions are compared in [Figure 3.6](#).

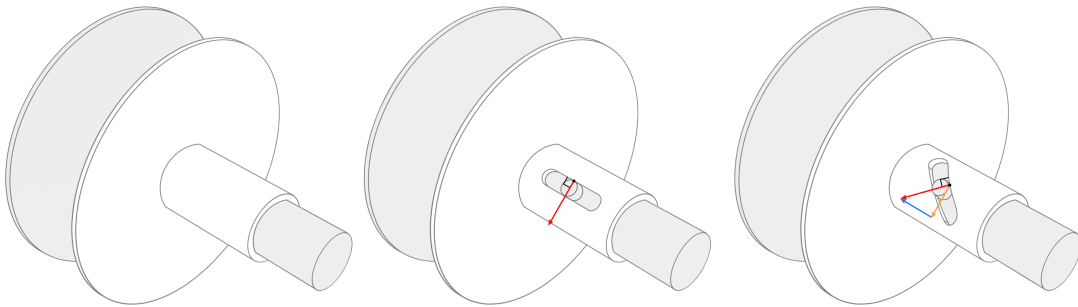


Figure 3.6: How the helix creates torque-reactive axial loading. With no connection, the two sheaves may rotate independently. A straight axial slot couples their rotation through a purely circumferential reaction; inclining the slot adds an axial component to that same reaction, coupling relative rotation to axial travel.

If the axial component of the helix reaction is balanced by the other axial loads, the movable sheave remains at its current position. Its relative angle then remains fixed, and the two sheaves rotate together with the shaft. If the axial loads do not balance, their resultant accelerates the movable sheave along the helix, and it rotates by the amount required by the slot geometry.

In the actual secondary, the belt and spring load this mechanism. The belt applies torque to the two sheave faces separately. The secondary spring is installed both compressed and twisted: its torsional preload tends to rotate the movable sheave relative to the fixed sheave, while its axial compression pushes the movable sheave toward closure. The torsional action loads the helix; the axial action enters the movable-sheave balance directly.

The axial free body gives the target for the derivation.

Axial Free Body

[Figure 3.7](#) isolates the movable assembly in the axial direction.

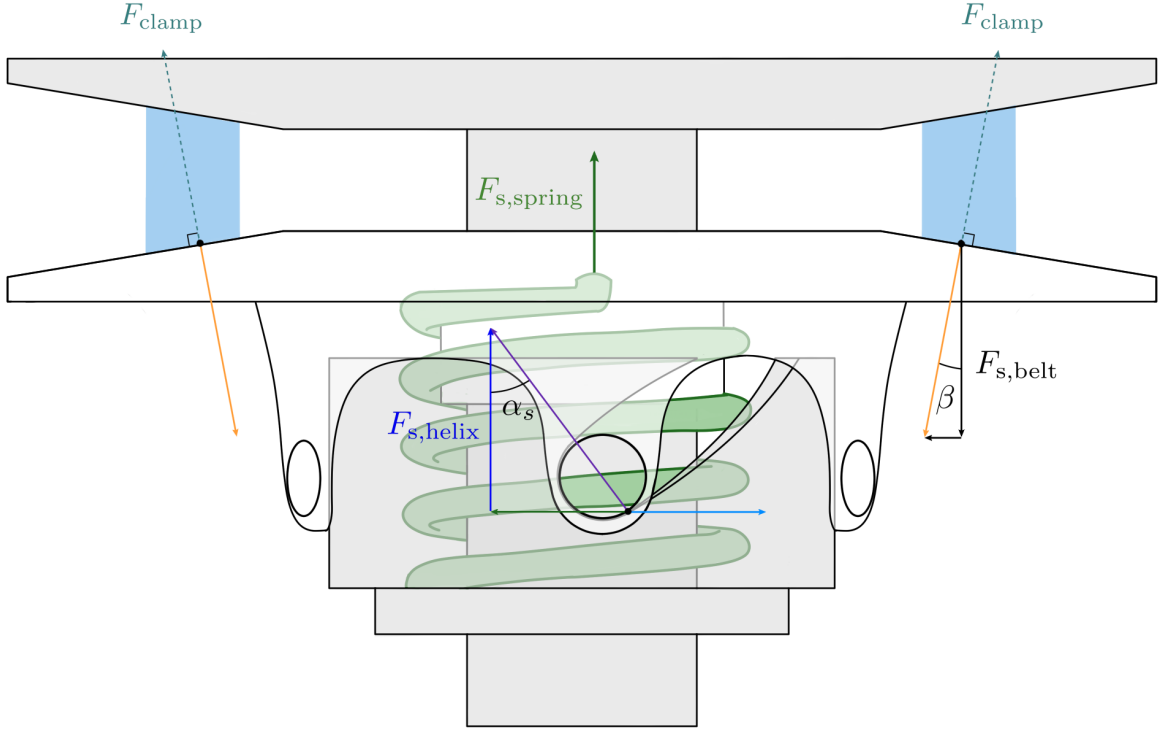


Figure 3.7: Axial free body of the complete movable secondary assembly. The axial spring force and belt reaction act directly on the translating assembly, while the secondary's torque-reactive behaviour enters through the pin-helix force $F_{s,\text{helix}}$.

Let m_s be the mass of the movable sheave and every component that translates rigidly with it. Three axial loads act on this assembly. The pin-helix contact contributes the signed force $F_{s,\text{helix}}$. The compressed spring pushes the movable sheave toward the fixed sheave in $+x_s$. The belt pushes the two sheave faces apart, so its reaction on the movable face acts in $-x_s$, just as on the primary. Newton's second law therefore gives

$$m_s \ddot{x}_s = F_{s,\text{helix}} + F_{s,\text{spring}} - F_{s,\text{belt}}. \quad (3.2.18)$$

This free body makes the remaining task clear. The direct spring and belt forces follow from compression and contact geometry. The helix force does not: the tilted slot has established that an axial reaction exists, but not its magnitude or sign. Because the same contact reaction must also carry the torque that constrains relative rotation, its rotational effect is resolved first. This sets the order of the derivation:

1. use the rotational balance to find the torque carried by the helix;
2. use the helix geometry to convert that torque into axial force; and
3. add this force to the direct spring and belt loads in the axial balance.

Rotational Loading of the Movable Sheave

As established in [Section 3.1](#), x_s is measured from the open secondary toward the fixed sheave, so $+x_s$ closes the pulley. Let θ_s measure the rotation of the movable sheave relative to the fixed sheave in the same positive rotational sense already assigned to ω_s (see [Section 2.3.1](#)). Only the conventional torque-reactive helix orientation is considered: as the secondary closes in $+x_s$, the movable sheave turns relative to the fixed sheave in $+\theta_s$.

With these directions set, the first question is what loads the helix circumferentially. The belt applies torque to the movable face, and the twisted spring applies torque between the two sheaves. The pin–helix contact supplies the torque that constrains the resulting relative motion to follow the slot.

Our goal is to find the torque reacted at the pin–helix contact. To do so, isolate the movable assembly and apply Newton’s second law for rotation. Use the following quantities:

- $I_{s,M}$: rotational inertia of the movable sheave and every component that rotates rigidly with it;
- $\omega_{s,M}$: absolute angular velocity of that assembly;
- $\tau_{s,M}$: signed belt torque applied to the movable face;
- $\tau_{s,\text{spring}}$: signed spring torque applied to the movable assembly; and
- $\tau_{s,\text{helix}}$: the contact torque at the pin–helix connection, produced by the circumferential component of the contact normal. A positive $\tau_{s,\text{helix}}$ acts on the movable assembly in $-\theta_s$.

The figures show one representative pin and track. The quantities listed above and later in this section are the totals over the complete set of loaded pin–track contacts, so no additional track multiplier is required.

The rotational balance is therefore

$$\tau_{s,M} + \tau_{s,\text{spring}} - \tau_{s,\text{helix}} = I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.19)$$

Solving for the helix torque gives

$$\tau_{s,\text{helix}} = \tau_{s,M} + \tau_{s,\text{spring}} - I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.20)$$

[Equation \(3.2.20\)](#) gives the contact torque we want, but its right-hand side still needs three relations:

- the share of belt torque applied to the movable face, $\tau_{s,M}$;
- the torque supplied by the wound spring, $\tau_{s,\text{spring}}$; and
- the absolute angular acceleration of the movable assembly, $\dot{\omega}_{s,M}$.

The belt term follows immediately from the face-torque assumption.

Movable-face belt torque Let τ_s be the total signed torque exerted by the belt on the secondary, so $\tau_s = \tau_{s,F} + \tau_{s,M}$. Under [Assumption 11 \(Symmetric Sheave-Face Load Sharing\)](#), the two faces receive equal torque shares. Only the movable-face share appears in its free body:

$$\tau_{s,M} = \frac{\tau_s}{2}. \quad (3.2.21)$$

This equal split is a reduced-contact assumption. During shift, the movable face rotates slightly faster or slower than the fixed face. The two belt–face contacts can therefore have different sliding conditions, so their torques need not divide equally.

Remark (Why the face-torque split can matter)

[Sorge and Cammalleri \(2011\)](#) resolve the two contacts separately and show that the face torques can depart substantially from an equal split; distributed tangential forces on the two faces can even act in opposite directions over parts of the wrap. Since $\tau_{s,M}$ enters [Equation \(3.2.20\)](#) directly, this is also a potential error in the torque reacted by the helix.

The remaining terms both depend on the helix kinematics: $\dot{\omega}_{s,M}$ depends on how the movable sheave rotates as it shifts, and the spring torque depends on how far that motion unwinds the spring. We therefore derive the relation between x_s and θ_s next.

Motion Imposed by the Helix

The helix geometry tells us how axial shift changes the relative angle of the movable sheave. Unwrap one track from its effective cylindrical radius r_h , as seen in [Figure 3.8](#). A small relative rotation $d\theta_s$ moves the pin by $r_h d\theta_s$ circumferentially while the sheave moves by dx_s axially. Let α_s be the local angle of the unwrapped track, measured from the positive circumferential direction toward the positive axial direction. For the conventional helix orientation considered here,

$$0 < \alpha_s < \frac{\pi}{2}.$$

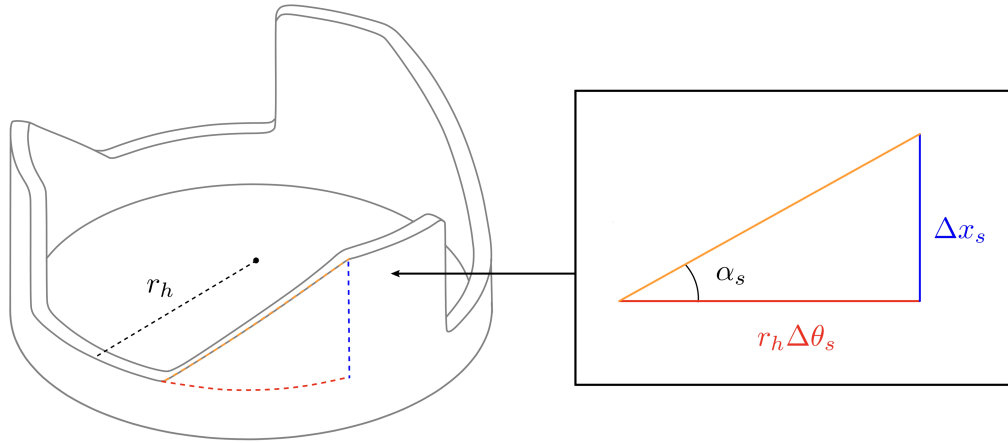


Figure 3.8: Unwrapped secondary-helix geometry. The same track angle relates relative rotation to axial travel and reaction torque to axial force.

Taking the infinitesimal limit of the triangle in [Figure 3.8](#),

$$\tan \alpha_s = \frac{dx_s}{r_h d\theta_s}, \quad \frac{d\theta_s}{dx_s} = \frac{1}{r_h \tan \alpha_s}. \quad (3.2.22)$$

Thus, $d\theta_s/dx_s$ is the relative rotation required per unit axial travel. It is positive for the conventional helix orientation assumed here. Returning to the straight axial slot from the opening example, $\alpha_s \rightarrow \pi/2$ and $d\theta_s/dx_s \rightarrow 0$, as expected.

Taking $\theta_s(0) = 0$ at the open secondary, integrate the local motion ratio to obtain

$$\theta_s(x_s) = \int_0^{x_s} \frac{1}{r_h \tan \alpha_s(\xi)} d\xi. \quad (3.2.23)$$

Equation (3.2.23) defines the helix as a function $\theta_s(x_s)$: for each axial position, it gives the relative angle enforced by the track. The function need not come from a constant helix angle. Any supplied function may be used provided it is single-valued and strictly increasing over the admissible shift range.

Rotation of the movable sheave We can now use this function to determine the two remaining quantities in the rotational balance. First consider the absolute motion of the movable sheave. The fixed sheave rotates with the secondary shaft at ω_s , while the movable sheave also has the relative rotation θ_s :

$$\omega_{s,M} = \omega_s + \dot{\theta}_s = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s. \quad (3.2.24)$$

Because θ_s was defined in the same positive rotational sense as ω_s , the relative speed enters with the plus sign shown here. Differentiating gives the angular acceleration required by the rotational balance:

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.2.25)$$

The shaft acceleration $\dot{\omega}_s$ remains unknown here because ω_s is a state of the coupled model. It is solved together with the axial motion rather than by the helix relation alone. The term proportional to $\ddot{x}_s$ is the angular acceleration caused by axial acceleration. The term proportional to $\dot{x}_s^2$ appears only when the motion ratio varies with position.

Torsional Spring Loading The same helix function tells us how much the spring has unwound. Let $k_{s,\theta}$ be the spring's torsional stiffness and $\theta_{s,\text{pre}}$ its signed twist at $x_s = 0$. Increasing θ_s reduces this twist, leaving $\theta_{s,\text{pre}} - \theta_s(x_s)$. The resulting torque on the movable assembly is

$$\tau_{s,\text{spring}} = k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)]. \quad (3.2.26)$$

For the installed winding assumed here, this quantity is positive while the spring remains preloaded. The opposite winding direction can be represented by a signed $\theta_{s,\text{pre}}$.

We now have relations for all three terms on the right-hand side of **Equation (3.2.20)**. Substituting them gives the pin-helix contact torque:

$$\begin{aligned}\tau_{s,\text{helix}} = & \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] \\ & - I_{s,M} \left[\dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right].\end{aligned}\tag{3.2.27}$$

This completes the rotational part of the derivation. The next step is to use the pin–helix contact geometry to find the axial force associated with this torque.

Axial Component of the Helix Reaction

The torque above is the moment produced by the circumferential component of the contact normal. Let $F_{s,\text{helix}}$ denote the signed axial force produced by the helix reaction, positive in $+x_s$.

Under **Assumption 3 (Idealized Actuator Mechanics)**, the pin–helix contact is treated as an ideal frictionless constraint. The same compatible-work argument used for the primary therefore converts the rotational contact load into its axial effect without contact dissipation. Consider the motion of the movable sheave relative to the helix carrier. The contact normal does no net work in this relative motion. Its axial component contributes $F_{s,\text{helix}} dx_s$, while its circumferential component contributes $-\tau_{s,\text{helix}} d\theta_s$. Their sum is zero, so

$$F_{s,\text{helix}} dx_s = \tau_{s,\text{helix}} d\theta_s.\tag{3.2.28}$$

Using the motion ratio gives

$$F_{s,\text{helix}} = \tau_{s,\text{helix}} \frac{d\theta_s}{dx_s}.\tag{3.2.29}$$

The motion ratio $d\theta_s/dx_s$ is therefore the mathematical origin of the axial effect: it converts the contact torque into axial force. In the straight-slot limit used in the opening example, this ratio is zero, so the slot can react torque while the axial conversion vanishes.

Substituting **Equation (3.2.27)** gives the helix force in terms of the model variables:

$$\boxed{F_{s,\text{helix}} = \frac{d\theta_s}{dx_s} \left\{ \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] - I_{s,M} \left[\dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right] \right\}}.\tag{3.2.30}$$

This gives $F_{s,\text{helix}}$. The belt and wound spring provide the driving torques inside the braces; part of that torque may be required to accelerate the movable assembly, and only the remainder is converted into axial thrust.

Remark (Admissible helix maps)

The completed force uses the first two derivatives of $\theta_s(x_s)$. Over the admissible shift range, the supplied helix map must therefore be single-valued and twice continuously differentiable, with a finite, nonzero motion ratio:

$$\theta_s \in C^2, \quad 0 < \frac{d\theta_s}{dx_s} < \infty.$$

A manufactured multi-angle helix may still be represented, but nominal segments must be joined smoothly rather than by a mathematical corner.

The difficult force in the preliminary axial balance is now known. The two remaining forces act directly on the translating assembly.

Direct Spring and Belt Forces

Direct axial spring force The spring force follows directly from its compression. Let $k_{s,x}$ be the axial stiffness and $x_{s,\text{pre}}$ the compression at $x_s = 0$. Closing the secondary by x_s lets the spring extend by the same amount, so its remaining compression is $x_{s,\text{pre}} - x_s$. Hence,

$$F_{s,\text{spring}} = k_{s,x} (x_{s,\text{pre}} - x_s). \quad (3.2.31)$$

This force and the torque in Equation (3.2.26) come from the same installed spring. Following the reduced decomposition of Kim et al. (2002), the model represents its axial compression and torsional twist with separate linear laws. This use of prescribed, uncoupled spring laws is part of Assumption 3 (Idealized Actuator Mechanics); compression–twist coupling, hysteresis, and end-contact friction are not retained.

Belt opening reaction The belt opening force follows the same projection already used on the primary. Let N_s be the integrated normal load over both secondary faces. Under Assumption 11 (Symmetric Sheave-Face Load Sharing), the movable face carries one half of this total load, giving

$$F_{s,\text{belt}} = \frac{N_s}{2} \cos \beta. \quad (3.2.32)$$

As on the primary, N_s is not determined by this local free body. It remains a coupled unknown supplied by the belt-contact and remaining system equations.

Completed Local Secondary Balance

We now have a relation for every term in the axial free body. Substitute them into Equation (3.2.18) and collect the terms multiplying $\ddot{x}_s$:

$$\begin{aligned}
\left[m_s + I_{s,M} \left(\frac{d\theta_s}{dx_s} \right)^2 \right] \ddot{x}_s = & \left[\frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \frac{d\theta_s}{dx_s} \\
& - I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \\
& + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta.
\end{aligned} \tag{3.2.33}$$

The completed equation separates into four physical pieces:

- $m_s + I_{s,M}(d\theta_s/dx_s)^2$ is the local axial inertia. Just as the flyweight pivot inertia appeared in the primary balance, the helix makes the secondary's relative-rotational inertia part of the load that axial motion must accelerate;
- the first bracket contains the belt and torsional-spring torques, less the portion required to accelerate the movable assembly with the secondary shaft;
- the term proportional to $\dot{x}_s^2$ accounts for a helix motion ratio that varies with position; and
- the last two terms are the direct spring closing force and belt opening force.

This is the axial equation of motion we set out to obtain. If its right-hand side remains zero and the sheave is initially stationary, x_s and θ_s remain fixed while the two sheaves rotate together. A nonzero right-hand side accelerates the sheave axially, and $\theta_s(x_s)$ fixes the accompanying relative rotation.

Classical Limit and Relation to Existing Models

As on the primary, the full dynamic balance should recover the established actuator equilibrium when the shift mechanism is stationary. Consider a fixed secondary configuration at constant shaft speed, $\dot{x}_s = \ddot{x}_s = \dot{\omega}_s = 0$. Under the equal face-torque split adopted here, the helix force reduces to

$$F_{s,\text{helix}} = \left[\frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s}. \tag{3.2.34}$$

Adding the direct spring and belt loads gives the corresponding equilibrium condition for the complete movable assembly:

$$\left[\frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s} + k_{s,x} (x_{s,\text{pre}} - x_s) = \frac{N_s}{2} \cos \beta. \tag{3.2.35}$$

These equations expose the classical mechanism directly. The axial spring supplies one closing force, while the belt torque and torsional-spring torque are converted into a second closing force by the helix motion ratio. The factor 1/2 is not a property of the helix itself; it follows from the equal face-torque closure used here. A face-resolved contact model would instead use the torque $\tau_{s,M}$ actually applied to the movable face.

The important comparison is how this equilibrium relation is used. In the present formulation, [Equations \(3.2.34\) and \(3.2.35\)](#) are stationary limits of a dynamic balance. In much of the mechanically actuated CVT literature, an equivalent algebraic relation remains the actuator closure while the transmission is shifting.

The formulation of Kim et al. (2002) is a direct example. Their driven thrust is the sum of a spring force and a torque-cam force obtained from instantaneous equilibrium. The cam term retains the applied torque, torsional-spring load, cam geometry, and cam friction. Suppressing that friction and adopting the equal face-torque split used here gives the same spring-plus-torque-cam structure as Equation (3.2.35). Their shift transient is not obtained from the movable secondary’s equation of motion, but from a separately identified first-order ratio law. Kim et al. therefore retain contact friction that is absent here, but do not use the secondary assembly’s inertia to determine its shift acceleration.

The design-oriented sources examined here apply the same configuration-by-configuration logic at different levels of detail. Messick (2018) calculates the secondary thrust required by a steady-state belt solution and uses it as a target for helix design, explicitly excluding the shift response time. Skinner (2020) calculates the compression-spring, torsional-spring, and torque-feedback contributions to secondary clamping algebraically. Arango and Muñoz Alzate (2020) construct the torque-sensing cam profile from the transmitted torque, spring force, required axial force, and axial displacement over the operating range. These treatments are useful for sizing and tuning the mechanism, but they do not determine its motion from a movable-sheave equation of motion.

Among the reviewed sources, Sorge and Cammalleri (2011) provide the most detailed treatment of the belt–face–helix interaction during shifting. Their distributed contact model allows the fixed and movable faces to rotate at different speeds and determines the torque $M_{N,s}$ carried by the movable face. The resulting driven-side actuator relation combines the spring load with a helix contribution having the same torque-times-motion-ratio structure as Equation (3.2.34). It therefore replaces the assumed $\tau_s/2$ with a face-resolved torque and can capture substantial departures from an equal split.

That contact calculation is shift-aware, but the secondary actuator itself is still closed quasi-statically. Sorge and Cammalleri prescribe the instantaneous radius and the dimensionless shift speed $\rho = \dot{r}/(\omega_f r)$, then solve for the corresponding belt-contact state. A nonzero ρ changes the relative face speeds, sliding directions, contact tractions, and face-torque split, but it is not produced by the forces on the movable secondary. Because no axial or relative-rotational equation of motion is written for the movable half-pulley, its translational inertia and helix-reflected rotational inertia are absent by construction.

The models therefore retain complementary parts of the physics. Kim et al. retain cam friction, while Sorge and Cammalleri resolve the two belt–face contacts and their unequal torques. The present reduced model retains neither capability. It instead carries the movable assembly’s rotational balance through the general constraint $\theta_s(x_s)$ and solves its free axial motion. This introduces the shaft-acceleration coupling, the reflected inertia $I_{s,M}(d\theta_s/dx_s)^2$, and the velocity-squared term generated when the helix motion ratio varies with position. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent explicit projection of the movable side’s rotational inertia through a mechanical secondary helix was identified.

This distinction is deliberately narrow. The rigid-hardware, ideal-actuator, and symmetric-face idealizations are stated explicitly in Assumption 2 (Rigid Pulley and Actuation Hardware), Assumption 3 (Idealized Actuator Mechanics), and Assumption 11 (Symmetric Sheave-Face Load Sharing). They leave the compliant and frictional actuator details, and the face-resolved load redistribution of the models above, outside the present balance. The comparison is therefore not a claim of greater local contact fidelity; it isolates the added treatment of the movable assembly’s inertia and helix-constrained dynamics.

3.2.3 Coupling Through the Shared Shift Coordinate

The primary and secondary balances have been derived in their natural local coordinates, x_p and x_s . The fixed-length belt geometry makes those motions compatible. Choosing the primary displacement as the shared shift coordinate leaves the primary balance unchanged and supplies the secondary motion through the map $x_s = x_s(s)$.

Local-to-Shared Kinematics

Because the shared coordinate is the primary displacement,

$$x_p = s, \quad \dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.2.36)$$

The secondary displacement follows from the fixed-length belt map. Its first two time derivatives are

$$x_s = x_s(s), \quad \dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2. \quad (3.2.37)$$

Positive x_p and x_s both denote sheave closing. Increasing s closes the primary but opens the secondary, so $dx_s/ds < 0$. If the belt geometry is nonlinear, this motion ratio changes through the shift and $d^2x_s/ds^2 \neq 0$. Both derivatives are determined by the belt-geometry relation established earlier.

Two Conditions on One Shift Motion

Substituting Equation (3.2.36) into Equation (3.2.14) gives the primary condition directly:

Primary Condition on the Shared Shift Motion (3.2.38) $ \begin{aligned} \left[m_p + I_f \left(\frac{dq_f}{ds} \right)^2 \right] \ddot{s} = & \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds} \\ & - I_f \frac{dq_f}{ds} \frac{d^2q_f}{ds^2} \dot{s}^2 \\ & - k_p (x_{p,\text{pre}} + s) - \frac{N_p}{2} \cos \beta. \end{aligned} $
--

Because $x_p = s$, this is simply the local primary balance written using the shared coordinate.

The secondary condition follows just as directly. Substituting Equation (3.2.37) into Equation (3.2.33) and collecting terms gives

$$\begin{aligned}
 & \left[m_s + I_{s,M} \left(\frac{d\theta_s}{dx_s} \right)^2 \right] \frac{dx_s}{ds} \ddot{s} \\
 &= \frac{d\theta_s}{dx_s} \left[\frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \\
 & \quad + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta \\
 & \quad - \left\{ \left[m_s + I_{s,M} \left(\frac{d\theta_s}{dx_s} \right)^2 \right] \frac{d^2 x_s}{ds^2} \right. \\
 & \quad \left. + I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2 \theta_s}{dx_s^2} \left(\frac{dx_s}{ds} \right)^2 \right\} \dot{s}^2.
 \end{aligned}$$

Here x_s , θ_s , and their local derivatives are evaluated at $x_s = x_s(s)$. Since $dx_s/ds < 0$, a force that closes the secondary drives the shift toward decreasing s , thereby opening the primary. Conversely, the secondary belt force acts in the opening direction and drives the shift toward increasing s . The kinematic map therefore carries the inverse coupling between the two pulleys directly into the acceleration balance.

The two terms proportional to $\dot{s}^2$ have different geometric origins. The term containing $d^2 x_s/ds^2$ appears because the belt geometry can require a changing amount of secondary travel per unit primary travel. The term containing $d^2 \theta_s/dx_s^2$ is inherited from the local secondary balance and appears when the helix motion ratio varies with position; substituting $\dot{x}_s = (dx_s/ds)\dot{s}$ supplies its squared local-to-shared motion ratio.

The two balances are not added into one axial free body. They remain separate conditions on the same shift acceleration: the same $\ddot{s}$ must satisfy both movable assemblies. The belt reactions N_p and N_s , together with the coupled shaft and contact quantities, allow both conditions to hold. Those unknowns are supplied by the rotational, belt, and contact relations developed later.

Relation to Existing Dynamic Mappings

Transient belt and pulley models provide broad precedent for dynamic ratio prediction. [Juli'o and Plante \(2011\)](#) discretize the rubber belt and integrate its motion, while [Ballew \(2015\)](#) allow both pulley speeds to evolve from the applied torques and inertias. In that model family, however, pulley axial force is prescribed and the radius follows from belt compression; a movable-sheave equation $m_j \ddot{x}_j = \sum F_{j,x}$ is not integrated.

Movable-sheave dynamics and shared-coordinate reductions also predate the present model. [Duan et al. \(2016\)](#) retain movable-pulley mass in a chain CVT and couple the pulley motions through chain deformation and tension. Most directly, [da Silva et al. \(2023\)](#) write axial equations for both movable sheaves of a mechanically actuated rubber-belt CVT, impose a linear relation between their displacements, and reduce the two translations to one shift equation. Thus transient shifting, axial sheave inertia, and a shared shift coordinate are not individually new.

The distinction here lies in how these ingredients are combined. The fixed-length belt geometry supplies the nonlinear map $x_s(s)$, so both dx_s/ds and $d^2 x_s/ds^2$ remain in the secondary acceleration.

At the same time, the flyweight pivot inertia and the movable secondary's relative-rotational inertia remain tied to their own mechanism maps, $q_f(x_p)$ and $\theta_s(x_s)$. The result is a pair of local free-body balances expressed as simultaneous conditions on one shift acceleration.

This reduction also defines the tradeoff. It does not retain the spatial belt deformation fields of [Juli'o and Plante \(2011\)](#) and [Ballew \(2015\)](#), or the prescribed-shift, face-resolved torque distribution of [Sorge and Cammalleri \(2011\)](#). It instead keeps the actuator and movable-sheave inertias explicit while closing the nominal belt geometry without spatial discretization. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent combination of these features was identified.

3.2.4 What the Axial Balances Establish

The primary and secondary now have complete axial equations of motion in their natural coordinates. On the primary, shaft speed produces a closing action because the flyweight centres of mass move outward as the movable sheave closes. Part of that action accelerates the flyweights about their pivots before the remainder is transmitted to the movable assembly. On the secondary, the belt and torsional-spring torques act through the helix. Part of their action changes the rotation of the movable sheave relative to the shaft-side assembly before the remainder appears as axial force.

Although the two actuators are physically different, their constrained motions produce the same underlying structure. The map $q_f(x_p)$ reflects the flyweights' pivot inertia into the primary axial coordinate, while the helix map $\theta_s(x_s)$ reflects the movable-sheave rotational inertia into the secondary coordinate. In each case, the local motion ratio contributes an effective axial inertia, and variation of that ratio with position produces a velocity-squared term. The primary closing action is additionally set by dr_f/dx_p , while the secondary balance couples to the shaft acceleration because its movable sheave rotates relative to an assembly that is itself rotating.

The shared belt geometry adds a second layer of constrained motion. The primary map $x_p(s) = s$ carries its local balance directly into the shift coordinate, whereas the secondary map $x_s(s)$ reflects its local motion into the same $\ddot{s}$ and contributes its own velocity-squared curvature term. Both free-body relations must therefore be satisfied by one shift motion.

They do not, however, determine that motion in isolation. At this stage, the two axial equations contain the five unknown quantities

$$\ddot{s}, \quad N_p, \quad N_s, \quad \tau_s, \quad \dot{\omega}_s.$$

The belt normal loads, transmitted secondary torque, and secondary shaft acceleration follow only when the rotational and belt-contact mechanics are added. The two balances derived here therefore form the axial part of one simultaneous dynamic solution for the engaged CVT.

This completes the axial force description. Attention now turns from translation along the pulley shafts to rotation about them.

3.3 Rotational Dynamics of the Pulley Assemblies

The preceding section established how forces along the pulley shafts move the sheaves, but its equations still contain quantities belonging to rotation about those shafts. The primary flyweight force depends on the primary speed ω_p , while the secondary balance contains the transmitted torque τ_s and the secondary acceleration $\dot{\omega}_s$. The axial equations can use these quantities, but cannot supply them.

The primary pulley is carried around the primary shaft at ω_p , while the secondary rotates about its own shaft at ω_s . These are separate rotations, so their speeds need not change together. The mechanisms inside each pulley also move relative to these underlying shaft rotations. The primary flyweights pivot as the primary rotates, while the secondary movable sheave translates and rotates relative to its shaft through the helix. These internal motions are already tied to the shift coordinate, but they also change how angular momentum is distributed within each pulley as the CVT shifts.

The remaining question is therefore what governs the evolution of ω_p and ω_s , and consequently how shifting affects that evolution.

3.3.1 Following the Torque

To determine how ω_p and ω_s evolve, each pulley side requires a rotational equation of motion. That equation must account for the angular momentum carried about the corresponding shaft and the torques capable of changing it. The first step is therefore to identify the rotating systems represented by the two shaft speeds.

The primary-side system contains the primary shaft and the complete primary pulley assembly. The movable assembly and flyweights remain part of this system even while they move relative to the fixed sheave, since they continue to carry angular momentum about the primary axis. The secondary-side system is defined in the same way: it contains the secondary shaft and the complete secondary pulley assembly, including the helix-guided motion of the movable sheave. The belt passes between these two systems and is not included in either one.

The resulting division is shown in [Figure 3.9](#).

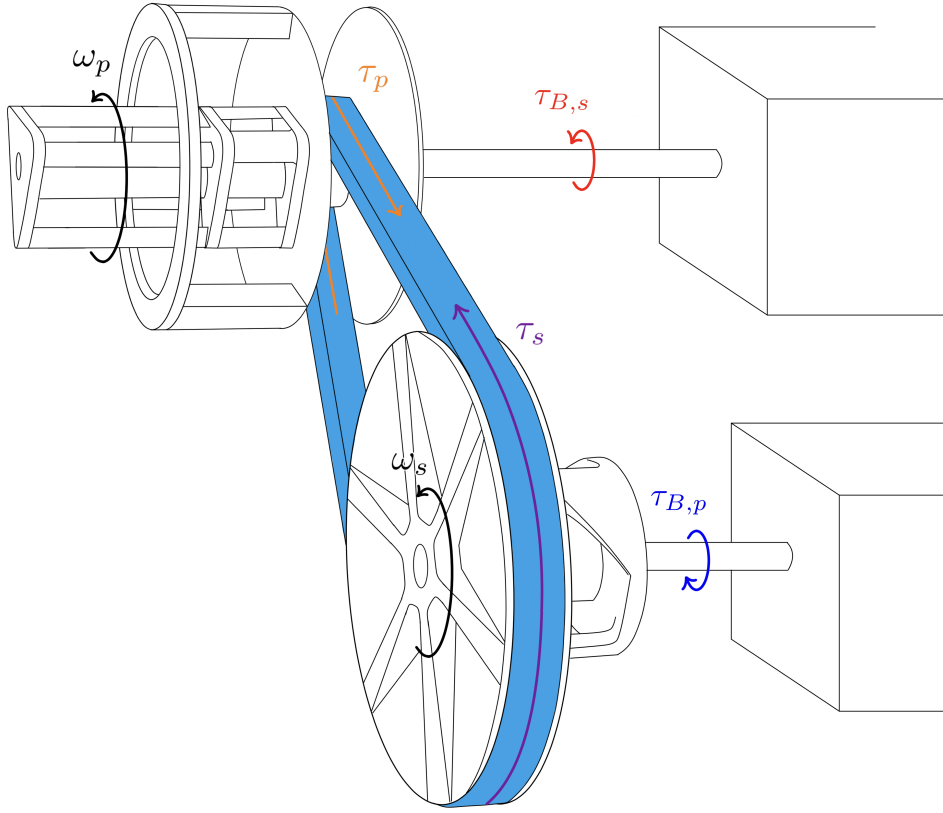


Figure 3.9: Rotating systems and torque path of the engaged CVT. The shaft-side boundaries represent the torque and equivalent inertia presented by the attached systems. The belt remains outside the two pulley-side systems and interacts with them through the contact torques τ_p and τ_s . The arrows show the signed directions associated with forward power transfer.

Begin with the primary. When its speed changes, the primary shaft must accelerate not only the pulley itself, but also the rotating hardware attached beyond the CVT boundary. As established in [Section 2.1.1](#), that external hardware is represented at the shaft by the equivalent inertia $I_{B,p}$, while $\tau_{B,p}$ is the torque it applies to the primary side. The remaining external interaction is the belt, which exerts the torque τ_p on the primary pulley.

The secondary follows the same construction. Its attached hardware contributes the equivalent inertia $I_{B,s}$ and applies the boundary torque $\tau_{B,s}$, while the belt exerts τ_s on the secondary pulley.

The signs follow directly from the convention established in [Section 2.3.1](#). During forward power transfer, the primary boundary drives the primary pulley and the belt resists it. The belt then drives the secondary pulley against the resisting secondary boundary. Thus,

$$\tau_{B,p} > 0, \quad \tau_p < 0, \quad \tau_s > 0, \quad \tau_{B,s} < 0.$$

Although the primary and secondary contain different mechanisms, their rotational balances now have the same outer structure. Let $j \in \{p, s\}$ denote the pulley side under consideration.

The angular momentum on side j has two contributions. The fixed equivalent boundary inertia rotates at ω_j , so it carries angular momentum $I_{B,j}\omega_j$. Let $H_{\text{CVT},j}$ denote the angular momentum carried by the corresponding shaft and pulley assembly, including the internal motions of its components. The total is therefore

$$H_j = I_{B,j}\omega_j + H_{\text{CVT},j}. \quad (3.3.1)$$

Bearing, seal, windage, and other unmodeled resisting torques are omitted under [Assumption 12 \(Negligible Unmodeled Parasitic Losses\)](#). Within the resulting pulley-side free body, the applied boundary torque and belt torque are therefore the only external torques retained. Newton's second law for rotation gives

$$\tau_{B,j} + \tau_j = \frac{dH_j}{dt}.$$

Substituting [Equation \(3.3.1\)](#) writes this as

$$\tau_{B,j} + \tau_j = \frac{d}{dt}(I_{B,j}\omega_j + H_{\text{CVT},j}).$$

The equivalent boundary inertia is fixed, so differentiating yields the common pulley-side equation of motion:

Common Pulley-Side Rotational Balance (3.3.2) $\tau_{B,j} + \tau_j = I_{B,j}\dot{\omega}_j + \frac{dH_{\text{CVT},j}}{dt}, \quad j \in \{p, s\}.$
--

The way shifting changes the angular momentum carried within a pulley is contained in $H_{\text{CVT},j}$. On the primary, the flyweights redistribute mass relative to the shaft as they pivot. On the secondary, the movable sheave rotates relative to the shaft-side assembly as it follows the helix. The two pulley sides therefore share the balance in [Equation \(3.3.2\)](#), but require separate expressions for their internal angular momentum. Beginning with the primary, each assembly can now be examined in turn.

3.3.2 Primary Pulley Assembly

The primary is the simpler of the two pulley assemblies because nearly all of its hardware shares the shaft speed ω_p . The shaft, sheaves, ramp carrier, and other coaxial components may translate during a shift, but that translation is along the shaft and therefore does not change their distance from its axis. Their combined shaft-axis inertia can consequently be collected into one constant quantity, $I_{p,\text{const}}$.

The flyweights are different. They are carried around the primary shaft at ω_p , but their mass moves radially as they pivot. The same motion was already used in [Section 3.2.1](#) to determine the centrifugal action that closes the primary. There, the flyweight geometry was summarized by the shaft-axis inertia map $J_{f,p}(x_p)$. Since $x_p = s$, that map is now $J_{f,p}(s)$.

It is useful to recall what this inertia represents. The quantity $J_{f,p}$ measures the flyweight mass distribution about the *primary shaft*. It is therefore distinct from the constant pivot-axis inertia I_f that appeared in the flyweight moment balance of Equation (3.2.3). The two quantities describe the same flyweights about different axes and enter different equations of motion.

At a given shift position, the angular momentum carried inside the primary CVT boundary is therefore

$$H_{\text{CVT},p} = [I_{p,\text{const}} + J_{f,p}(s)] \omega_p. \quad (3.3.3)$$

If the shift position were fixed, differentiating this expression would simply multiply the current inertia by $\dot{\omega}_p$. During a shift, however, $J_{f,p}$ changes as well. Because its only configuration variable is s , differentiating Equation (3.3.3) gives

$$\frac{dH_{\text{CVT},p}}{dt} = [I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \quad (3.3.4)$$

The two terms describe different ways of changing the same angular momentum. The first accelerates the primary with the flyweights held at their current configuration. The second changes the angular momentum because the flyweight mass itself moves toward or away from the shaft while already rotating.

Substituting this result into the common pulley-side balance from Equation (3.3.2) gives the primary equation of motion:

Primary Pulley Rotational Balance (3.3.5) $\begin{aligned} \tau_{B,p} + \tau_p = & [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p \\ & + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \end{aligned}$

The connection to the axial flyweight balance is now explicit. In Equation (3.2.38), the derivative $dJ_{f,p}/ds$ determined how shaft rotation produces a centrifugal closing action. Here, the same derivative determines how that closing motion changes the shaft-axis angular momentum. The two appearances are not separate flyweight corrections: they are the axial and rotational consequences of the same radial redistribution of flyweight mass.

For the conventional primary mechanism, outward flyweight motion increases $J_{f,p}$. During an upshift, $\dot{s} > 0$, so the rotating primary must supply the angular momentum required by that outward motion in addition to accelerating the shaft. During a backshift the redistribution reverses, and the flyweights return angular momentum to the shaft-side motion.

If the shift is held fixed, $\dot{s} = 0$, and the flyweight mass distribution no longer changes. The additional term vanishes and the familiar fixed-inertia form is recovered:

$$\tau_{B,p} + \tau_p = [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p, \quad \dot{s} = 0. \quad (3.3.6)$$

3.3.3 Secondary Pulley Assembly

The secondary differs from the primary in one important way. During a shift, the complete secondary remains one mechanical assembly, but not every part of that assembly has the same angular speed.

The secondary shaft, fixed sheave, and hardware locked to them rotate at ω_s . Let their combined shaft-axis inertia inside the CVT boundary be $I_{s,F}$. The movable sheave and the components rigidly attached to it have inertia $I_{s,M}$. As established while deriving the secondary axial balance, the helix makes this movable assembly rotate relative to the shaft-side assembly as it translates. Its absolute angular speed is therefore

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s, \quad (3.3.7)$$

which is the relation already obtained in Equation (3.2.24). The angular momentum carried by the secondary must therefore retain the two angular speeds separately:

$$H_{\text{CVT},s} = I_{s,F}\omega_s + I_{s,M}\omega_{s,M}. \quad (3.3.8)$$

The free body here is larger than the movable-sheave free body used in Section 3.2.2. There, only the movable assembly was isolated, so the spring and helix torques crossed the boundary and had to appear explicitly. Here, the shaft-side and movable-side hardware are enclosed together. The torque exerted by the helix on the movable side is accompanied by an equal-and-opposite torque on the shaft-side assembly, and both actions lie inside the same free body. The torsional spring produces the same kind of internal pair. These torques therefore cancel from the moment balance of the complete secondary. Their mechanical effect has not disappeared: the helix still allows the movable side to rotate at $\omega_{s,M}$ rather than ω_s , and that difference remains in Equation (3.3.8).

The common pulley-side balance in Equation (3.3.2) requires the time rate of change of the angular momentum inside this boundary. Because Equation (3.3.8) contains two angular velocities, differentiating it requires the corresponding two angular accelerations:

$$\frac{dH_{\text{CVT},s}}{dt} = I_{s,F}\dot{\omega}_s + I_{s,M}\dot{\omega}_{s,M}. \quad (3.3.9)$$

The first acceleration, $\dot{\omega}_s$, is the shaft acceleration governed by the present rotational balance. The second is not another independent degree of freedom. The helix already tied the movable-side rotation to the axial sheave motion in Section 3.2.2. In particular, differentiating the helix kinematics there gave Equation (3.2.25):

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.3.10)$$

The rotational balance has therefore led back to a kinematic relation already established by the axial mechanism. Substituting Equation (3.3.10) into Equation (3.3.9) gives

$$\begin{aligned} \frac{dH_{\text{CVT},s}}{dt} = & (I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[\frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned} \quad (3.3.11)$$

Substituting this rate into the common pulley-side rotational balance in Equation (3.3.2) produces the local secondary equation of motion:

Secondary Pulley Rotational Balance (3.3.12) $\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[\frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned}$
--

The first term accelerates the rotation shared by the complete secondary. The bracketed terms change only the movable side's rotation relative to the shaft. They are the rotational counterpart of the helix-reflected inertia already encountered in the secondary axial balance.

The complete CVT uses the shared shift coordinate rather than x_s as an independent coordinate. The required mapping was already established in Equation (3.2.37). Substituting

$$\dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2$$

into Equation (3.3.12) gives the same rotational balance in the coordinate used by the complete model:

$$\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} \ddot{s} \\ & + I_{s,M} \left[\frac{d\theta_s}{dx_s} \frac{d^2x_s}{ds^2} + \frac{d^2\theta_s}{dx_s^2} \left(\frac{dx_s}{ds} \right)^2 \right] \dot{s}^2. \end{aligned} \quad (3.3.13)$$

All local helix quantities in this expression are evaluated at $x_s = x_s(s)$. Its signs follow directly from the kinematics already chosen. For the conventional helix, $d\theta_s/dx_s > 0$, while the belt geometry gives $dx_s/ds < 0$ over the engaged shift range. Hence

$$\frac{d\theta_s}{ds} = \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} < 0.$$

During an upshift, $\dot{s} > 0$, so $\dot{\theta}_s < 0$: the secondary opens and the movable side rotates more slowly than the shaft-side assembly. The sign of the corresponding angular-momentum term is therefore carried by the mechanism maps rather than assigned separately in the rotational balance.

If the shift is held fixed, $\dot{s} = \ddot{s} = 0$. The helix then produces no relative motion, and the fixed and movable sides rotate together. The balance reduces to

$$\tau_{B,s} + \tau_s = (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s. \quad (3.3.14)$$

3.3.4 What the Rotational Balances Establish

The primary and secondary now have complete shaft-axis equations of motion. Both remain direct statements of $\sum \tau = dH/dt$. The additional terms appear because the hardware that actuates the pulleys continues to move while the pulley assemblies rotate, so shifting changes the angular momentum stored inside the assemblies themselves.

On the primary, this connection is carried by the same flyweight inertia map $J_{f,p}(s)$ that appeared in the axial derivation. There, $dJ_{f,p}/ds$ contributes to the centrifugal closing action because the flyweights move outward as the primary closes. Here, $J_{f,p}(s)$ and its derivative determine how much shaft-axis angular momentum those same flyweights carry as their radial mass distribution changes. The axial and rotational balances are therefore two views of the same flyweight motion: one asks how rotation drives the movable sheave, while the other asks how that moving flyweight mass changes the rotational response of the pulley.

The secondary shows the same reciprocity through $I_{s,M}$ and the helix map $\theta_s(x_s)$. In the axial balance, the helix reflects the movable assembly's rotational inertia into the sheave motion, so part of the available torque is required to change the movable side's rotation before the remainder appears as axial force. In the rotational balance, that same relative motion means the movable side carries angular momentum at a speed different from the shaft-side assembly, so the axial shift acceleration enters the shaft-axis balance. Again, the two equations are describing one constrained piece of hardware from different free bodies.

Mechanical actuation therefore creates a direct coupling between the axial and rotational dynamics inside each pulley assembly. This is distinct from the coupling carried by the belt reaction. Belt contact can connect the two kinds of balance because its normal action loads the sheaves axially while its tangential action transmits torque about the shafts. The present terms add a second path: the flyweight and helix mechanisms that generate the clamping response also carry inertia, so their own motion appears directly in the rotational equations. The actuator is therefore part of both balances, rather than acting only as a source of clamping force whose influence reaches shaft rotation through the belt.

This distinction gives a precise comparison with earlier transient models. In the rubber-belt formulation of [Ballew \(2015\)](#), belt-contact torque changes pulley speed through a fixed lumped pulley inertia, while pulley axial force is supplied to the belt model. Axial and rotational behaviour can still interact through the belt solution, but motion of the clamping hardware does not alter the pulley angular momentum itself. [Duan et al. \(2016\)](#) provide a stronger adjacent precedent in a chain CVT: rotational dynamics, movable-pulley axial dynamics, chain motion, and contact forces are solved as a coupled system. Their shaft balance nevertheless retains a fixed pulley inertia, while the movable-pulley mass appears in a separate axial equation. The coupling is therefore carried through the chain, contact forces, and changing radius rather than through actuator motion entering both balances.

A mechanically actuated rubber-belt comparison is provided by [da Silva et al. \(2023\)](#), who retain axial equations for both movable sheaves and reduce their translations to a shared shift coordinate. Their rotational treatment uses a fixed primary-pulley inertia. This establishes direct precedent for mechanical clamping, movable-sheave inertia, and coupled shift motion, but not for the additional

shaft-axis angular-momentum terms produced by the moving actuator hardware.

The helical model of [Sorge and Cammalleri \(2011\)](#) provides a complementary secondary comparison. Their formulation resolves the relative motion of the driven sheaves and the generally unequal belt torques acting on the two faces in substantially greater local detail. That work therefore captures secondary helix and face-contact mechanics that are reduced here. The distinction in the present balance is different: the helix-constrained movable assembly is retained as part of the complete secondary angular momentum, so the same relative motion that appears in its axial dynamics also appears in the shaft-axis equation.

Taken together, these comparisons show that axial-rotational coupling itself is not the new ingredient. Existing formulations can produce that coupling through belt or chain reactions, changing radii, and shared shift motion. Among the reviewed reduced-order formulations for mechanically actuated rubber-belt CVTs, however, no equivalent pair of pulley balances was identified in which the moving clamping hardware itself provides this additional inertial coupling on both sides: the primary flyweights change their shaft-axis mass distribution as they generate closing action, while the secondary movable assembly changes its shaft-axis angular momentum through the same helix motion that produces its axial response.

The result is a single mechanical picture of each pulley rather than separate axial and rotational descriptions joined only through transmitted load. The same mechanism maps that determine how the pulley clamps also determine how its internal motion changes the shaft dynamics. When those actuator motions cease, the additional terms vanish and the familiar fixed-configuration inertia balances are recovered.

3.4 Belt Transport and Pulley Contact Mechanics

The preceding sections established the axial and rotational equations of motion of the two pulley assemblies, but in both cases the belt still appears through actions that those equations cannot determine. The axial balances contain the belt normal loads N_p and N_s , while the rotational balances contain the belt torques τ_p and τ_s . The belt also carries the transport state v_b , whose acceleration has not yet been established.

These quantities cannot be assigned independently. The contact that loads the sheave faces is also the contact through which torque is exchanged with the belt, and the same forces must account for the changing momentum of the belt itself. The remaining question is therefore how the motion and tension of one closed belt determine its own transport dynamics while producing the normal reactions and torques acting on both pulley assemblies.

3.4.1 Following Tension Around the Closed Belt

The belt transfers power between the pulleys through forces exchanged at the two belt-sheave interfaces. For power to cross either interface, the pulley and belt must exert tangential forces on one another. Those forces transmit torque, and the tension entering a pulley wrap will therefore generally differ from the tension leaving it.

The pulley contacts are not the only places where tension can change. The straight spans also contain belt mass. If the belt transport speed changes, a net tangential force is required to accelerate the belt contained within each span, so its two endpoint tensions can differ as well. The closed belt thus contains two repeated types of mechanical region: two pulley wraps, where load is exchanged

with the sheaves through contact, and two free spans, where belt mass is carried between those contacts. The tangent points between the regions provide convenient boundaries at which to record the tension, as shown in [Figure 3.10](#).

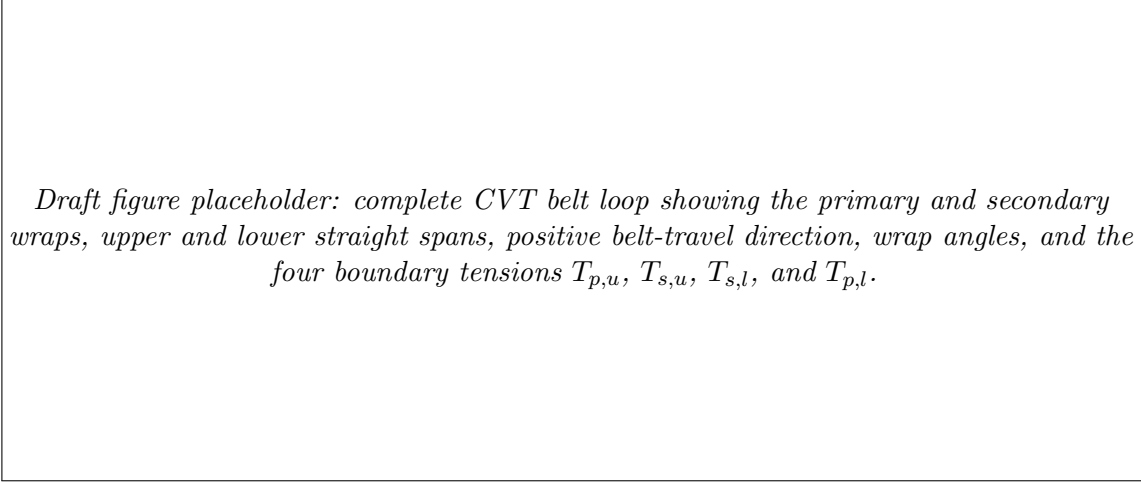


Figure 3.10: Complete belt loop and the four tension values at the boundaries between the pulley wraps and straight spans. The arrows indicate the positive belt-transport direction.

The four tensions in [Figure 3.10](#) are not additional dynamic states. They are the values taken by one continuous belt-tension field at the four boundaries between the spans and wraps. With the positive belt path introduced in [Section 2.3.1](#), the upper span is traversed from the secondary to the primary, from $T_{s,u}$ to $T_{p,u}$, while the lower span is traversed from the primary back to the secondary, from $T_{p,l}$ to $T_{s,l}$.

Begin with a free span, where the belt is no longer in contact with either pulley and is acted on tangentially only through the tensions at its ends. Consider either span $i \in \{u, l\}$, with instantaneous length l_i , and orient its local tangential direction with positive belt travel. Under [Assumption 7 \(Single Belt Transport Coordinate\)](#), the retained motion along the span is the common belt transport speed v_b . Under [Assumption 8 \(Prescribed Planar Belt Path\)](#), the span remains the instantaneous straight tangent joining the two pulley wraps. Its mass per unit length is $\rho_b A_b$.

Isolate the belt occupying this span. Its tangential equation of motion follows from Newton's second law in momentum form,

$$\sum F_{i,t} = \frac{dP_{i,t}^{\text{sys}}}{dt}. \quad (3.4.1)$$

The force side is immediate from the free body: the two endpoint tensions pull in opposite directions, so $\sum F_{i,t} = T_{i,\text{out}} - T_{i,\text{in}}$. The momentum side requires more care. The tangent points that bound the span move as the pulley radii change, so the spatial region between them does not contain a fixed set of belt material. Its length changes while belt enters and leaves through the moving endpoint stations.

Let $v_{\text{st},i}^{\text{in}}$ and $v_{\text{st},i}^{\text{out}}$ denote the velocities of those stations projected along the instantaneous span direction. Evaluating the system momentum rate in [Equation \(3.4.1\)](#) with that moving control volume gives

$$T_{i,\text{out}} - T_{i,\text{in}} = \frac{d}{dt} (\rho_b A_b l_i v_b) + \rho_b A_b v_b [(v_b - v_{\text{st},i}^{\text{out}}) - (v_b - v_{\text{st},i}^{\text{in}})] . \quad (3.4.2)$$

The first term is the change of tangential momentum stored between the two stations; the second is the net tangential momentum carried through those moving boundaries. The span-length rate is set by the same endpoint motion,

$$\dot{l}_i = v_{\text{st},i}^{\text{out}} - v_{\text{st},i}^{\text{in}} . \quad (3.4.3)$$

Substituting Equation (3.4.3) into Equation (3.4.2) cancels the term produced by the changing amount of belt stored in the span with the corresponding momentum flux through its moving endpoints. What remains is

$$\boxed{T_{i,\text{out}} - T_{i,\text{in}} = \rho_b A_b l_i \dot{v}_b} . \quad (3.4.4)$$

The compact result therefore does not require fixed tangent points. Their motion was retained in Equation (3.4.2); it simply makes no additional contribution to the final tangential balance once the associated momentum flux is included. The balance is only along the instantaneous span. As the pulley radii change, the span direction also changes in the pulley plane; the transverse dynamics associated with that prescribed reorientation are not retained under Assumption 8 (Prescribed Planar Belt Path).

Applying Equation (3.4.4) to the two spans in Figure 3.10 gives

$$\boxed{\begin{aligned} T_{p,u} - T_{s,u} &= \rho_b A_b l_u \dot{v}_b, \\ T_{s,l} - T_{p,l} &= \rho_b A_b l_l \dot{v}_b. \end{aligned}} \quad (3.4.5)$$

The straight spans can therefore be related from their endpoint tensions and the common belt transport acceleration. A pulley wrap is different. Between its two tangent points the belt is continuously redirected while the sheave faces apply distributed contact forces, so the entry and exit tensions cannot be related from the endpoints alone. Determining how the tension changes through the wrap requires a free body taken from within the contact itself.

3.4.2 Local Equations of Motion on a Pulley Wrap

The straight-span balances in Equation (3.4.5) describe the parts of the belt whose direction is constant between their two tangent points. Inside a pulley wrap, the belt direction changes continuously and the sheave faces act on it throughout the contact arc. Consider therefore the infinitesimal belt element shown in Figure 3.11 on a generic pulley $j \in \{p, s\}$. Let θ increase through the wrap in the direction of belt travel, and let the element subtend the angle $d\theta$.

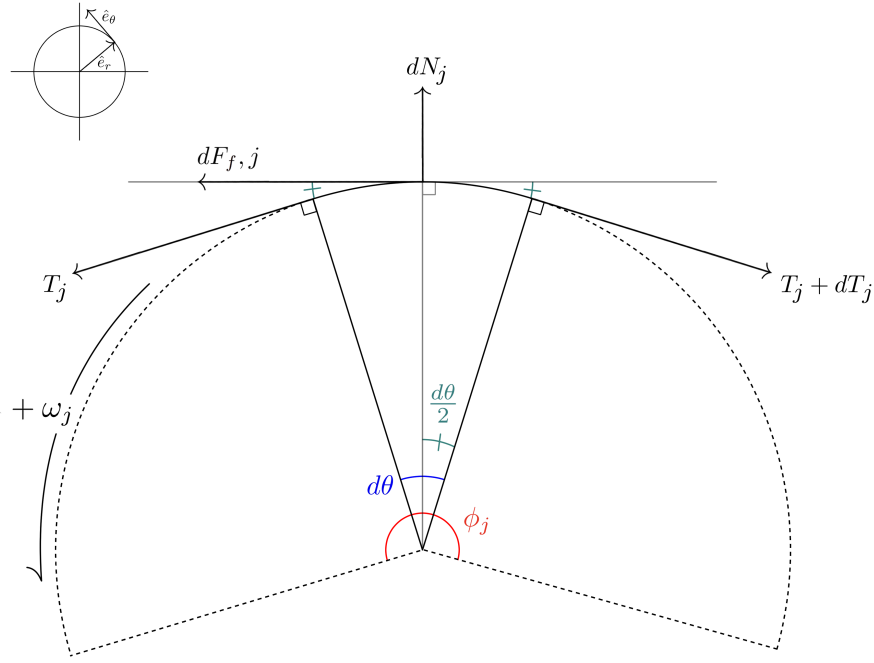


Figure 3.11: Free-body diagram of an infinitesimal belt element within the wrap of a generic pulley j . The local radial direction is positive outward from the pulley axis and the tangential direction is positive with belt travel. The belt tensions act at the two cut faces, while the sheave faces supply the normal contact and tangential traction.

The local axes are taken through the midpoint of the element. The belt entering it pulls with tension T_j , and the belt leaving it pulls with $T_j + dT_j$. Because the two tension directions are tangent to the wrap at opposite ends of the element, each is inclined by $d\theta/2$ from the midpoint tangent. The sheave faces apply a combined physical normal load dN_j . Under **Assumption 11 (Symmetric Sheave-Face Load Sharing)**, each face carries $dN_j/2$; their axial components oppose one another on the belt, while their radial components add to $dN_j \sin \beta$. The retained tangential contact force is denoted by $dF_{t,j}$, taken as positive in the direction of belt travel.

The belt element can therefore accelerate in two in-plane directions. Applying Newton's second law in the midpoint radial and tangential directions gives, directly from the free body,

$$dN_j \sin \beta - T_j \sin \left(\frac{d\theta}{2} \right) - (T_j + dT_j) \sin \left(\frac{d\theta}{2} \right) = dm_j a_{r,j}, \quad (3.4.6)$$

$$(T_j + dT_j) \cos \left(\frac{d\theta}{2} \right) - T_j \cos \left(\frac{d\theta}{2} \right) + dF_{t,j} = dm_j a_{\theta,j}. \quad (3.4.7)$$

These balances expose what the free body alone does not determine. Both right-hand sides require the mass and acceleration of the belt element, and the tangential balance contains the additional contact force $dF_{t,j}$. The inertial quantities are common to both directions, so they can be established first.

Mass and motion of the belt element The radius r_j defined in Equation (3.1.8) follows the outer surface of the belt. The mass of the belt element instead travels with the centroid of its trapezoidal cross-section. Let y measure radially inward from the outer surface through the belt height h_b . Under Assumption 4 (Fixed Belt Cross-Section), the cross-section retains the outer and inner widths b_{out} and b_{in} , so its width at y is

$$b(y) = b_{\text{out}} - \frac{b_{\text{out}} - b_{\text{in}}}{h_b} y. \quad (3.4.8)$$

The cross-sectional area and its first moment about the outer surface are

$$A_b = \int_0^{h_b} b(y) dy = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}), \quad (3.4.9)$$

$$A_b y_{\text{cm}} = \int_0^{h_b} y b(y) dy. \quad (3.4.10)$$

Substitution of Equation (3.4.8) gives

$$y_{\text{cm}} = h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \quad (3.4.11)$$

The belt mass therefore follows the radius

$$r_{j,\text{cm}} = r_j - y_{\text{cm}}. \quad (3.4.12)$$

Since the cross-section is fixed, y_{cm} is constant. The centroid consequently inherits the radius rate and acceleration already obtained from the pulley geometry,

$$\dot{r}_{j,\text{cm}} = \dot{r}_j, \quad \ddot{r}_{j,\text{cm}} = \ddot{r}_j. \quad (3.4.13)$$

This is the first place where the radius derivatives developed in Section 3.1 enter the belt dynamics directly: the same changing pulley geometry that moves the belt radially also accelerates its mass.

Under Assumption 8 (Prescribed Planar Belt Path), the centroid follows a circular arc through the wrap. The element length is therefore $r_{j,\text{cm}} d\theta$, and its mass is

$$dm_j = \rho_b A_b r_{j,\text{cm}} d\theta. \quad (3.4.14)$$

The same element moves radially with the changing wrap and tangentially with the belt transport. Under Assumption 7 (Single Belt Transport Coordinate), its velocity is

$$\mathbf{v}_{b,j} = \dot{r}_j \hat{\mathbf{e}}_r + v_b \hat{\mathbf{e}}_\theta. \quad (3.4.15)$$

As the element travels around the pulley, the local basis rotates at

$$\dot{\theta}_j = \frac{v_b}{r_{j,\text{cm}}}, \quad \dot{\mathbf{e}}_r = \frac{v_b}{r_{j,\text{cm}}} \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\frac{v_b}{r_{j,\text{cm}}} \mathbf{e}_r. \quad (3.4.16)$$

Differentiating the radial part of Equation (3.4.15) gives

$$\frac{d}{dt}(\dot{r}_j \mathbf{e}_r) = \ddot{r}_j \mathbf{e}_r + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}} \mathbf{e}_\theta, \quad (3.4.17)$$

while differentiating its tangential part gives

$$\frac{d}{dt}(v_b \mathbf{e}_\theta) = \dot{v}_b \mathbf{e}_\theta - \frac{v_b^2}{r_{j,\text{cm}}} \mathbf{e}_r. \quad (3.4.18)$$

Adding the two contributions separates the acceleration into

$$a_{r,j} = \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}, \quad (3.4.19)$$

$$a_{\theta,j} = \dot{v}_b + \frac{\dot{r}_j}{r_{j,\text{cm}}} v_b. \quad (3.4.20)$$

The radial acceleration therefore contains both the motion of the shifting wrap and the centripetal acceleration required to turn the belt. The tangential acceleration contains the direct change in belt transport together with the contribution created as the belt moves radially through the rotating local basis.

Radial balance With the mass and radial acceleration known, return to Equation (3.4.6). Combining the two inward tension projections gives

$$\begin{aligned} dN_j \sin \beta - (2T_j + dT_j) \sin\left(\frac{d\theta}{2}\right) \\ = \rho_b A_b r_{j,\text{cm}} d\theta \left(\ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}} \right). \end{aligned} \quad (3.4.21)$$

Dividing through by $d\theta$ gives

$$\sin \beta \frac{dN_j}{d\theta} - (2T_j + dT_j) \frac{\sin(d\theta/2)}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2). \quad (3.4.22)$$

As the belt element shrinks,

$$\frac{\sin(d\theta/2)}{d\theta} = \frac{1}{2} \frac{\sin(d\theta/2)}{d\theta/2} \longrightarrow \frac{1}{2}. \quad (3.4.23)$$

Over the same vanishing interval, $dT_j = (dT_j/d\theta)d\theta \rightarrow 0$. The dT_j contribution to the radial projection therefore disappears, while the two leading tension terms leave one T_j . The radial equation of motion becomes

$$\sin \beta \frac{dN_j}{d\theta} = T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j). \quad (3.4.24)$$

The tension provides part of the inward force needed to curve the belt. The normal contact supplies the remainder required by the actual radial acceleration of the moving belt path.

Tangential balance and traction The tangential balance in [Equation \(3.4.7\)](#) begins more simply. The common T_j terms cancel, leaving only the tension change across the element,

$$dT_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}. \quad (3.4.25)$$

The remaining contact term is frictional. Under [Assumption 9 \(Gross Tangential Coulomb Traction\)](#), the usual Coulomb description bounds its magnitude by the available static friction while sticking and fixes it at the kinetic-friction magnitude while sliding,

$$\begin{cases} |dF_{t,j}| \leq \mu_s dN_j, & \text{sticking,} \\ |dF_{t,j}| = \mu_k dN_j, & \text{sliding.} \end{cases} \quad (3.4.26)$$

That statement is naturally written as a magnitude inequality, but the force balance requires a signed force. A single signed traction utilization λ_j carries both pieces of information without introducing separate positive- and negative-friction balances:

Signed Traction Utilization	(3.4.27)
$dF_{t,j} = \lambda_j dN_j, \quad \lambda_j = \begin{cases} \lambda_{j,\text{st}}, & \lambda_{j,\text{st}} \leq \mu_s, & \text{sticking,} \\ \sigma_j \mu_k, & \sigma_j \in \{-1, +1\}, & \text{sliding.} \end{cases}$	

For sliding contact, σ_j takes the sign that makes the traction oppose the relative sliding motion. Under [Assumption 10 \(Wrap-Level Contact Resolution\)](#), the same λ_j represents the complete wrap at a given instant.

Substituting [Equations \(3.4.14\)](#) and [\(3.4.20\)](#) and ?? into [Equation \(3.4.25\)](#) and dividing by $d\theta$ gives

$$\frac{dT_j}{d\theta} \cos\left(\frac{d\theta}{2}\right) + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.28)$$

As $d\theta \rightarrow 0$, $\cos(d\theta/2) \rightarrow 1$. The tangential equation of motion therefore reduces to

$$\frac{dT_j}{d\theta} + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.29)$$

Generic pulley-wrap equation The radial and tangential equations now describe the same belt element in terms of the same normal-load gradient. Solving Equation (3.4.24) for that gradient gives

$$\frac{dN_j}{d\theta} = \frac{1}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] . \quad (3.4.30)$$

Substitution into Equation (3.4.29) removes the internal normal load and leaves a single differential equation for the belt tension through the wrap,

$$\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) . \quad (3.4.31)$$

Collecting the tension term gives the standard linear first-order form,

Generic Pulley-Wrap Tension Equation (3.4.32) $\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} T_j = \rho_b A_b \left[r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b + \frac{\lambda_j}{\sin \beta} (v_b^2 - r_{j,\text{cm}} \ddot{r}_j) \right] , \quad j \in \{p, s\} .$
--

The coordinate θ increases with belt travel on either pulley, and the sign of the tangential contact is carried by λ_j . The same equation therefore governs tension evolution through the primary and secondary wraps without a separate sign convention for each pulley.

3.5 Rotational and Belt Transport Dynamics

The axial force model developed in ?? gives one equation for the shift coordinate $s(t)$. That equation depends on the transmitted secondary torque τ_s and on the secondary angular acceleration $\dot{\omega}_s$. Those quantities are determined by the torque path through the CVT.

This section derives the common equations of motion for the rotating and translating bodies in that torque path. The primary pulley, belt, and secondary pulley are treated as coupled dynamic bodies. Engine torque enters at the primary pulley, torque is exchanged between the primary pulley and the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain.

The equations derived here are independent of whether the belt is sticking or slipping at either pulley. They describe the bodies being accelerated; the contact state supplies the remaining closure equations later.

3.5.1 System Definition and Torque Path

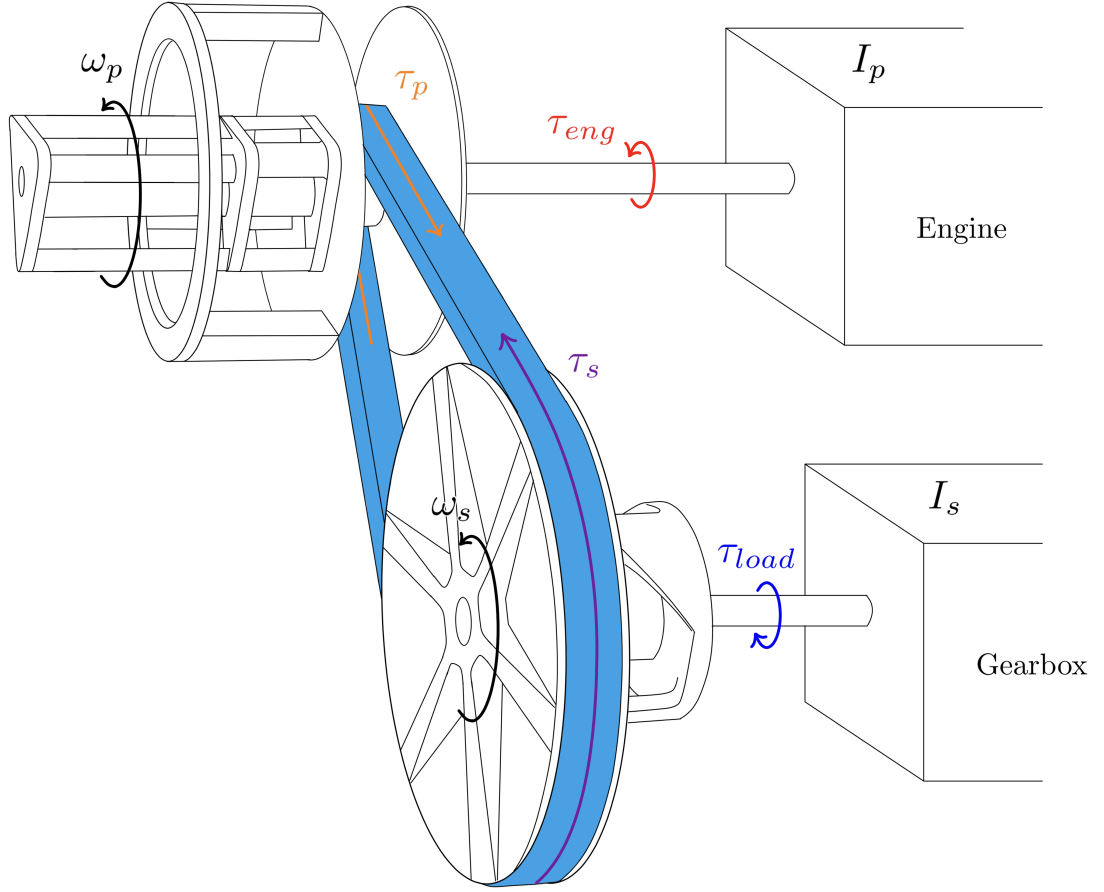


Figure 3.12: Torque path through the CVT. Engine torque τ_{eng} drives the primary pulley with angular speed ω_p . The primary pulley exchanges torque with the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain while opposing the reflected load torque τ_{load} .

The torque path shown in [Figure 3.12](#) is described using three contact torques and three dynamic motions:

- the primary pulley rotates with angular speed ω_p ,
- the belt translates with transport speed v_b ,
- the secondary shaft and fixed sheave rotate with angular speed ω_s .

The primary pulley receives engine torque τ_{eng} . Let τ_p denote the torque exerted by the belt contact on the primary pulley. With the rotational sign convention from [Section 2.3.1](#), τ_p opposes positive engine-driven primary rotation.

The belt receives torque from the primary contact and applies torque to the secondary contact. Let τ_s denote the total signed torque exerted by the belt on the secondary pulley. This torque acts in

the positive secondary rotational direction, while the reflected drivetrain load τ_{load} opposes it.

3.5.2 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley. Under Assumption ?? (??), engine torque is represented as a prescribed full-throttle function of primary angular speed:

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}(\omega_p(t)). \quad (3.5.1)$$

No specific analytic form is required. The model only requires that the mapping $\omega_p \mapsto \tau_{\text{eng}}$ be finite over the operating range used in simulation. Outside the supplied range, the torque curve is either saturated or the state is constrained to remain within the admissible engine-speed domain.

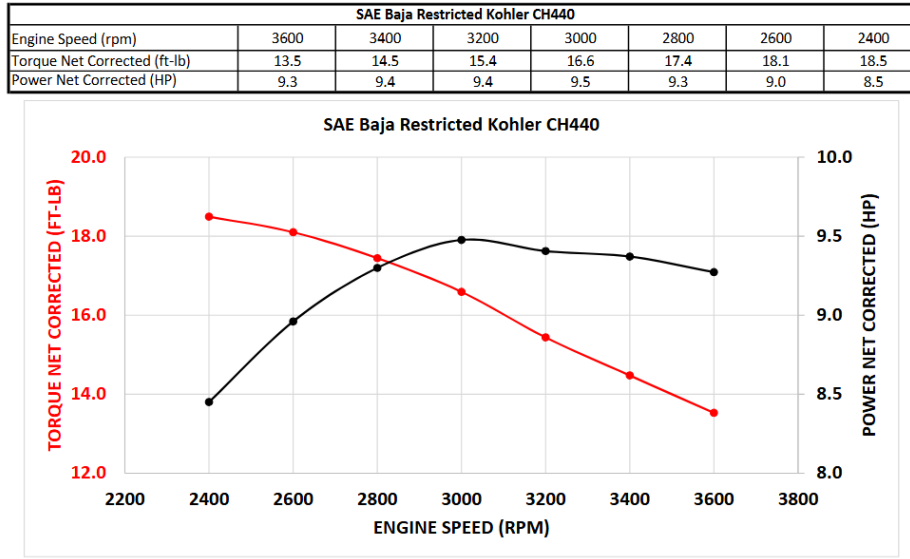


Figure 3.13: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (3.5.1).

In simulation, the discrete torque curve is interpolated to provide a continuous input torque over the operating range.

3.5.3 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ represents the longitudinal vehicle load reflected back to the secondary pulley.

Let r_w denote the effective wheel rolling radius and let $G > 0$ denote the fixed gear ratio defined by

$$\omega_s = G \omega_w.$$

The vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G} \omega_s(t). \quad (3.5.2)$$

The total longitudinal road force is modeled as the sum of rolling resistance, grade resistance, and aerodynamic drag:

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \quad (3.5.3)$$

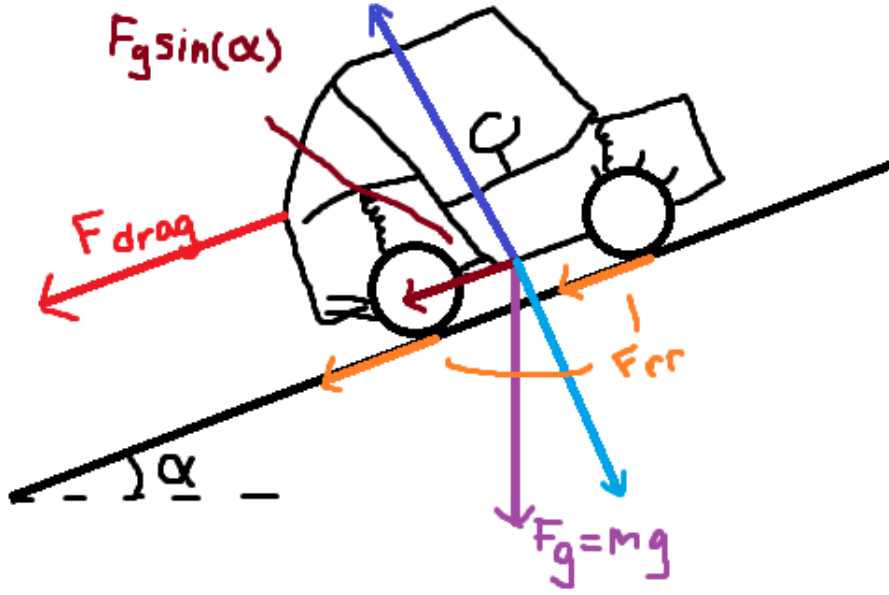


Figure 3.14: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg \sin \alpha$ along the slope and $mg \cos \alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Using Theory ?? (??), Theory ?? (??), and Theory ?? (??), the road-load components are

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos \alpha \operatorname{sgn}(v), \quad (3.5.4)$$

$$F_{\text{grade}}(\alpha) = mg \sin \alpha, \quad (3.5.5)$$

and

$$F_{\text{drag}}(v) = \frac{1}{2} \rho C_d A_f v^2 \operatorname{sgn}(v). \quad (3.5.6)$$

Substituting these terms into Equation (3.5.3) gives

$$F_{\text{road}}(t) = C_{\text{rr}}mg \cos \alpha \operatorname{sgn}(v(t)) + mg \sin \alpha + \frac{1}{2}\rho C_d A_f v(t)^2 \operatorname{sgn}(v(t)). \quad (3.5.7)$$

The wheel torque associated with this road load is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \quad (3.5.8)$$

With the downstream fixed-ratio drive treated as ideal under Assumption ?? (??), the resisting torque reflected to the secondary pulley is

$$\tau_{\text{load}}(t) = \frac{1}{G} \tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \quad (3.5.9)$$

Using $v = (r_w/G)\omega_s$, this becomes

$$\tau_{\text{load}}(t) = \frac{r_w}{G} \left[C_{\text{rr}}mg \cos \alpha \operatorname{sgn}\left(\frac{r_w}{G}\omega_s(t)\right) + mg \sin \alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \operatorname{sgn}\left(\frac{r_w}{G}\omega_s(t)\right) \right]. \quad (3.5.10)$$

3.5.4 Inertia Definitions

The rotational equations require inertias expressed about the primary and secondary axes. Under Assumption ?? (??), these inertias are treated as lumped scalar quantities about their corresponding shafts.

Primary-Side Inertia

The primary pulley is rigidly connected to the engine side. All components in this lump rotate with angular speed ω_p , so the primary-side inertia is

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \quad (3.5.11)$$

where I_{eng} is the engine rotational inertia and I_{prim} is the primary pulley inertia.

Secondary Fixed-Side and Downstream Inertia

The secondary side requires one additional distinction. The fixed secondary sheave and downstream drivetrain rotate with ω_s . The movable secondary sheave does not generally rotate with exactly ω_s during shift, because the helix couples axial motion to relative rotation. Its rotational motion is therefore handled separately using the helix kinematics from ?? and ??.

Let $I_{s,F}$ denote the inertia that rotates directly with the fixed secondary sheave and shaft, excluding the movable secondary sheave inertia I_M . This fixed-side inertia includes the fixed secondary

components, gearbox-side inertia, wheel inertia reflected through the fixed ratio, and the vehicle translational inertia reflected to the secondary axis.

The vehicle translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2 = \frac{1}{2}m(r_w\omega_w)^2,$$

so the vehicle mass appears as an equivalent wheel-axis inertia mr_w^2 . Since $\omega_s = G\omega_w$, wheel-axis inertias reflect to the secondary axis by division by G^2 . Thus,

$$I_{s,F} = I_{\text{sec},F} + I_{\text{gb}} + \frac{I_{\text{wheel}} + mr_w^2}{G^2}. \quad (3.5.12)$$

Here $I_{\text{sec},F}$ includes the fixed secondary sheave and other secondary components that rotate directly with the secondary shaft. The movable secondary sheave inertia I_M is not included in $I_{s,F}$.

3.5.5 Common Equations of Motion

The primary pulley, belt, secondary fixed-side assembly, and shift coordinate provide the four common dynamic equations used in every contact state. The contact state will later determine the two additional closure equations.

Primary Pulley Rotational Dynamics

Applying Theory ?? (??) to the primary pulley gives

Primary Pulley Rotational Dynamics (3.5.13) $\tau_{\text{eng}}(t) - \tau_p(t) = I_p\dot{\omega}_p(t)$
--

The engine torque drives positive primary rotation, while the belt contact torque opposes it.

Belt Transport Dynamics

The belt is represented by the transport speed $v_b(t)$ introduced in [Section 2.4](#). Under [Assumption 6 \(Fixed Belt Length\)](#) and [Assumption 7 \(Single Belt Transport Coordinate\)](#), the belt is modeled as a single moving path with one lumped transport speed.

A contact torque produces an equivalent belt-line force when divided by the effective torque radius from ??. The primary contact contributes $\tau_p/r_{p,\text{eff}}$ to the belt-line motion, while the secondary contact contributes $\tau_s/r_{s,\text{eff}}$ in the opposing direction. Applying Theory ?? (??) to the belt gives

Belt Transport Dynamics (3.5.14) $m_b\dot{v}_b(t) = \frac{\tau_p(t)}{r_{p,\text{eff}}(s)} - \frac{\tau_s(t)}{r_{s,\text{eff}}(s)}$

where m_b is the total belt mass from ??.

Secondary Rotational Dynamics

The secondary side is not treated as a single rigid body during shift. The fixed secondary sheave and downstream drivetrain rotate with the secondary shaft speed ω_s , while the movable secondary sheave rotates with ω_M . These two angular speeds are equal only when the secondary is not shifting. When the helix is active, axial motion of the movable sheave is accompanied by relative rotation, so the movable sheave must be counted with its own angular acceleration.

The cleanest way to write the secondary rotational balance is to consider the fixed-side secondary assembly and the movable sheave together as one system. The external torques on this combined secondary system are the total belt-applied secondary torque τ_s and the reflected load torque τ_{load} . The helix and spring torques between the fixed and movable sheaves are internal to this combined system, so they do not appear in the net external torque balance.

Let $I_{s,F}$ denote the inertia of the fixed-side secondary assembly, excluding the movable secondary sheave inertia I_M . The angular momentum of the secondary system about the shaft axis is then

$$L_s = I_{s,F}\omega_s + I_M\omega_M.$$

Applying Theory ?? (??) in angular-momentum form gives

$$\tau_s - \tau_{\text{load}} = \frac{dL_s}{dt} = I_{s,F}\dot{\omega}_s + I_M\dot{\omega}_M. \quad (3.5.15)$$

The movable-sheave acceleration is not independent. From the helix kinematics derived in ??,

$$\dot{\omega}_M = \dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}.$$

Substituting this relation into Equation (3.5.15) gives the secondary rotational equation in terms of the common dynamic variables:

Secondary Rotational Dynamics (3.5.16) $\tau_s - \tau_{\text{load}} = (I_{s,F} + I_M)\dot{\omega}_s - I_M\frac{d^2\theta_s}{ds^2}\dot{s}^2 - I_M\frac{d\theta_s}{ds}\ddot{s}$

This form preserves the distinction between the fixed secondary components and the movable sheave. If the CVT is not shifting, then $\dot{s} = 0$ and $\ddot{s} = 0$, so the movable sheave rotates with the fixed sheave and the secondary inertia reduces to $I_{s,F} + I_M$. During shift, the helix introduces the additional coupling terms shown above.

3.5.6 Common Dynamic Relations and Remaining Closure

At this point, the model has four dynamic relations that are used regardless of the belt contact state. These are:

- primary rotational dynamics, Equation (3.5.13),
- belt transport dynamics, Equation (3.5.14),

- secondary rotational dynamics, [Equation \(3.5.16\)](#),
- axial shift dynamics, ??.

The first three equations describe the torque path through the primary pulley, belt, and secondary pulley. The fourth equation carries the axial force balance into the same coupled system.

Together, the common dynamics involve the six unknown quantities

$$\dot{\omega}_p, \quad \dot{\omega}_s, \quad \dot{v}_b, \quad \ddot{s}, \quad \tau_p, \quad \tau_s.$$

The four equations above therefore describe the body dynamics, but they do not yet determine all six unknowns. Two additional relations are still needed to specify how torque is exchanged at the primary and secondary belt–pulley contacts.

This is where the belt contact state enters the model. The equations derived so far describe how the bodies respond to applied forces and torques; the contact relations determine what forces and torques can actually be exchanged between the belt and each pulley. The following section begins with the gross sticking case, where the belt is assumed to remain kinematically compatible with both pulley contacts.

3.6 Gross Stick Compatibility and No-Slip Candidate

The previous section established the common equations of motion for the primary pulley, belt, secondary assembly, and shift coordinate. These equations describe how the bodies respond to applied forces and torques, but they do not yet specify the contact state between the belt and the pulleys.

This section considers the gross sticking case. In this contact state, the belt is assumed to remain adhered to both pulleys at their effective torque radii. The resulting solution is the *no-slip candidate*: the accelerations and contact torques that would be required if both belt–pulley contacts remained stuck.

This candidate is not automatically admissible. It is first computed from the sticking constraints, then checked against the available traction limits in the following section.

3.6.1 Gross Stick Compatibility

Let $j \in \{p, s\}$ denote either the primary or secondary pulley. Under [Assumption 7 \(Single Belt Transport Coordinate\)](#), the belt is represented by a single transport speed v_b . Under [Assumption 8 \(Prescribed Planar Belt Path\)](#), each pulley contact is represented by a single effective torque radius $r_{j,\text{eff}}$.

If the belt is grossly stuck to pulley j , the belt transport speed equals the tangential surface speed of that pulley at the effective torque radius:

$$v_b(t) = r_{j,\text{eff}}(s(t)) \omega_j(t), \quad j \in \{p, s\}. \quad (3.6.1)$$

Differentiating [Equation \(3.6.1\)](#) gives the acceleration-level stick compatibility condition:

Gross Stick Compatibility**(3.6.2)**

$$\dot{v}_b(t) = \frac{dr_{j,\text{eff}}}{ds} \dot{s}(t) \omega_j(t) + r_{j,\text{eff}}(s(t)) \dot{\omega}_j(t), \quad j \in \{p, s\}.$$

This equation is the contact closure for a stuck belt–pulley interface. It states that the belt-line acceleration must match the acceleration of the pulley surface at the effective radius, including the contribution from changing pulley radius during shift.

Solving Equation (3.6.2) for the pulley angular acceleration gives

$$\dot{\omega}_j(t) = \frac{\dot{v}_b(t) - \frac{dr_{j,\text{eff}}}{ds} \dot{s}(t) \omega_j(t)}{r_{j,\text{eff}}(s(t))}, \quad j \in \{p, s\}. \quad (3.6.3)$$

For the no-slip candidate, Equation (3.6.3) is applied at both the primary and secondary contacts.

3.6.2 No-Slip Candidate Solve

For compactness, define

$$r_p = r_{p,\text{eff}}(s), \quad r_s = r_{s,\text{eff}}(s),$$

and

$$\dot{r}_p = \frac{dr_{p,\text{eff}}}{ds} \dot{s}, \quad \dot{r}_s = \frac{dr_{s,\text{eff}}}{ds} \dot{s}.$$

Also define the effective shift inertia from ?? as

$$M_{\text{shift}} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4} + I_M \left(\frac{d\theta_s}{ds} \right)^2. \quad (3.6.4)$$

This is not a new model assumption; it is the denominator of the axial shift equation already derived.

Stick-Compatible Angular Accelerations

Applying the stick condition from Equation (3.6.3) at the primary pulley gives

$$\dot{\omega}_{p,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p}. \quad (3.6.5)$$

Applying the same condition at the secondary pulley gives

$$\dot{\omega}_{s,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_s \omega_s}{r_s}. \quad (3.6.6)$$

Thus, once the no-slip belt acceleration $\dot{v}_{b,\text{ns}}$ is known, both pulley angular accelerations are known.

Primary No-Slip Torque Demand

The primary rotational equation is

$$\tau_{\text{eng}} - \tau_p = I_p \dot{\omega}_p.$$

Substituting Equation (3.6.5) gives the primary contact torque required to maintain no slip:

Primary No-Slip Torque Demand Before Belt Solve	(3.6.7)
--	----------------

$$\tau_{p,\text{ns}} = \tau_{\text{eng}} - I_p \left(\frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p} \right)$$

This is already expressed in terms of known state quantities and the single remaining unknown $\dot{v}_{b,\text{ns}}$.

Secondary No-Slip Torque Demand

The secondary torque demand is coupled to the axial shift equation because the secondary helix couples τ_s , $\dot{\omega}_s$, and $\ddot{s}$.

From ??, the shift acceleration can be written directly as

$$\ddot{s}_{\text{ns}} = \frac{F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} \left[\frac{\tau_{s,\text{ns}}}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left(\dot{\omega}_{s,\text{ns}} - \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{\text{shift}}}. \quad (3.6.8)$$

The secondary rotational equation from Equation (3.5.16) is

$$\tau_{s,\text{ns}} - \tau_{\text{load}} = (I_{s,F} + I_M) \dot{\omega}_{s,\text{ns}} - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - I_M \frac{d\theta_s}{ds} \ddot{s}_{\text{ns}}. \quad (3.6.9)$$

Substituting Equation (3.6.8) into Equation (3.6.9), then collecting terms in $\tau_{s,\text{ns}}$, gives

$$\tau_{s,\text{ns}} = \frac{\tau_{\text{load}} + \left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{\text{shift}}} \right] \dot{\omega}_{s,\text{ns}} - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{I_M \frac{d\theta_s}{ds}}{M_{\text{shift}}} \left[F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{\text{shift}}}}. \quad (3.6.10)$$

Finally, substituting the secondary stick-compatible acceleration from [Equation \(3.6.6\)](#) gives the secondary torque demand in terms of the belt acceleration:

$$\tau_{\text{load}} + \left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{\text{shift}}} \right] \left(\frac{\dot{v}_{b,\text{ns}} - \dot{r}_s \omega_s}{r_s} \right) - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2$$

$$\tau_{s,\text{ns}} = \frac{- \frac{I_M}{M_{\text{shift}}} \frac{d\theta_s}{ds} \left[F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{\text{shift}}}}. \quad (3.6.11)$$

This expression is longer than the primary torque demand because the secondary torque is coupled through the helix to both secondary rotation and axial shift.

Explicit No-Slip Belt Acceleration

The belt transport equation is

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting [Equation \(3.6.7\)](#) and [Equation \(3.6.11\)](#), then collecting terms in $\dot{v}_{b,\text{ns}}$, gives the explicit no-slip belt acceleration:

No-Slip Belt Acceleration

(3.6.12)

$$\dot{v}_{b,ns} = \frac{\frac{\tau_{eng}}{r_p} + \frac{I_p \dot{r}_p \omega_p}{r_p^2} - \tau_{load} - \left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \frac{\dot{r}_s \omega_s}{r_s} - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2 - \frac{1}{r_s} \frac{\frac{I_M \frac{d\theta_s}{ds}}{M_{shift}} \left[F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}}{m_b + \frac{I_p}{r_p^2} + \frac{1}{r_s} \frac{\left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \frac{1}{r_s}}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}}$$

This equation contains only quantities known from the current state, evaluated geometry, force models, and inertial parameters. It therefore determines the belt acceleration required by the no-slip candidate.

Recovered No-Slip Accelerations and Candidate Torques

Once $\dot{v}_{b,ns}$ has been computed from Equation (3.6.12), the remaining no-slip quantities follow in direct order.

The pulley angular accelerations are

$$\dot{\omega}_{p,ns} = \frac{\dot{v}_{b,ns} - \dot{r}_p \omega_p}{r_p}, \quad (3.6.13)$$

and

$$\dot{\omega}_{s,ns} = \frac{\dot{v}_{b,ns} - \dot{r}_s \omega_s}{r_s}. \quad (3.6.14)$$

The primary torque candidate is

Primary No-Slip Torque Demand**(3.6.15)**

$$\tau_{p,ns} = \tau_{eng} - I_p \left(\frac{\dot{v}_{b,ns} - \dot{r}_p \omega_p}{r_p} \right)$$

The secondary torque candidate is

Secondary No-Slip Torque Demand**(3.6.16)**

$$\tau_{load} + \left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \left(\frac{\dot{v}_{b,ns} - \dot{r}_s \omega_s}{r_s} \right) - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2$$

$$\tau_{s,ns} = \frac{- \frac{I_M}{M_{shift}} \frac{d\theta_s}{ds} \left[F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}$$

Finally, the no-slip shift acceleration is obtained from the axial shift equation:

No-Slip Shift Acceleration**(3.6.17)**

$$\ddot{s}_{ns} = \frac{F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} \left[\frac{\tau_{s,ns}}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left(\dot{\omega}_{s,ns} - \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{shift}}$$

Together, these equations explicitly determine the no-slip candidate:

$$\dot{v}_{b,ns}, \quad \dot{\omega}_{p,ns}, \quad \dot{\omega}_{s,ns}, \quad \tau_{p,ns}, \quad \tau_{s,ns}, \quad \ddot{s}_{ns}.$$

If both belt–pulley contacts were guaranteed to remain stuck, these equations would close the CVT dynamics. In that idealized case, the belt would act as a kinematic constraint between the two pulleys while the shift coordinate changes the effective ratio.

For a frictional belt drive, however, the required no-slip torques must still be checked against the available traction capacity.

3.6.3 Transition to Traction Limits

The no-slip candidate answers one question: what accelerations and contact torques are required if the belt remains stuck to both pulleys? It does not yet answer whether the contacts can physically support those torques.

The next section derives the tangential traction limits for the primary and secondary belt–pulley contacts. The candidate torques $\tau_{p,ns}$ and $\tau_{s,ns}$ can then be compared with those limits. If both

contact torques are admissible, the no-slip candidate is used. If either required torque exceeds the available traction capacity, the corresponding sticking constraint is replaced by a slip relation.

3.7 Traction Limits and Slip Conditions

The previous section derived the no-slip candidate, including the contact torque demands $\tau_{p,ns}$ and $\tau_{s,ns}$. These are the torques that the belt–pulley interfaces would have to exchange if the belt remained grossly stuck to both pulleys.

Those torque demands are not automatically attainable. Torque is transmitted through frictional traction between the belt and the sheave faces, and that traction is limited by the normal load generated by axial clamping. Therefore, each belt–pulley contact can support only a finite contact torque before gross slip must occur.

The purpose of this section is to derive the traction checks used to decide whether the no-slip candidate is admissible. The important distinction is that the dynamic variables τ_p and τ_s are treated as contact torques, not as tight–slack span-tension differences. The corresponding span tension change can still be reconstructed after the contact torque is known, but the stick/slip admissibility check is posed directly in terms of frictional contact torque.

3.7.1 Contact-Torque Capacity

The same traction argument applies to both pulleys. Let

$$j \in \{p, s\},$$

where $j = p$ denotes the primary pulley and $j = s$ denotes the secondary pulley. At pulley j , the belt–pulley contact is represented by an effective torque radius $r_{j,\text{eff}}$. The signed contact torque at that pulley is denoted τ_j .

Under **Assumption 9 (Gross Tangential Coulomb Traction)** and **Assumption 9 (Gross Tangential Coulomb Traction)**, the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ . Under **Assumption 9 (Gross Tangential Coulomb Traction)**, the available friction is resolved in the gross tangential torque-transfer direction. Thus, at the differential level,

$$|dF_{t,j}| \leq \mu dN_j, \tag{3.7.1}$$

where $dF_{t,j}$ is the tangential contact force and dN_j is the local normal load.

Integrating over the wrap gives the bound on total tangential contact force,

$$|F_{t,j}| \leq \mu N_{\phi,j}, \tag{3.7.2}$$

where

$$N_{\phi,j} \equiv \int_{\text{wrap}} dN_j \tag{3.7.3}$$

is the total normal load carried by the belt–pulley contact on pulley j .

The corresponding contact torque magnitude is

$$|\tau_j| = r_{j,\text{eff}} |F_{t,j}|. \quad (3.7.4)$$

Therefore, for a prescribed normal load, the maximum gross contact torque that can be supported without slip is

Gross Contact-Torque Capacity	(3.7.5)
$ \tau_j \leq r_{j,\text{eff}} \mu N_{\phi,j}$	

This is the contact-torque capacity used for stick/slip admissibility. It bounds the torque exchanged directly at the belt–pulley interface.

Remark (Contact Torque Versus Span Tension)

The contact torque τ_j should not be confused with the torque-equivalent of the tight–slack tension difference across the wrap. During transient motion, some contact traction may be used to accelerate belt material on the wrap, so the contact force and the span tension rise need not be equal. The present model uses contact torque as the dynamic coupling variable. A span-tension reconstruction is given in ??, but it is not used as the primary stick/slip admissibility condition.

Remark (Other Friction Models)

Although the present simulator adopts a Coulomb friction law for simplicity and analytic transparency, richer friction descriptions are available if greater fidelity near the stick–slip transition is desired. A common next step is a Stribeck-type law, in which the effective friction level varies with relative slip speed so that breakaway friction near zero speed can exceed the kinetic friction observed at larger slip speeds. Such a model can better represent engagement and low-slip behaviour than a purely constant Coulomb coefficient, while still remaining relatively simple to implement [Armstrong-H'elouvry et al. \(1994\)](#); [Canudas de Wit et al. \(1995\)](#).

For still higher fidelity, dynamic friction models introduce internal state variables that represent the evolving contact condition. The LuGre model is a standard example and can capture presliding displacement, hysteresis, friction lag, and smoother stick–slip transitions, while related multistate models such as the generalized Maxwell–slip model can represent more detailed microslip and memory effects [Canudas de Wit et al. \(1995\)](#); [Al-Bender et al. \(2005\)](#). These models are often more physically expressive, but they also require additional parameter identification and increase numerical and modelling complexity. For the present work, Coulomb friction is therefore retained as a deliberate low-complexity approximation, while Stribeck- or state-based friction models remain natural extensions for future refinement [Armstrong-H'elouvry et al. \(1994\)](#).

3.7.2 Normal Force from Axial Clamping

The torque capacity in Equation (3.7.5) depends on the total normal load $N_{\phi,j}$. We now express that normal load in terms of the axial clamping force generated by pulley j .

The belt is pressed between two opposing conical sheaves. The two inclined faces apply a net radial normal load on the belt determined by the axial clamp. Using the groove geometry from ??,

$$F_{\text{rad},j} = 2F_{\text{ax},j} \tan \beta. \quad (3.7.6)$$

The radial load in Equation (3.7.6) represents the total inward force applied by the sheave faces onto the belt. This load is carried by the distributed normal contact forces along the wrap, so

$$N_{\phi,j} = F_{\text{rad},j}. \quad (3.7.7)$$

Only compressive axial clamp can create normal load. To prevent an unphysical negative traction capacity, define the compressive part of any axial force as

$$\langle F \rangle_+ = \max(F, 0). \quad (3.7.8)$$

Therefore,

Wrap-Normal Load from Axial Clamp	(3.7.9)
$N_{\phi,j} = 2 \langle F_{\text{ax},j} \rangle_+ \tan \beta$	

Substituting Equation (3.7.9) into Equation (3.7.5) gives the gross contact-torque capacity

Pulley Contact-Torque Capacity	(3.7.10)
$ \tau_j \leq 2r_{j,\text{eff}} \mu \langle F_{\text{ax},j} \rangle_+ \tan \beta$	

For a prescribed axial clamp, this bound is symmetric in positive and negative contact torque. However, the admissible torque range of a complete pulley mechanism may still be asymmetric if the axial clamp itself depends on the signed contact torque. This distinction is important for the torque-reactive secondary pulley.

Remark (Wrap Angle in the Reduced Capacity Model)

The wrap angle does not appear explicitly in Equation (3.7.10) because $N_{\phi,j}$ is the total integrated normal load imposed by the clamp. Under the averaged-contact approximation, increasing wrap angle spreads this same integrated normal load over more contact length rather than increasing the total available gross friction force. Wrap angle still matters for local contact pressure, heat, wear, pressure distribution, and span-tension reconstruction, but it is not an independent multiplier in the reduced gross contact-torque capacity.

3.7.3 Primary Pulley No-Slip Admissibility

The generic contact-torque capacity law can now be specialized to the primary pulley. At this stage, the goal is only to test whether the primary contact torque demanded by the no-slip candidate can be supported by the available primary traction.

For compactness, write

$$r_p = r_{p,\text{eff}}(s).$$

The primary axial force is known explicitly from ??:

$$F_{\text{ax},p}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds} - k_p(x_{p,0} + s). \quad (3.7.11)$$

Using the static friction coefficient μ_s , the primary no-slip contact is admissible if

Primary No-Slip Admissibility (3.7.12) $ \tau_{p,\text{ns}} \leq 2r_p \mu_s \langle F_{\text{ax},p}(s, \omega_p) \rangle_+ \tan \beta$

This condition is symmetric in positive and negative primary contact torque because the primary axial force does not depend on the signed primary torque. If [Equation \(3.7.12\)](#) is satisfied, the primary contact can support the torque required by the no-slip candidate. If it is not satisfied, the primary contact cannot remain grossly stuck under the demanded contact torque.

3.7.4 Secondary Pulley No-Slip Admissibility

The secondary pulley follows the same contact-torque capacity law as the primary, but with one important difference: the secondary axial clamp is torque-reactive. Through the helix, the same signed secondary contact torque being checked for admissibility also contributes to the axial force that generates the available traction.

To keep this dependence explicit, write the secondary axial force as the function

$$F_{\text{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}) = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[\frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right) \right]. \quad (3.7.13)$$

For compactness, write

$$r_s = r_{s,\text{eff}}(s).$$

On the no-slip candidate branch, the quantities $\tau_{s,\text{ns}}$, $\dot{\omega}_{s,\text{ns}}$, and $\ddot{s}_{\text{ns}}$ have already been determined by the no-slip solve. Therefore, the secondary clamp used in the traction check is evaluated at the candidate values:

$$F_{\text{ax},s}^{\text{ns}} = F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}).$$

Using the static friction coefficient μ_s , the secondary no-slip contact is admissible if

$$\text{Secondary No-Slip Admissibility} \quad (3.7.14)$$

$$|\tau_{s,\text{ns}}| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}) \rangle_+ \tan \beta$$

This condition is written as a magnitude check, but it is not assuming a symmetric secondary torque range. The axial force inside the right-hand side is evaluated using the signed no-slip torque $\tau_{s,\text{ns}}$. Therefore, positive and negative secondary torques of equal magnitude generally produce different clamp forces and different available traction. The condition above is a pointwise admissibility test: it asks whether the specific signed no-slip candidate torque creates enough clamp to support itself.

If Equation (3.7.14) is satisfied, the secondary contact can support the torque required by the no-slip candidate. If it is not satisfied, the secondary contact cannot remain grossly stuck under the demanded contact torque.

Remark (Optional Secondary Torque Envelope)

For branch selection, the pointwise check in Equation (3.7.14) is sufficient because the no-slip solve has already produced a specific value of $\tau_{s,\text{ns}}$. However, it can also be useful to reconstruct the full secondary torque envelope at a fixed instantaneous state, for example when plotting how the torque-reactive helix changes the available torque range.

At a fixed state, the secondary stick envelope is the set of torques satisfying

$$|\tau_s| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}) \rangle_+ \tan \beta. \quad (3.7.15)$$

Because $F_{\text{ax},s}$ contains the signed torque τ_s , the set defined by Equation (3.7.15) is generally not symmetric about zero. One torque direction may increase clamp through the helix, while the opposite direction may reduce clamp. This envelope is useful for diagnostic plots or design interpretation, but the branch-selection logic only needs the pointwise candidate check in Equation (3.7.14).

3.7.5 Admissibility of the No-Slip Candidate

The full no-slip candidate is admissible only if both pulley contacts satisfy their static traction checks:

$$|\tau_{p,\text{ns}}| \leq 2r_p \mu_s \langle F_{\text{ax},p}(s, \omega_p) \rangle_+ \tan \beta, \quad (3.7.16)$$

and

$$|\tau_{s,\text{ns}}| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}) \rangle_+ \tan \beta. \quad (3.7.17)$$

When both conditions hold, the no-slip candidate can be accepted as the contact state for that instant. If either condition fails, at least one contact cannot maintain gross sticking and the corresponding contact relation must be replaced by a slip relation.

3.7.6 Kinetic Traction Bound for Slipping Contacts

The same contact-torque capacity structure is used when a contact is sliding, but the kinetic friction coefficient μ_k replaces the static coefficient μ_s . Thus, the magnitude of the slipping contact torque at pulley j is

Kinetic Contact-Torque Magnitude	(3.7.18)
$ \tau_{j,\text{slip}} = 2r_{j,\text{eff}}\mu_k \langle F_{\text{ax},j} \rangle_+ \tan \beta$	

The sign of the kinetic torque is chosen so that the friction force opposes the relative sliding motion between the belt and pulley surface. The exact sign convention for τ_p and τ_s is applied in the contact-state selection section, where the primary and secondary slip branches are written explicitly.

3.7.7 Section Conclusion and Transition

This section derived the traction checks needed to test the no-slip candidate from [Section 3.6.2](#). The stick/slip admissibility check was posed directly in terms of contact torque because the rotational and belt transport equations use τ_p and τ_s as the torques exchanged at the belt–pulley contacts.

The primary check is symmetric because the primary axial force is independent of the signed primary contact torque. The secondary check is not generally symmetric in torque because the helix makes the secondary axial clamp depend on the signed secondary contact torque. For branch selection, this is handled by evaluating the secondary clamp directly on the signed no-slip candidate.

If both pulley contacts satisfy their admissibility conditions, the no-slip candidate is accepted. If one or both contacts cannot support their demanded contact torques, the corresponding contact must be described by a slipping traction law. The next section develops the contact-state selection rules and the resulting slip-branch dynamics.

3.8 Slip Branch Dynamics and Contact-State Selection

The previous section determined when the no-slip torque demands can be supported by the available belt–pulley traction. If the primary and secondary contacts both satisfy their admissibility conditions, the no-slip candidate is physically consistent and the belt remains adhered to both pulleys.

When one of these admissibility conditions fails, the no-slip candidate can no longer be used as the final torque-transfer law. The contact that exceeds its available traction cannot supply the demanded torque while remaining stuck, so relative motion begins at that belt–pulley interface. Once this occurs, the contact torque is no longer determined by the no-slip demand. Instead, it is set by the kinetic traction available at the slipping contact.

This section develops the rules used to select the active contact state and the corresponding equations of motion. The same three bodies introduced in [Section 3.5.1](#) are retained: the primary pulley, the belt, and the secondary pulley. What changes from one branch to another is the contact condition

imposed at each pulley. A stuck contact enforces belt-speed compatibility, while a slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion.

The goal is therefore to complete the torque-transfer closure. We first define the relative belt–pulley speeds used to identify sliding, then state the rules for entering, maintaining, and exiting slip. The remaining subsections derive the branch dynamics for primary slip, secondary slip, and simultaneous slip at both contacts.

3.8.1 Relative Contact Speeds and Slip-State Logic

To determine whether a belt–pulley contact is sticking or slipping, the model compares the belt transport speed to the local surface speed imposed by each pulley.

For the primary pulley, define the relative contact speed as

$$v_{\text{rel},p} = r_p \omega_p - v_b, \quad (3.8.1)$$

where $r_p = r_{p,\text{eff}}(s)$. Thus, $v_{\text{rel},p} > 0$ means the primary pulley surface is moving faster than the belt at the contact radius and tends to drive the belt forward.

For the secondary pulley, define

$$v_{\text{rel},s} = v_b - r_s \omega_s, \quad (3.8.2)$$

where $r_s = r_{s,\text{eff}}(s)$. Thus, $v_{\text{rel},s} > 0$ means the belt is moving faster than the secondary pulley surface and tends to drive the secondary forward.

These definitions are chosen so that positive relative speed corresponds to the usual forward torque-transfer direction at each contact: primary-to-belt on the primary side, and belt-to-secondary on the secondary side.

A contact can remain stuck only if its relative contact speed is zero, or within a small numerical tolerance:

$$|v_{\text{rel},j}| \leq \varepsilon_v, \quad j \in \{p, s\}. \quad (3.8.3)$$

If both contacts satisfy [Equation \(3.8.3\)](#), the model first tests the no-slip candidate derived in [??](#). This candidate is accepted only if both pulley contacts also satisfy the traction admissibility conditions derived in [Section 3.7](#):

$$|\tau_{p,\text{ns}}| \leq \tau_{p,\text{max}}^{\text{stick}} \quad (3.8.4)$$

and

$$\tau_{s,-}^{\text{stick}} \leq \tau_{s,\text{ns}} \leq \tau_{s,+}^{\text{stick}}. \quad (3.8.5)$$

If these conditions hold, the no-slip candidate becomes the active torque-transfer branch:

$$\tau_p = \tau_{p,\text{ns}}, \quad \tau_s = \tau_{s,\text{ns}}, \quad \dot{v}_b = \dot{v}_{b,\text{ns}}.$$

If either contact fails its admissibility check, that contact cannot support the required no-slip torque and must enter slip. The direction of slip is determined by the direction in which the demanded torque exceeds the admissible interval. For the primary pulley,

$$\tau_{p,\text{ns}} > \tau_{p,\text{max}}^{\text{stick}} \Rightarrow \text{positive primary slip,}$$

while

$$\tau_{p,ns} < -\tau_{p,max}^{stick} \Rightarrow \text{negative primary slip.}$$

For the secondary pulley,

$$\tau_{s,ns} > \tau_{s,+}^{stick} \Rightarrow \text{positive secondary slip,}$$

while

$$\tau_{s,ns} < \tau_{s,-}^{stick} \Rightarrow \text{negative secondary slip.}$$

Once a contact is already sliding, its state is determined primarily by its relative contact speed. If

$$|v_{rel,j}| > \varepsilon_v,$$

then contact j is treated as slipping, and its torque is set by kinetic traction rather than by the no-slip demand. The slip direction is recorded by

$$\sigma_j = \text{sgn}(v_{rel,j}), \quad j \in \{p, s\}. \quad (3.8.6)$$

Slip persists while the relative contact speed remains nonzero. During this time, the slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion. The no-slip candidate is not reaccepted until the relative speed returns to the sticking tolerance and the static admissibility conditions can again be satisfied.

This gives three possible slipping branches:

- primary slip, where the primary contact slides and the secondary contact remains stuck;
- secondary slip, where the secondary contact slides and the primary contact remains stuck;
- two-contact slip, where both pulley contacts slide simultaneously.

The following subsections derive the equations of motion for each branch.

Remark (Power Flow During Slip)

In the no-slip branch, belt-speed compatibility constrains the primary pulley, belt, and secondary pulley to move together kinematically. Once slip occurs, that compatibility is lost at the slipping contact. The contact torque is then limited by kinetic traction, while the pulley and belt speeds evolve according to their own equations of motion.

As a result, the instantaneous mechanical power at the primary contact, $\tau_p \omega_p$, does not need to match the instantaneous mechanical power at the secondary contact, $\tau_s \omega_s$. The difference is associated with changes in stored kinetic energy of the rotating components and belt, together with slip dissipation. Thus, an apparent short-term excess of output-side power does not imply energy creation; it reflects energy being drawn from inertia while the slipping contact breaks the ideal no-slip power constraint.

3.8.2 Kinetic Traction Law for a Slipping Contact

Once a belt–pulley contact is slipping, the torque at that contact is no longer determined by the no-slip torque demand. Instead, the contact applies the largest kinetic traction torque available in the direction opposing relative motion.

The generic torque-capacity expression from Equation (3.7.10) still applies, but the static coefficient μ_s is replaced by the kinetic coefficient μ_k . For a slipping contact on pulley $j \in \{p, s\}$, define

$$\tau_{j,\max}^{\text{slip}}(\dot{v}_b) = r_j [2\mu_k F_{\text{ax},j} \tan \beta - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_{j,\text{cm}} v_b)], \quad (3.8.7)$$

where $r_j = r_{j,\text{eff}}(s)$. The notation $\tau_{j,\max}^{\text{slip}}(\dot{v}_b)$ emphasizes that the available slipping torque is not generally a fixed number; it depends on the belt acceleration through the inertial correction term.

The direction of the slipping torque is determined by the relative contact speed defined in Section 3.8.1. Let

$$\sigma_j = \text{sgn}(v_{\text{rel},j}), \quad j \in \{p, s\}. \quad (3.8.8)$$

With the relative-speed definitions adopted in Equation (3.8.1) and Equation (3.8.2), $\sigma_j > 0$ corresponds to the usual forward torque-transfer direction at contact j . The slipping contact torque is then modeled as the saturated kinetic traction torque

$$\tau_j = \sigma_j \tau_{j,\max}^{\text{slip}}(\dot{v}_b). \quad (3.8.9)$$

Equation (3.8.9) should not be interpreted as a completed solution for τ_j , because the right-hand side still contains $\dot{v}_b$. In a slip branch, the belt acceleration must be solved simultaneously with the saturated contact torque using the belt equation of motion Equation (3.5.14). The following subsections carry out this solve for primary slip, secondary slip, and simultaneous slip at both contacts.

3.8.3 Primary-Slip Branch

We first consider the branch where the primary contact is slipping while the secondary contact remains stuck. In this case, the primary pulley no longer enforces belt-speed compatibility. Instead, the primary contact torque is set by the saturated kinetic traction law from Section 3.8.2.

The secondary contact, however, is still adhered. Therefore, the secondary pulley remains kinematically compatible with the belt:

$$\dot{\omega}_s = \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s}, \quad (3.8.10)$$

Using the secondary rotational equation of motion Equation (3.5.16), the secondary contact torque is then

$$\tau_s = \tau_{\text{load}} + I_s \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \quad (3.8.11)$$

At the slipping primary contact, let

$$\sigma_p = \text{sgn}(v_{\text{rel},p}).$$

Applying Equation (3.8.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p [2\mu_k F_{\text{ax},p} \tan \beta - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b)], \quad (3.8.12)$$

where $r_p = r_{p,\text{eff}}(s)$.

The remaining unknown is the belt acceleration $\dot{v}_b$. This is found by substituting Equation (3.8.11) and Equation (3.8.12) into the belt equation of motion Equation (3.5.14):

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}. \quad (3.8.13)$$

Substituting gives

$$m_b \dot{v}_b = \sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p (r_{p,cm} \dot{v}_b + \dot{r}_{p,cm} v_b)] - \frac{\tau_{load}}{r_s} - \frac{I_s}{r_s} \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \quad (3.8.14)$$

Collecting the $\dot{v}_b$ terms yields the primary-slip belt acceleration:

Primary-Slip Belt Acceleration (3.8.15) $\dot{v}_{b,p\text{-slip}} = \frac{\sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p \dot{r}_{p,cm} v_b] - \frac{\tau_{load}}{r_s} + \frac{I_s \dot{r}_s \omega_s}{r_s^2}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,cm} + \frac{I_s}{r_s^2}}$

Once $\dot{v}_{b,p\text{-slip}}$ is known, the branch torques follow by substitution into Equation (3.8.12) and Equation (3.8.11). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\tau_{eng} - \tau_p}{I_p}, \quad (3.8.16)$$

and

$$\dot{\omega}_s = \frac{\dot{v}_{b,p\text{-slip}} - \dot{r}_s \omega_s}{r_s}. \quad (3.8.17)$$

Thus, in the primary-slip branch, the primary contact torque is set by kinetic traction, the secondary contact still enforces belt-speed compatibility, and the belt equation of motion determines the resulting belt acceleration. This branch remains valid only while the secondary contact can remain adhered. If the secondary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.8.4 Secondary-Slip Branch

We next consider the branch where the secondary contact is slipping while the primary contact remains stuck. In this case, the secondary contact torque is set by saturated kinetic traction, while the primary contact still enforces belt-speed compatibility.

The primary contact remains adhered, so the primary pulley must satisfy

$$\dot{\omega}_p = \frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p}, \quad (3.8.18)$$

where $r_p = r_{p,\text{eff}}(s)$. Using the primary rotational equation of motion [Equation \(3.5.13\)](#), the primary contact torque is therefore

$$\tau_p = \tau_{\text{eng}} - I_p \left(\frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p} \right). \quad (3.8.19)$$

At the slipping secondary contact, let

$$\sigma_s = \text{sgn}(v_{\text{rel},s}).$$

Applying [Equation \(3.8.9\)](#) with $j = s$ gives

$$\tau_s = \sigma_s r_s [2\mu_k F_{\text{ax},s}(s, \tau_s) \tan \beta - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b)], \quad (3.8.20)$$

where $r_s = r_{s,\text{eff}}(s)$.

Unlike the primary slip branch, this expression is implicit in τ_s because the secondary axial force is torque-reactive. Substituting the secondary axial force model from ?? gives

$$\begin{aligned} \tau_s = \sigma_s r_s \left[\mu_k \tan \beta (\tau_s + k_{s,\theta}(\theta_{s,0} + \theta_s(s))) \frac{d\theta_s}{ds} \right. \\ \left. + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]. \end{aligned} \quad (3.8.21)$$

Collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan \beta k_{s,\theta}(\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}. \quad (3.8.22)$$

The remaining unknown is the belt acceleration $\dot{v}_b$. Substituting [Equation \(3.8.19\)](#) and [Equation \(3.8.22\)](#) into the belt equation of motion [Equation \(3.5.14\)](#),

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s},$$

and collecting the $\dot{v}_b$ terms gives the secondary-slip belt acceleration:

Secondary-Slip Belt Acceleration

(3.8.23)

$$\dot{v}_{b,s-\text{slip}} = \frac{\frac{\tau_{\text{eng}}}{r_p} + \frac{I_p \dot{r}_p \omega_p}{r_p^2} - \frac{\sigma_s \left[\mu_k \tan \beta k_{s,\theta}(\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\text{cm}} v_b \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}{m_b + \frac{I_p}{r_p^2} - \frac{\sigma_s \rho_b A_b \phi_s r_{s,\text{cm}}}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}$$

Once $\dot{v}_{b,s-\text{slip}}$ is known, the branch torques follow by substitution into Equation (3.8.19) and Equation (3.8.22). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\dot{v}_{b,s-\text{slip}} - \dot{r}_p \omega_p}{r_p}, \quad (3.8.24)$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \quad (3.8.25)$$

Thus, in the secondary-slip branch, the secondary contact torque is set by kinetic traction, including the torque-reactive helix feedback, while the primary contact still enforces belt-speed compatibility. This branch remains valid only while the primary contact can remain adhered. If the primary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.8.5 Two-Contact Slip Branch

Finally, consider the branch where both belt-pulley contacts are slipping. In this case, neither pulley enforces belt-speed compatibility. Both contact torques are instead determined by saturated kinetic traction, and the belt acceleration is obtained directly from the belt equation of motion.

Let

$$\sigma_p = \text{sgn}(v_{\text{rel},p}), \quad \sigma_s = \text{sgn}(v_{\text{rel},s}).$$

At the primary contact, applying Equation (3.8.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p [2\mu_k F_{\text{ax},p} \tan \beta - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b)], \quad (3.8.26)$$

where $r_p = r_{p,\text{eff}}(s)$.

At the secondary contact, applying Equation (3.8.9) with $j = s$ gives

$$\tau_s = \sigma_s r_s [2\mu_k F_{\text{ax},s}(s, \tau_s) \tan \beta - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b)], \quad (3.8.27)$$

where $r_s = r_{s,\text{eff}}(s)$. As in Section 3.8.4, the secondary expression is implicit because the secondary axial force is torque-reactive. Substituting ?? and collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan \beta k_{s,\theta} (\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x} (x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}. \quad (3.8.28)$$

The belt equation of motion is still

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.8.26) and Equation (3.8.28), then collecting the $\dot{v}_b$ terms, gives the two-contact-slip belt acceleration:

Two-Contact-Slip Belt Acceleration (3.8.29)

$$\dot{v}_{b,ps\text{-slip}} = \frac{\sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p \dot{r}_{p,cm} v_b] - \frac{\sigma_s \left[\mu_k \tan \beta k_{s,\theta} (\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x} (x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,cm} v_b \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,cm} - \frac{\sigma_s \rho_b A_b \phi_s r_{s,cm}}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}$$

Once $\dot{v}_{b,ps\text{-slip}}$ is known, the branch torques follow by substitution into Equation (3.8.26) and Equation (3.8.28). Since neither contact is adhered, the pulley accelerations are obtained directly from the rotational equations of motion:

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_p}{I_p}, \quad (3.8.30)$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \quad (3.8.31)$$

Thus, in the two-contact slip branch, both pulley contacts are governed by kinetic traction, and the belt is not kinematically constrained to either pulley. The belt acceleration is determined only by the balance of the two saturated contact torques through the belt equation of motion.

3.8.6 Contact-State Selection

The preceding subsections define the possible torque-transfer closures. The active branch is selected by checking which contacts can remain adhered.

Let the primary sticking condition be

$$\mathcal{S}_p \equiv (|v_{\text{rel},p}| \leq \varepsilon_v) \wedge \left(|\tau_{p,ns}| \leq \tau_{p,\text{max}}^{\text{stick}} \right),$$

and let the secondary sticking condition be

$$\mathcal{S}_s \equiv (|v_{\text{rel},s}| \leq \varepsilon_v) \wedge \left(\tau_{s,-}^{\text{stick}} \leq \tau_{s,ns} \leq \tau_{s,+}^{\text{stick}} \right).$$

The branch closure for the belt acceleration and contact torques is then

$$(\dot{v}_b, \tau_p, \tau_s) = \begin{cases} (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}), & \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,p\text{--slip}}, \tau_p \text{ from Equation (3.8.12)}, \tau_s \text{ from Equation (3.8.11)}), & \neg \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,s\text{--slip}}, \tau_p \text{ from Equation (3.8.19)}, \tau_s \text{ from Equation (3.8.22)}), & \mathcal{S}_p \wedge \neg \mathcal{S}_s, \\ (\dot{v}_{b,ps\text{--slip}}, \tau_p \text{ from Equation (3.8.26)}, \tau_s \text{ from Equation (3.8.28)}), & \neg \mathcal{S}_p \wedge \neg \mathcal{S}_s. \end{cases} \quad (3.8.32)$$

The corresponding belt accelerations are given by Equation (3.6.12), Equation (3.8.15), Equation (3.8.23), and Equation (3.8.29), respectively.

Thus, when both contacts can remain adhered, the model uses the no-slip torque demands. When one or both contacts cannot remain adhered, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves the belt acceleration from the belt equation of motion.

3.8.7 Section Summary

This section completed the torque-transfer closure for cases where the no-slip candidate cannot be maintained. Relative contact speeds were introduced to identify whether each belt–pulley interface is adhered or sliding, and kinetic traction was used to define the torque applied by a slipping contact.

The resulting branch structure separates the possible contact states into no-slip, primary slip, secondary slip, and two-contact slip. In each slipping case, the saturated contact torque replaces the no-slip demand, and the belt equation of motion determines the corresponding belt acceleration.

Together with the no-slip demand from ?? and the traction admissibility conditions from Section 3.7, the branch selection law in Equation (3.8.32) provides the final torque-transfer rule needed by the complete CVT dynamic model.

3.9 Complete CVT Dynamic Model

This section collects the results of the preceding derivations into one closed nonlinear dynamical model. No new physics is introduced here. The purpose is to summarize, in one place,

1. the state variables,
2. the geometric and contact quantities computed from those states,
3. the torque-transfer closure,
4. and the final governing equations of motion.

3.9.1 State Definition

The state vector is

$$\mathbf{x}(t) = (\omega_p, \omega_s, s, \dot{s}, v_b), \quad (3.9.1)$$

where

- ω_p is the primary pulley angular velocity,

- ω_s is the secondary pulley angular velocity,
- s is the axial shift coordinate,
- $\dot{s}$ is the axial shift velocity,
- v_b is the belt transport speed.

3.9.2 Geometry and Contact Kinematics

At each instant, the shift coordinate s determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived in [Section 3.1](#). In particular,

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s$$

are functions of s . For compactness in the complete model, write

$$r_p = r_{p,\text{eff}}(s), \quad r_s = r_{s,\text{eff}}(s).$$

The geometric CVT ratio remains

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \quad (3.9.2)$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \quad (3.9.3)$$

as given by ??.

The relative contact speeds at the two pulley interfaces are

$$v_{\text{rel},p} = r_p \omega_p - v_b, \quad v_{\text{rel},s} = v_b - r_s \omega_s, \quad (3.9.4)$$

as defined in [Equation \(3.8.1\)](#) and [Equation \(3.8.2\)](#). These quantities determine whether each contact is eligible to remain adhered or must be treated as sliding.

3.9.3 Torque-Transfer Closure

The torque-transfer subsystem is closed by determining the primary contact torque τ_p , the secondary contact torque τ_s , and the belt acceleration $\dot{v}_b$. These quantities are computed algebraically from the current state and contact branch.

The no-slip candidate derived in [Section 3.6.2](#) provides

$$\dot{v}_{b,\text{ns}}, \quad \tau_{p,\text{ns}}, \quad \tau_{s,\text{ns}},$$

from [Equation \(3.6.12\)](#), [Equation \(3.6.15\)](#), and [Equation \(3.6.16\)](#). These are the belt acceleration and contact torque demands required for adhered motion at both pulleys.

The admissibility conditions derived in [Section 3.7.3](#) and [Section 3.7.4](#) determine whether those no-slip demands can be supported. The primary contact is admissible when

$$|\tau_{p,\text{ns}}| \leq \tau_{p,\text{max}}^{\text{stick}}, \quad (3.9.5)$$

as in [Equation \(3.7.12\)](#). The secondary contact is admissible when

$$\tau_{s,-}^{\text{stick}} \leq \tau_{s,\text{ns}} \leq \tau_{s,+}^{\text{stick}}, \quad (3.9.6)$$

as in [Equation \(3.7.14\)](#).

The active contact branch is selected using the closure law from [Section 3.8.6](#). In compact form,

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the branch output selected by [Equation \(3.8.32\)](#).} \quad (3.9.7)$$

If both contacts remain adhered, the branch output is

$$(\dot{v}_b, \tau_p, \tau_s) = (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}).$$

If one or both contacts slide, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves $\dot{v}_b$ from the belt equation of motion. The primary-slip, secondary-slip, and two-contact-slip belt accelerations are given by [Equation \(3.8.15\)](#), [Equation \(3.8.23\)](#), and [Equation \(3.8.29\)](#), respectively.

3.9.4 Governing ODEs

Once the branch closure has supplied $\dot{v}_b$, τ_p , and τ_s , the primary pulley equation of motion is

$$\dot{\omega}_p = \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}. \quad (3.9.8)$$

The secondary pulley equation of motion is

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}. \quad (3.9.9)$$

The shift-position kinematics are

$$\frac{d}{dt}s = \dot{s}. \quad (3.9.10)$$

The axial shift acceleration is obtained from ??:

$$\ddot{s} = \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\right) - \left(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}. \quad (3.9.11)$$

Finally, the belt transport speed evolves according to the branch-selected belt acceleration:

$$\frac{d}{dt}v_b = \dot{v}_b, \quad \dot{v}_b \text{ from [Equation \(3.9.7\)](#).} \quad (3.9.12)$$

3.9.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = (\omega_p, \omega_s, s, \dot{s}, v_b),$$

the model is evaluated in the following order:

1. Compute the pulley geometry from s , including r_p , r_s , $r_{p,\text{cm}}$, $r_{s,\text{cm}}$, ϕ_p , and ϕ_s .
2. Compute the engine torque $\tau_{\text{eng}}(\omega_p)$ and the reflected load torque $\tau_{\text{load}}(\omega_s)$.

3. Compute the no-slip candidate

$$\dot{v}_{b,\text{ns}}, \quad \tau_{p,\text{ns}}, \quad \tau_{s,\text{ns}},$$

using Equation (3.6.12), Equation (3.6.15), and Equation (3.6.16).

4. Compute the primary and secondary no-slip admissibility quantities from Section 3.7.3 and Section 3.7.4.
5. Compute the relative contact speeds $v_{\text{rel},p}$ and $v_{\text{rel},s}$ using Equation (3.9.4).
6. Select the active torque-transfer branch using Equation (3.8.32). This determines the applied values of

$$\dot{v}_b, \quad \tau_p, \quad \tau_s.$$

7. Evaluate the ODE right-hand sides Equation (3.9.8), Equation (3.9.9), Equation (3.9.10), Equation (3.9.11), and Equation (3.9.12).

3.9.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$\begin{aligned} \dot{\omega}_p &= \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}, \\ \dot{\omega}_s &= \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}, \\ \frac{d}{dt}s &= \dot{s}, \\ \frac{d}{dt}\dot{s} &= \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\right) - \left(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}, \\ \frac{d}{dt}v_b &= \dot{v}_b. \end{aligned} \tag{3.9.13}$$

with the algebraic closure

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the active branch output from Equation (3.8.32)}.$$

In summary, the state determines the pulley geometry and relative contact speeds. The no-slip candidate determines the torque demand required for adhered motion. The traction admissibility conditions determine whether that candidate can be maintained. If not, the appropriate slip branch supplies saturated contact torques and the resulting belt acceleration. These branch-selected quantities then feed back into the rotational, belt, and axial shift dynamics, closing the complete CVT state-space model.

Chapter 4

Results and Discussion

4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows ?? : low ratio means larger R (greater speed reduction), and high ratio means smaller R . following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in [Section 4.2](#).

4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in [Figure 4.1](#). Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800 \text{ rpm}, \quad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

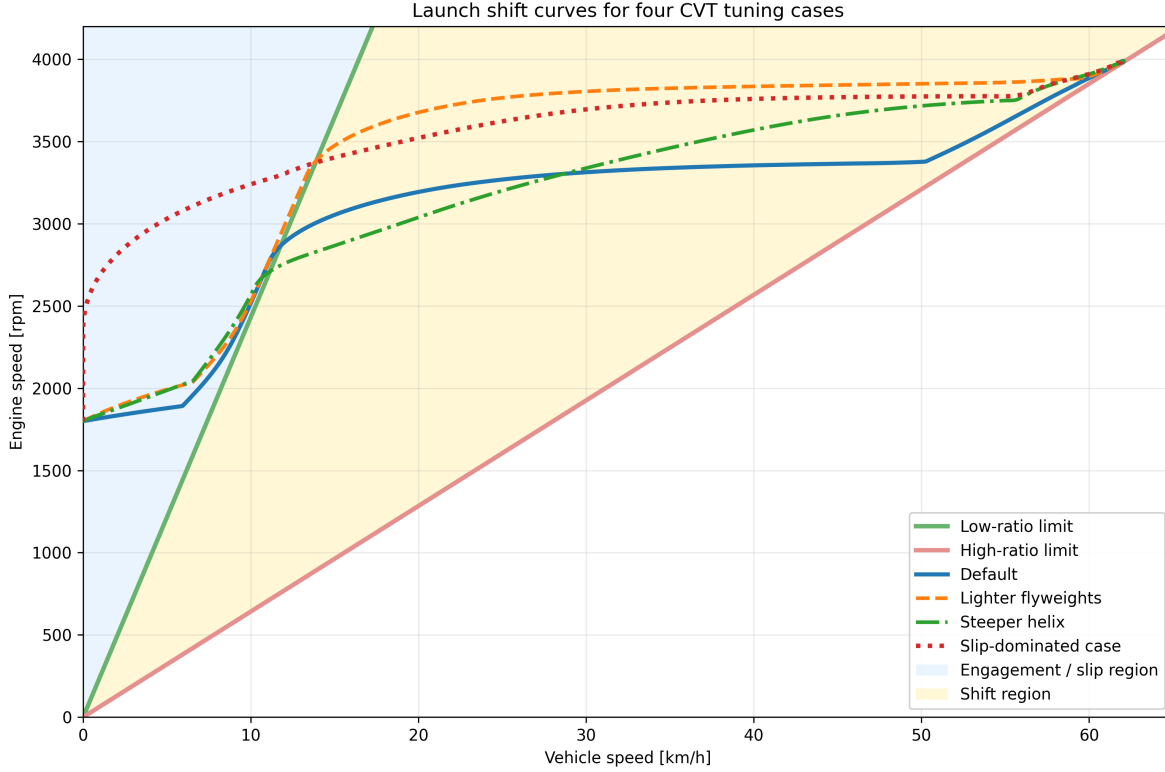


Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio

boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, [Figure 4.1](#) is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of [Figure 4.1](#) indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, [Figure 4.2](#) compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

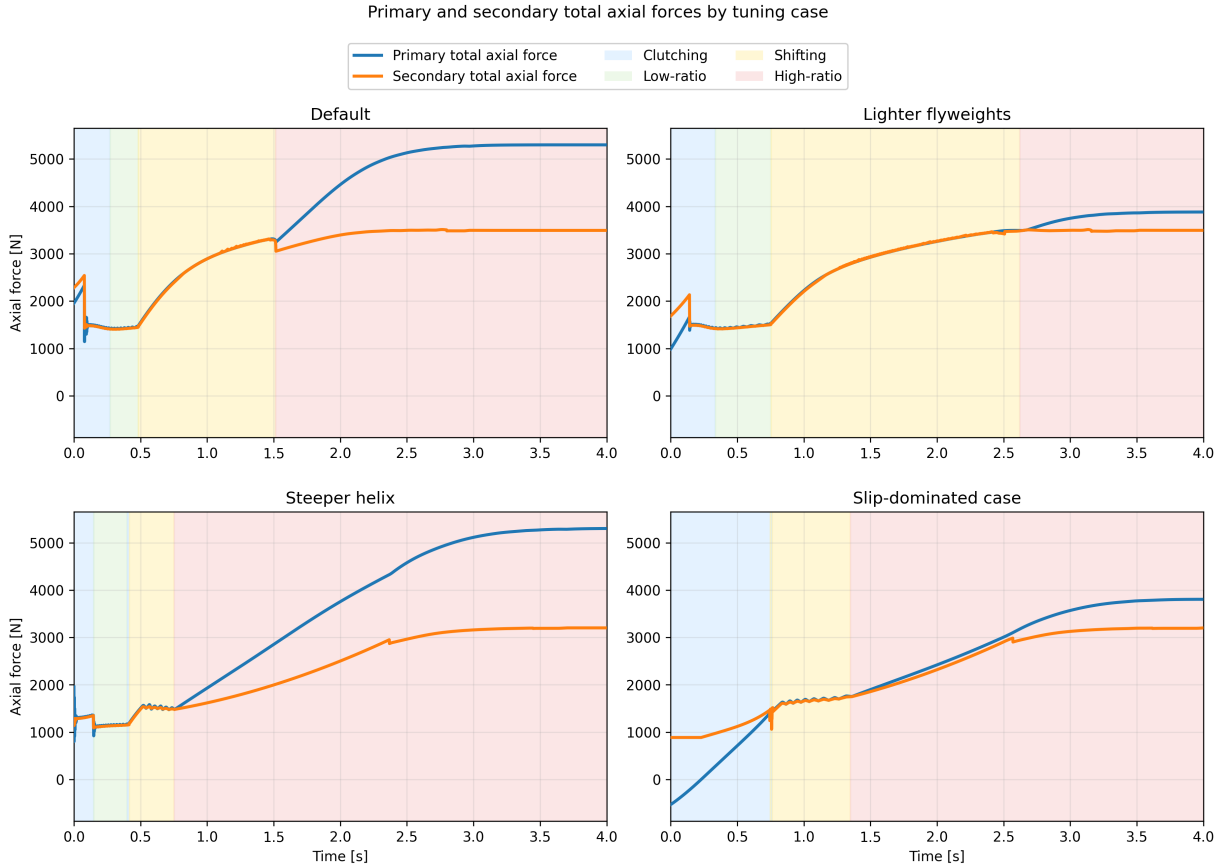


Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in [Figure 4.1](#). As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, [Figure 4.2](#) provides the mechanical explanation for the differences observed in [Figure 4.1](#). A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque τ_{ns} , and the actual transmitted coupling torque τ_c .

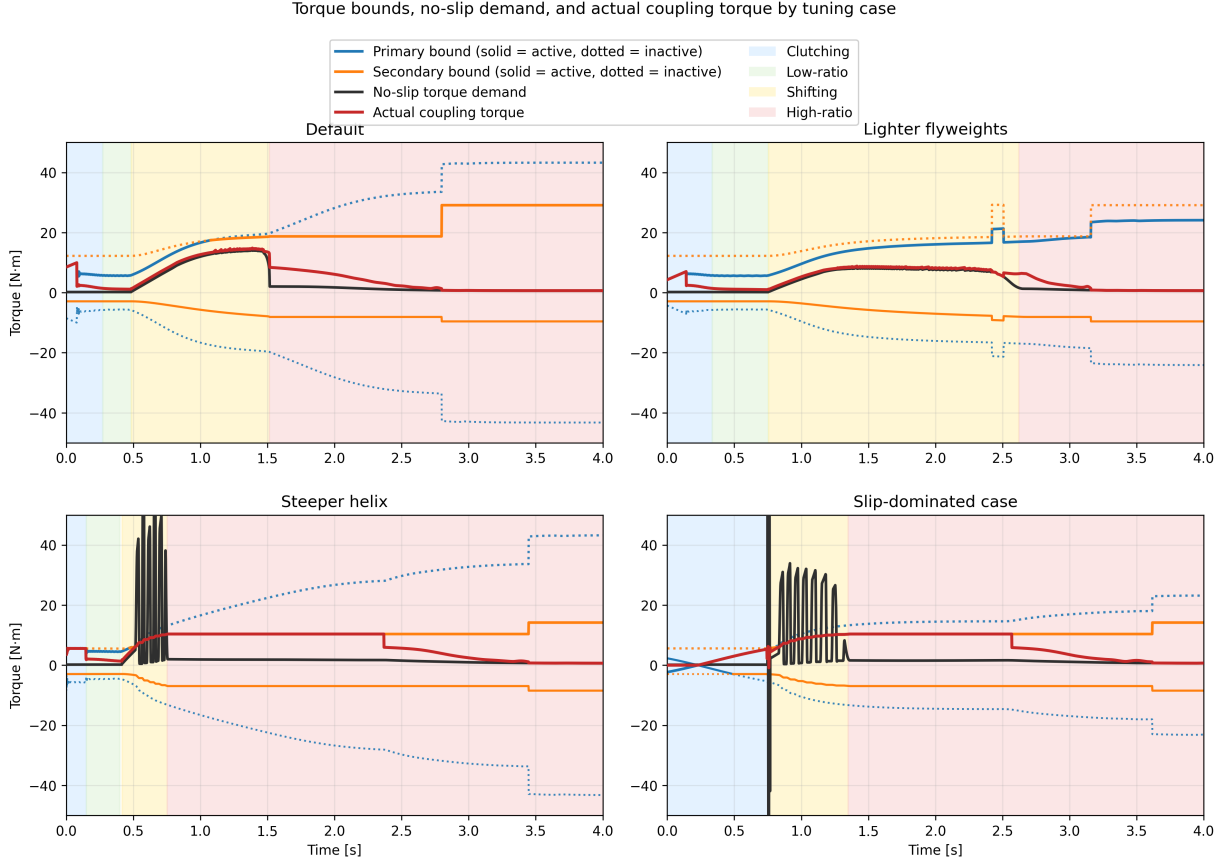


Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand τ_{ns} , and the red curve shows the actual transmitted coupling torque τ_c . Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When τ_{ns} lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When τ_{ns} exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a

relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c . In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns} , in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns} : if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick-slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix C.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick-slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, [Figure 4.3](#) closes the interpretation begun in the previous two subsections. The

shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model’s clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

Case	Time to 50 m [s]	Time to 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps “kick-start” vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed

primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at <https://github.com/gr812b/CVT-Simulator>. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in [Section C.3](#).

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): <https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z>,
- archival DOI landing page (placeholder): <https://doi.org/XX.XXXX/zenodo.XXXXXXX>.

Unless otherwise stated, simulation cases in [Section 4.1](#) correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: <https://cvt.kaiarseneau.dev>. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

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Appendix A

Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint ?? reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (\text{A.1.1})$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1.1) and solve for C .

Step 1: Isolate the square root. Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \quad (\text{A.2.1})$$

Step 2: Square both sides.

$$[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})]^2 = 4[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2]. \quad (\text{A.2.2})$$

Step 3: Solve for C^2 .

$$C^2 = \frac{[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2. \quad (\text{A.2.3})$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2,$$

the approximate center distance becomes

$$C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2}. \quad (\text{A.2.4})$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	Value	SI Value
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer},\text{min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer},\text{max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer},\text{min}} = 0.8125 + 0.613 = 1.4255$ in.

Computed Center Distances Solving the exact belt-length constraint ?? numerically yields

$$C_{\text{exact}} = 0.251\,74\text{ m} \approx 9.911\text{ in.}$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\text{ m} \approx 10.566\text{ in.}$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\text{ m} = 0.655\text{ in,} \quad (\text{A.3.1})$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \quad (\text{A.3.2})$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C , the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation [Equation \(A.1.1\)](#), this yields a closed-form expression.

Step 1: Isolate the square root. From [Equation \(A.1.1\)](#):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \quad (\text{A.4.1})$$

Step 2: Square both sides.

$$[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}]^2 = 4[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2]. \quad (\text{A.4.2})$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2 + 4)r_{s,\text{outer}}^2 - 2[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}]r_{s,\text{outer}} + [(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2] = 0. \quad (\text{A.4.3})$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$r_{s,\text{approx}} = \frac{\pi L_b + (4 - \pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2 + 4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2 + 4}. \quad (\text{A.4.4})$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx} , so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from ??:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer},\text{min}} = 0.036\,21\text{ m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\text{ m}.$$

Evaluating the approximate formula [Equation \(A.4.4\)](#) with $C_{\text{approx}} = 0.268\,37\text{ m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\text{ m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\text{ m}$:

Quantity	Exact	Approximate
$r_{p,\text{eff}}$	0.0284 m	0.0284 m
$r_{s,\text{eff}}$	0.0938 m	0.0938 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using ?? with $s = s_{\max} = 0.75 \text{ in} = 0.01905 \text{ m}$ and deadzone $s_{\text{dz}} = 0.0024892 \text{ m}$:

$$\begin{aligned}
r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer},\min} + \frac{s_{\max} - s_{\text{dz}}}{2 \tan \beta} \\
&= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5^\circ)} \\
&= 0.03621 + \frac{0.01656}{0.4070} \\
&= 0.0769 \text{ m.}
\end{aligned} \tag{A.4.5}$$

Secondary Radius Comparison Solving the exact belt-length constraint ?? numerically with $C_{\text{exact}} = 0.25174 \text{ m}$ yields

$$r_{s,\text{exact}} = 0.06673 \text{ m.}$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.26837 \text{ m}$ yields

$$r_{s,\text{approx}} = 0.05626 \text{ m.}$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (??):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.00779 \text{ m}$:

Quantity	Exact	Approximate
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	Ratio Error
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\text{max}}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 3.9), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

Appendix B

Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From ??, the centrifugal axial force is

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds}, \quad (\text{B.0.1})$$

where m_f is the flyweight mass, $r_{f,0}$ is the initial flyweight radius, and $r_f(s)$ is the radial displacement along the ramp as a function of axial shift s .

The profile $r_f(s)$ must satisfy the admissibility conditions from ??: continuously differentiable, non-negative slope, and finite slope throughout $[0, s_{\max}]$.

B.1 Linear Ramp

A linear ramp with slope k_r (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \quad \frac{dr_f}{ds} = k_r = \text{const.} \quad (\text{B.1.1})$$

Substituting into [Equation \(B.0.1\)](#):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 k_r (r_{f,0} + k_r s) = m_f \omega_p^2 k_r r_{f,0} + m_f \omega_p^2 k_r^2 s. \quad (\text{B.1.2})$$

At constant ω_p , axial force increases linearly with s because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles α_1 and α_2 at the start and end of the segment, where $|dr_f/ds| = \tan \alpha$ specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius R_c centered at (s_c, r_c) , the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \quad (\text{B.2.1})$$

Solving for $r_f(s)$ on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \quad (\text{B.2.2})$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \quad (\text{B.2.3})$$

Parameterization from slope angles. At any point on the arc where the local slope angle is α (i.e. $dr_f/ds = \tan \alpha$), the circle constraint combined with Equation (B.2.3) gives the parametric form:

$$s = s_c - R_c \sin \alpha, \quad r_f = r_c + R_c \cos \alpha. \quad (\text{B.2.4})$$

The slope angle α decreases monotonically from α_1 (entry) to $\alpha_2 < \alpha_1$ (exit) as s increases along the ramp. Applying Equation (B.2.4) at both boundary shifts s_1 and s_2 and subtracting:

$$s_2 - s_1 = R_c(\sin \alpha_1 - \sin \alpha_2). \quad (\text{B.2.5})$$

With $\Delta s = s_2 - s_1$ as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin \alpha_1 - \sin \alpha_2}, \quad s_c = s_1 + R_c \sin \alpha_1. \quad (\text{B.2.6})$$

The boundary condition $r_f(s_1) = 0$ (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos \alpha_1. \quad (\text{B.2.7})$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \quad (\text{B.2.8})$$

To see the cancellation explicitly, split $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$:

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[\underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 (s - s_c) \left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \quad (\text{B.2.9})$$

This form reveals the structure clearly. The dominant factor is $-(s - s_c)$, which is linear in shift: the $(r_f - r_c)$ terms cancel exactly in the first part of the split. The bracket $\left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]$ is a nonlinear correction multiplier. Using the parametric form Equation (B.2.4) with Equation (B.2.7), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos \alpha_1}{R_c \cos \alpha}, \quad (\text{B.2.10})$$

where α is the local slope angle at shift s , sweeping from α_1 at entry to α_2 at exit. This makes the effective proportionality between force and $(s - s_c)$ position-dependent.

The numerator $r_{f,0} - R_c \cos \alpha_1$ is the key residual: it is controlled entirely by the choice of α_1 , α_2 , and Δs through $R_c = \Delta s / (\sin \alpha_1 - \sin \alpha_2)$. Increasing either α_1 or Δs raises R_c , making $R_c \cos \alpha_1$ larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing α_1 and α_2 adjusts both simultaneously.

The condition $r_{f,0} = R_c \cos \alpha_1$ makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed ω_p , the centrifugal force is approximately proportional to $(s - s_c)$, growing linearly with shift. This directly matches the linear restoring force of the primary spring (force $\propto k_s \cdot s$). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset $(r_{f,0} + r_c)$ is small relative to the arc radius R_c , but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.2.10), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\text{cent}}}{-m_f \omega_p^2 (s - s_c)} = 1 + \frac{r_{f,0} - R_c \cos \alpha_1}{R_c \cos \alpha}. \quad (\text{B.3.1})$$

This varies with position because α changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call K , and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s)) \frac{dr_f}{ds} = Ks, \quad (\text{B.3.2})$$

where $K > 0$. This produces centrifugal force exactly proportional to shift:

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 K s. \quad (\text{B.3.3})$$

K is not a free parameter any more than R_c is for the circular arc: just as R_c is uniquely determined by α_1 , α_2 , and Δs (Equation (B.2.6)), K is uniquely determined by α_1 , α_2 , s_1 , and s_2 (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes K exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

Deriving the profile. Equation (B.3.2) is separable. Substituting $u = r_{f,0} + r_f$, so $du/ds = dr_f/ds$:

$$u du = Ks ds. \quad (\text{B.3.4})$$

Integrating both sides and writing the constant of integration as $\frac{1}{2}Ks_0^2$:

$$\frac{1}{2}u^2 = \frac{1}{2}K(s^2 + s_0^2). \quad (\text{B.3.5})$$

Substituting back $u = r_{f,0} + r_f$:

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2)}, \quad (\text{B.3.6})$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \quad (\text{B.3.7})$$

In the (s, r_f) plane, the locus $(r_{f,0} + r_f)^2 = Ks^2$ (taking $s_0 = 0$) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in [Equation \(B.2.9\)](#) could not be removed, this profile forces the product $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$ exactly for all s .

Local slope and its variation. To see how K relates to actual ramp angles, differentiate [Equation \(B.3.6\)](#) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K} s}{\sqrt{s^2 + s_0^2}}. \quad (\text{B.3.8})$$

Since $dr_f/ds = \tan \alpha(s)$ by definition of the ramp slope angle, the local angle satisfies:

$$\tan \alpha(s) = \frac{\sqrt{K} s}{\sqrt{s^2 + s_0^2}}. \quad (\text{B.3.9})$$

Two observations follow. First, the slope is zero at $s = 0$ and grows monotonically toward the asymptote $\sqrt{K}$ as $s \rightarrow \infty$. This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm $(r_{f,0} + r_f)$ at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product Ks linear. Second, the asymptotic slope $\lim_{s \rightarrow \infty} \tan \alpha(s) = \sqrt{K}$ gives a simple interpretation: $K = \tan^2 \alpha_\infty$ where α_∞ is the slope the profile approaches at large shift.

Parameterization by entry and exit angles. Following the same convention as [Section B.2](#), specify desired slope angles α_1 and α_2 at entry shift s_1 and exit shift s_2 . From [Equation \(B.3.9\)](#):

$$\tan^2 \alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \quad \tan^2 \alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \quad (\text{B.3.10})$$

Dividing the two equations eliminates K :

$$\frac{\tan^2 \alpha_1}{\tan^2 \alpha_2} = \frac{s_1^2 (s_2^2 + s_0^2)}{s_2^2 (s_1^2 + s_0^2)}. \quad (\text{B.3.11})$$

Solving for s_0^2 :

$$s_0^2 = \frac{s_1^2 s_2^2 (\tan^2 \alpha_2 - \tan^2 \alpha_1)}{\tan^2 \alpha_1 s_2^2 - \tan^2 \alpha_2 s_1^2}. \quad (\text{B.3.12})$$

Then K follows from either condition in Equation (B.3.10):

$$K = \frac{\tan^2 \alpha_1 (s_1^2 + s_0^2)}{s_1^2}. \quad (\text{B.3.13})$$

For $s_0^2 > 0$ a consistent solution requires the numerator and denominator of Equation (B.3.12) to have the same sign. Since the hyperbolic slope increases with s (Eq. (B.3.8)), the physically consistent case is $\alpha_2 > \alpha_1$ (exit angle larger than entry), with $\tan^2 \alpha_1 s_2^2 > \tan^2 \alpha_2 s_1^2$ ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ($\tan^2 \alpha_1 s_2^2 = \tan^2 \alpha_2 s_1^2$).

Practical impact. In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc’s partial cancellation is not a fundamental limitation but rather a small correction from the constant offset ($r_{f,0} + r_c$).

Limitations. Both profiles address only the centrifugal force term. The secondary pulley torque feedback ($F_{s,\text{ax}}(s, \tau_s)$ in Equation (3.9.11)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts $s_1, s_2, \dots, s_{n-1}$. This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

Admissibility at junctions. The admissibility conditions from ?? require $r_f(s)$ to be continuously differentiable (C^1). A piecewise ramp satisfies C^0 by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at s_k differs from the right-segment slope, Eq. (B.0.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

Numerical stiffness considerations. A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative $d^2 r_f / ds^2$ near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores C^1 continuity and bounds $d^2 r_f / ds^2$ throughout the ramp.

Appendix C

Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C , the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from ?? into the residual form:

$$g(r_2; r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1) \arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \quad (\text{C.1.1})$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\min}, r_{2,\max})$ with C as the unknown; this is exactly the initialization step implied by ?? before the ratio law [Equation \(3.9.2\)](#) can be evaluated over the shift range.

Bracketing strategy. The arcsin term in [Equation \(C.1.1\)](#) is real only when $|r_2 - r_1| \leq C$, so the natural domain for r_2 is the open interval $(r_1 - C, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \rightarrow r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \rightarrow 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \rightarrow r_1 + C$, by symmetry $g \rightarrow 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon, L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent’s method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent’s algorithm [Brent \(1973\)](#). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order- ≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

C.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in [Equation \(3.9.13\)](#), with rotational sub-equations [Equation \(3.9.8\)](#) and [Equation \(3.9.9\)](#), axial dynamics [Equation \(3.9.11\)](#), belt-speed dynamics for v_b , and algebraic geometry closures [Equation \(3.9.2\)](#) and [Equation \(3.9.3\)](#). Although [Equation \(3.9.11\)](#) is second-order in s , introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the RK45 method — the Dormand–Prince explicit embedded Runge–Kutta pair [Dormand and Prince \(1980\)](#). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\text{atol} + \text{rtol} \cdot |y_i|} \leq 1, \quad (\text{C.2.1})$$

and the next step size is scaled proportionally to $(\text{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\text{atol} = 10^{-6}$ and $\text{rtol} = 10^{-4}$ ([Table C.1](#)). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	Value	Interpretation
atol	10^{-6}	absolute floor: 1 μm on s , 1 prad/s on ω
rtol	10^{-4}	relative tolerance: 0.01% of current state magnitude
t_eval	10 000 pts	uniform output grid over 30 s

Table C.1: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`’s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in [Equation \(3.9.11\)](#) and the full state evolution in [Equation \(3.9.13\)](#). After each accepted step the solver evaluates the event functions on

a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping [Equation \(3.9.11\)](#) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in [Equation \(3.8.9\)](#) is physically clear but discontinuous at $v_{\text{rel}} = 0$, which can induce step-size chatter near stick-slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \quad (\text{C.3.1})$$

where $v_{\text{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\max}, \tau_{\max}). \quad (\text{C.3.2})$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}}, 0, 1\right), \quad (\text{C.3.3})$$

with $v_{\text{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha) \tau_{\text{req}} + \alpha \tau_{\text{sat}}. \quad (\text{C.3.4})$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \rightarrow 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth} . Choose the smoothing scale small enough that $\tanh(v_{\text{rel}}/v_{\text{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\text{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth} , so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

Appendix D

Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from [Section 3.1.4](#). Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from [??](#):

$$R_g = \frac{r_s}{r_p}, \quad (\text{D.0.1})$$

The individual primary and secondary radius rates and accelerations have already been established in [Section 3.1.4](#). This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

D.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in [Equation \(D.0.1\)](#) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt} \left(\frac{r_s}{r_p} \right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2} \dot{r}_p. \end{aligned} \quad (\text{D.0.2})$$

The secondary-radius result in [Equation \(3.1.51\)](#) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p. \quad (\text{D.0.3})$$

Substituting this into Equation (D.0.2) gives

$$\begin{aligned}
\dot{R}_g &= \frac{1}{r_p} \left(-\frac{\phi_p}{\phi_s} \dot{r}_p \right) - \frac{r_s}{r_p^2} \dot{r}_p \\
&= -\frac{\dot{r}_p}{r_p} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \\
&= -\frac{\dot{r}_p}{r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right).
\end{aligned} \tag{D.0.4}$$

Finally, Equation (3.1.30) gives $\dot{r}_p = \dot{s}/(2 \tan \beta)$ in the active-shift branch. Therefore,

Rate of Change of Geometric CVT Ratio	(D.0.5)
$ \dot{R}_g = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right), & s_{dz} < s \leq s_{max}. \end{cases} $	

For an upshift, $\dot{s} > 0$, so the available geometric reduction decreases as the belt moves outward on the primary pulley.

D.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{D.0.6}$$

Thus, every point in the (r_p, r_s) plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant- R_g contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length L_b or center-to-center distance C —is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure D.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both

panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

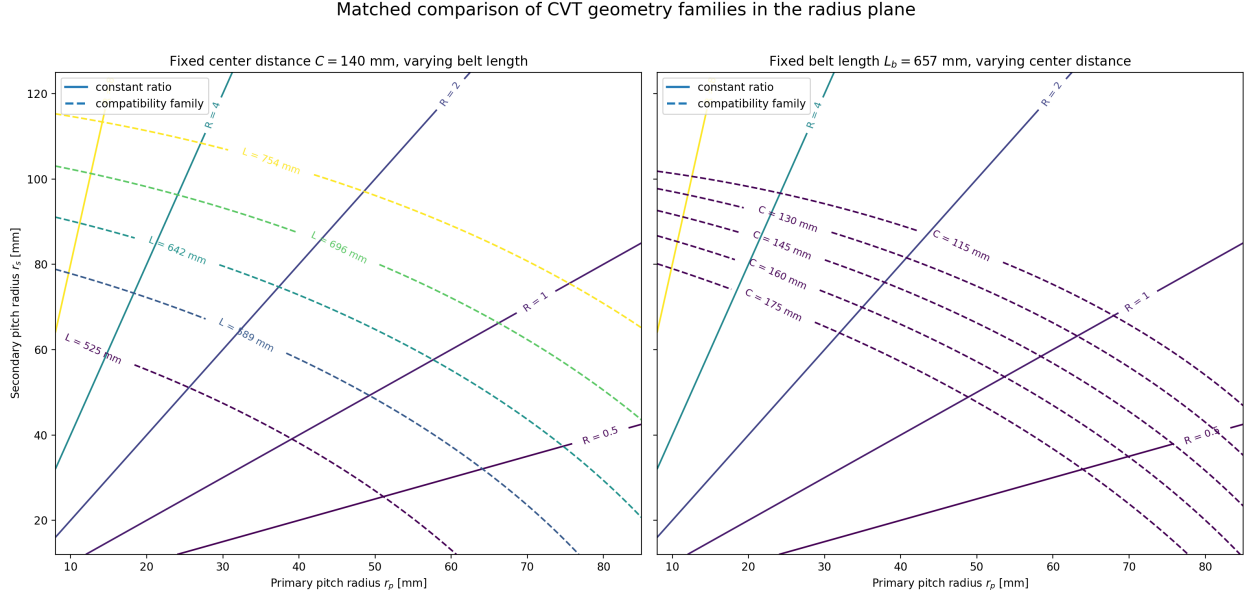


Figure D.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio $R_g = r_s/r_p$. In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing L_b or C changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, [Figure D.1](#) is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

D.0.3 Exploring the Local Ratio-Change Map

The expression for $\dot{R}_g$ gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide

out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right). \quad (\text{D.0.7})$$

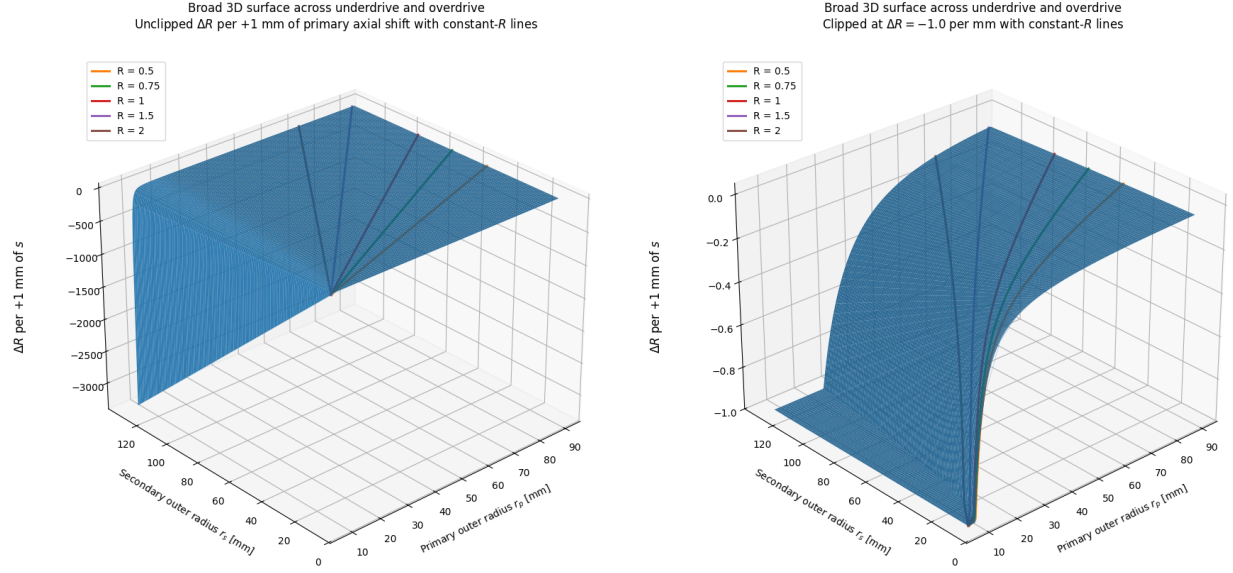
For a small positive increment Δs ,

$$\Delta R_g \approx \frac{dR_g}{ds} \Delta s. \quad (\text{D.0.8})$$

The plots below take $\Delta s = 1 \text{ mm}$, so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing s reduces the geometric reduction ratio.

A sensitivity field over the radius plane The first view is a broad surface over the radius plane. Here, r_p and r_s are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

Figure D.2 shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit $1/r_p$ factor in **Equation (D.0.7)**, and is reinforced by the fact that the geometric ratio $R_g = r_s/r_p$ is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the $R_g = 1$ locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small r_p . (b) The same domain with values below -1 per 1 mm clipped, exposing the curvature through the mid-range.

Figure D.2: Local geometric-ratio change for a 1 mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the $R_g = 1$ locus.

How belt choice and installed geometry select a path The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the (r_p, r_s) plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure D.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small- r_p region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

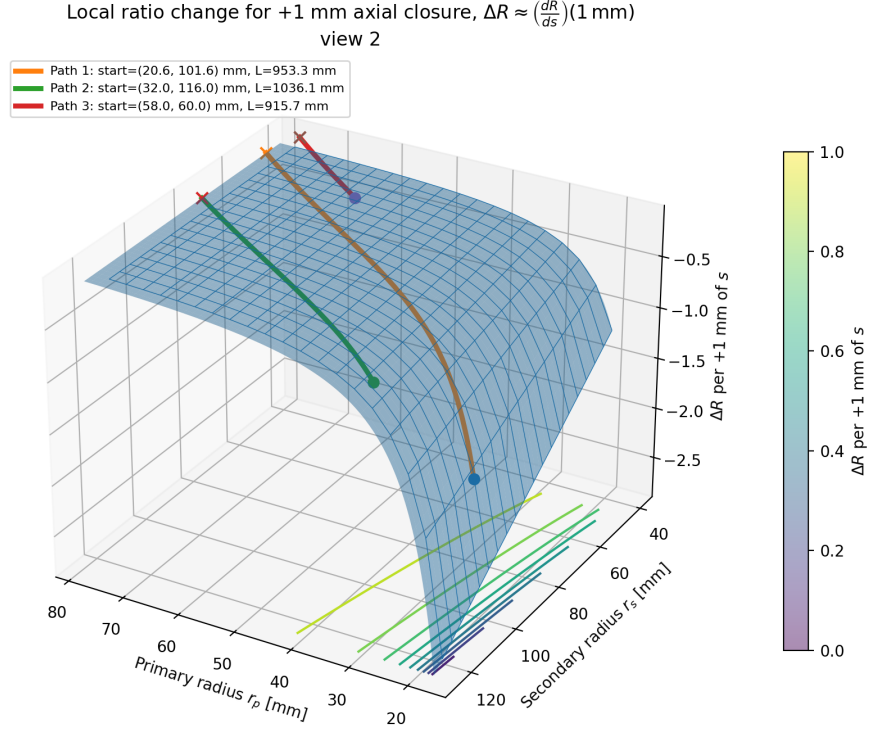
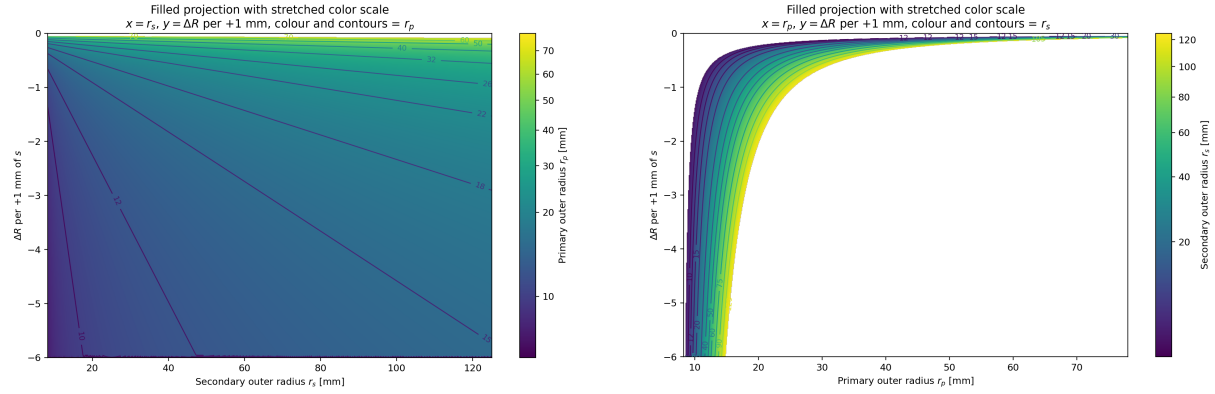


Figure D.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

Separating the roles of the two pulley radii The two projections in [Figure D.4](#) help isolate the individual effects of r_p and r_s . In [Figure D.4\(a\)](#), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing r_s increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

[Figure D.4\(b\)](#) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as r_p approaches its lower limit, then flattens at larger radius. This is the visual form of the $1/r_p$ dependence in [Equation \(D.0.7\)](#). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. (b) Local ratio change against primary radius. Colour and contours identify the primary radius. Colour and contours identify the secondary radius.

Figure D.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

Design interpretation Taken together, [Figure D.1](#) and [Figure D.2](#)–[Figure D.4](#) give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small r_p can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible r_p is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by [Chen et al. \(1998\)](#) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, [Childs and Cowburn \(1987\)](#) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small- r_p region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable ([Gerbert, 1996](#); [Bertini et al., 2014](#); [Messick, 2018](#)).

D.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (D.0.2) term by term to capture both effects.

For the first term, $\dot{r}_s/r_p$, use the quotient rule:

$$\frac{d}{dt} \left(\frac{\dot{r}_s}{r_p} \right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2}. \quad (\text{D.0.9})$$

For the second term, $r_s \dot{r}_p / r_p^2$, all three factors may change. Using the product rule,

$$\frac{d}{dt} \left(\frac{r_s \dot{r}_p}{r_p^2} \right) = \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3}. \quad (\text{D.0.10})$$

Subtracting Equation (D.0.10) from Equation (D.0.9) gives

$$\begin{aligned} \ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2} - \left[\frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3} \right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s \dot{r}_p}{r_p^2} - \frac{r_s \ddot{r}_p}{r_p^2} + \frac{2r_s \dot{r}_p^2}{r_p^3}. \end{aligned} \quad (\text{D.0.11})$$

The main derivation gives the required coupled secondary motion in ????:

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p, \quad (\text{D.0.12})$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2. \quad (\text{D.0.13})$$

Substituting these into Equation (D.0.11) gives

$$\begin{aligned} \ddot{R}_g &= \frac{1}{r_p} \left[-\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2 \right] - \frac{2}{r_p^2} \left[-\frac{\phi_p}{\phi_s} \dot{r}_p \right] \dot{r}_p \\ &\quad - \frac{r_s}{r_p^2} \ddot{r}_p + \frac{2r_s}{r_p^3} \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \ddot{r}_p + \left[\frac{2}{r_p^2} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right) \ddot{r}_p + \left[\frac{2}{r_p^2} \left(\frac{\phi_p}{\phi_s} + R_g \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2. \end{aligned} \quad (\text{D.0.14})$$

Finally, use the primary radius rate and acceleration from [Equations \(3.1.30\)](#) and [\(3.1.59\)](#):

Acceleration of Geometric CVT Ratio

(D.0.15)

$$\ddot{R}_g = \begin{cases} 0, & 0 \leq s < s_{\text{dz}}, \\ -\frac{\ddot{s}}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right) + \frac{\dot{s}^2}{2 \tan^2 \beta r_p^2} \left(\frac{\phi_p}{\phi_s} + R_g \right) - \frac{2\pi^2 \dot{s}^2}{\phi_s^3 r_p \tan^2 \beta \sqrt{C^2 - (r_s - r_p)^2}}, & s_{\text{dz}} < s \leq s_{\text{max}}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to $\ddot{s}$ is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity dR_g/ds , so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to $\dot{s}^2$ arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation $R_g(s)$: as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.

Appendix E

Primary Flyweight Geometry Examples

Placeholder